Ramsey Alignment and the Postwar Tax System

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How far is the tax system from optimal, and how much does it matter? I develop a sufficient statistic that answers both questions locally: the cosine of the angle between the welfare and revenue gradients. I show that this sufficient statistic applies across a wide range of economies, including those featuring dynamics, cross-base spillovers, and nonlinear tax instruments. In an application to federal tax policy in the United States, I estimate a general equilibrium revenue gradient for a planner with access to a personal income tax, a corporate tax, a tariff, and a nonlinear income tax instrument. I find that the tax system became substantially more aligned with the local Ramsey optimum over time across these margins, a result largely driven by the long-run decline in the corporate tax rate.

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1 Introduction

The case for tax reform rests on how far the system is from whatever policymakers perceive as optimal. And yet, without a structural model, it is difficult to assess the scope for reform or the direction reform should take. However, structural models also tend to disagree on both of those questions, which creates a difficulty for both policymakers and economists in deciding what ought to be reformed and what the gains would be from reform. In this paper, I develop a set of sufficient statistics that answers both questions locally with minimal structure.

The sufficient statistics emerge from a simple insight: at a local Ramsey optimum, no revenue-neutral tax reform can improve welfare. From the Ramsey planner's first-order condition, the marginal welfare cost per dollar of revenue raised must be equal across tax instruments. The contribution is that, to first order, this condition can be represented geometrically by the angle θ between the revenue and welfare gradients, and this angle can be used as an aggregate diagnostic for the tax system. While the classic tax reform literature used this geometry to identify the optimal direction of reform (Dixit 1979; Tirole and Guesnerie 1981), I use the angle to measure how closely the existing system satisfies the first-order condition. When $\cos\theta = 1$, the welfare and revenue gradients point in the same direction and the system is perfectly aligned. When $\cos\theta = 0$, the two gradients are orthogonal, so the unconstrained welfare-improving direction is already revenue-neutral. A simple transformation then shows that, for a reform of a given size, $\sin\theta$ measures the share of the largest local welfare gain that remains available under revenue neutrality. Thus, the cosine and sine of the same angle state how closely the system satisfies the Ramsey condition locally and how much can be gained from reform.

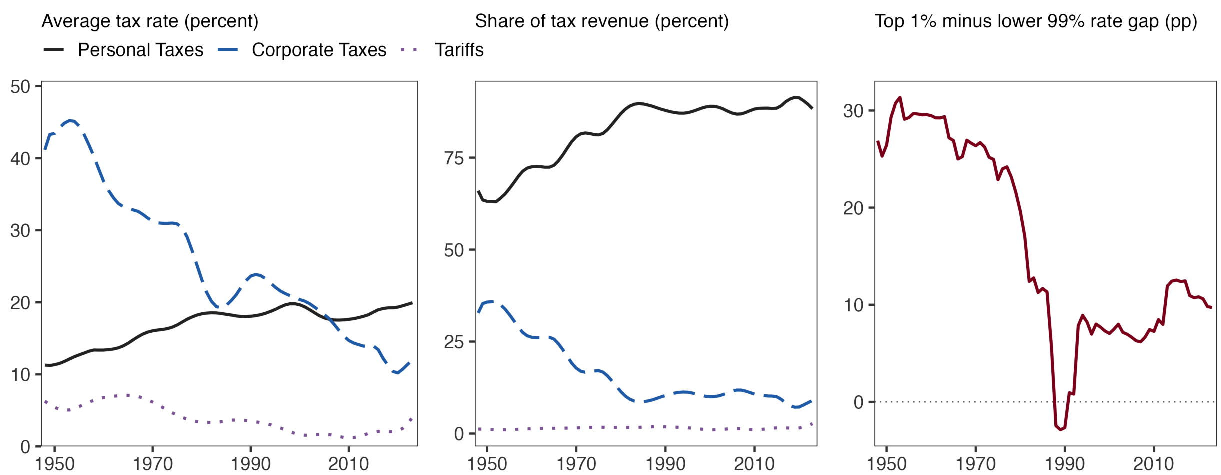
The sufficient statistics are closely related to the literature on the marginal value of public funds (MVPF), which is also exactly what makes them measurable (Mayshar 1990; Bastani 2025). For any particular tax change, the MVPF is the marginal welfare cost relative to the marginal revenue it raises, which just means that equality across MVPFs is another representation of the planner's first-order condition. The cosine provides an aggregate diagnostic of whether that equality holds across the included tax changes, while remaining well-defined when a marginal revenue effect is zero or negative. The same local logic makes the required objects measurable. As Hendren (2016) and Hendren and Sprung-Keyser (2020) show, the envelope theorem allows the welfare cost of a marginal policy change to be recovered from the mechanical burden it imposes, once its incidence is specified. Under the equal incidence benchmark, the welfare gradient therefore depends only on tax bases. On the other hand, the revenue gradient can be recovered from causal estimates of the elasticities of taxable income across estimates, which may include dynamics and spillovers across tax bases. Those dynamic general equilibrium effects are especially important in macroeconomic applications, where a tax change may affect both future revenue from its own base and revenue collected through other instruments. The next example illustrates this logic in a familiar Ramsey setting.

To get at the economics behind the geometry, consider the canonical neoclassical model in which a government chooses a broad labor tax and a broad capital tax for a representative household with time-separable isoelastic preferences. In this representation, reform concerns the division of the tax burden between two broad income bases; it does not change the distribution of rates within either one. A permanent capital tax distorts saving in every subsequent period, affecting revenue not only from capital income but also from the future labor and capital income generated by accumulated capital. These dynamic fiscal effects make capital taxation especially costly in the Chamley (1986) and Chari, Nicolini, and Teles (2020) models. At the long-run Ramsey allocation, the capital tax is zero, and the welfare and revenue gradients are aligned across the two broad instruments. Away from that allocation, a positive capital tax creates scope for a revenue-neutral shift of the burden toward labor income more broadly.

However, policymakers and economists are typically just as interested in the distribution of a tax burden within an instrument—like the progressivity of the income tax system—as they are in its average burden, and I show that the same alignment statistics apply to both margins. Alongside a broad reform, which changes how much revenue is raised through a tax base, a policymaker can shift tax payments within a schedule while holding total payments fixed before behavioral responses. That shift can nevertheless change total revenue. For example, increasing the top tax rate may induce passthrough businesses to become corporations, thereby increasing corporate tax revenue. These spillovers affect the revenue gradient, while any distributional weight the planner places on the burden affects the welfare gradient. Under equal incidence, the first channel remains even when the second is absent: the planner may prefer a more progressive schedule not because redistribution has direct social value, but because its revenue effects permit a welfare-improving change in broad tax rates. Thus, the alignment metrics can measure alignment for systems with both linear and nonlinear instruments using the same types of welfare and revenue gradients for each.

The postwar transformation of the federal tax system provides a natural setting for this framework. At the end of World War II, corporate and personal income taxes together accounted for nearly all of federal tax revenue, while the tariff share was negligible. Until 2018, almost all reform happened on the income tax margin, with the corporate share falling from nearly 50 percent to below 15 percent and personal taxes absorbing the difference. Since then, tariff increases in 2018 and 2025 brought the customs share of tax revenue to nearly ten percent—well above the typical postwar share of 2-3 percent. At the same time, the personal tax schedule became less progressive, as the marginal tax rate faced by the top 1 percent fell relative to the lower 99 percent. Those developments prompt two questions. First, was the shift from corporate to personal taxation welfare-improving? Second, how large a role should tariffs play in the federal tax system? Third, can changing the progressivity of the personal tax schedule create additional scope for welfare-improving, revenue-neutral reform?

Figure 1: Changes in Postwar Sources of Revenue and Tax Rates



Note: The left panel plots the averagae tax rates for personal income, corporate income, and imports. The middle panel plots the corresponding revenue shares. The right panel, using Mertens and Montiel Olea (2018), plots the difference between the top income percentile's marginal tax rate and the marginal tax rate for the bottom 99%.

I study the alignment of the federal tax system by constructing welfare and revenue gradients for four margins: the broad personal, corporate, and tariff rates, as well as the progressivity of the personal tax schedule. Under an equal incidence benchmark, an application of the envelope theorem yields a welfare gradient for each of the three broad instruments equal to its incidence-adjusted observed tax base, while a change in progressivity has no direct first-order welfare effect. The bigger empirical challenge is the revenue gradient. In this setting, the central models and questions pertaining to the broad structure of the tax system imply that reforms may have important spillovers effects on other tax bases and instruments, as well as effects over time. I therefore use time series methods, together with exogenous variation in tax policy, to estimate the general equilibrium responses of tax bases to each instrument and construct the corresponding revenue semi-elasticities.

I use standard methods and narrative instruments to construct the revenue gradient, which can be decomposed into the three broad components and the progressivity entry. I estimate these components separately because the broad analysis uses quarterly variation in average tax rates, while the progressivity analysis uses annual variation in tax rates across the income distribution. First, I use narrative instruments for personal and corporate tax changes from Mertens and Ravn

(2013), as well as tariff instruments from Besten and Känzig (2026), to estimate the impulse responses of each tax base to each broad tax instrument using Bayesian local projections (Cloyne et al. 2022). Cumulating those impulse responses and using the complete matrix of discounted tax-rate responses to re-express the matching base responses as responses to a change in one rate while holding the others fixed yields a 3×3 matrix of present-value elasticities of the three tax bases with respect to the three broad rates. These nine elasticities are a key input to the broad revenue gradient. Evaluating progressivity is similar, but requires annual data. I split the income distribution into the top 1% and the lower 99% using data from Mertens and Montiel Olea (2018). I then use the joint variation in those two rates across exogenous tax reforms reforms to construct the semi-elasticities of the two personal income bases, as well as the corporate and tariff bases, with respect to a change that raises the top-1 tax rate relative to the lower-99 rate while holding personal tax revenue fixed. It is then straightforward to construct the revenue semi-elasticity for each of the four reform margins by combining the observed tax rates and tax bases with the estimated broad elasticities and progressivity semi-elasticities.

The resulting semi-elasticities of revenue for each instrument change substantially over the postwar era. The revenue semi-elasticity of personal taxation is by far the largest of the four margins and remains significantly positive throughout the sample. It starts around 2 and grows to roughly 3.5 by the 1980s, staying stable until after then. By contrast, the corporate tax and tariff margins have modestly negative point estimates in the early postwar era, though neither is precisely estimated then. Both rise over time to just below one: by the 1970s, the corporate tax margin is positive, and the tariff margin is positive at the 68-percent credible level. The progressivity margin moves in the opposite direction. Its revenue semi-elasticity is around one initially, but falls to zero from 1980 onward. All four movements are largely attributable to the long-run postwar decline in the corporate tax rate, most of which had occurred by the 1980s. Cuts in all three broad instruments increase corporate profits, so the fiscal externality of higher taxes on any of those three declines when the corporate tax rate itself declines. By contrast, higher progressivity increases corporate profits, likely in part through income shifting, so a lower corporate tax rate reduces the revenue-raising potential of marginally higher progressivity.

Finally, I evaluate the evolution of the federal tax system's local alignment with the Ramsey optimum over time. Under the equal incidence benchmark, I show that when we only consider personal and corporate taxes, the tax system was initially somewhat aligned in the 1950s, but the gains from reform on that margin fell to almost nothing by the 1980s. If we add in tariffs, then there is considerably more scope for alignment early in the period, but again, it becomes almost fully aligned by the 1980s, in part because tariff revenue is so small that it cannot affect the system locally. However, progressivity considerably changes the picture. In the full four-margin system, the median cosine draw is only 0.25 in 1950 and rises to rougly 0.8 by 1990, where it has stayed since, albeit with considerable posterior uncertainty. Thus, the system is largely aligned on the broad margins of how revenue is raised, but not necessarily when accounting for progressivity. That is reflected in the locally recommended reforms: throughout the postwar era, the optimal reform is to modestly cut personal taxes, while there is currently no indication that raising corporate taxes or tariffs is optimal. Under equal incidence, progressivity has no local effect on revenue after 1980, nor on welfare, so there is little reason to alter it either.

In short, the application shows how the framework developed in the first part of the paper converts observable tax bases and causal responses to tax policy into an aggregate diagnostic for the proximity of a tax system to its local Ramsey condition, and for the corresponding direction of reform, without committing to a complete structural model. Under the equal incidence benchmark, the broad federal tax mix is close to alignment in the later postwar period. Nevertheless, the precise level of alignment is somewhat sensitive to a number of choices, but the framework makes them explicit and permits their consequences for the local Ramsey condition to be assessed directly rather than buried.

Related Literature. This paper primarily contributes to a classic literature on tax reform. The central insight of that literature is that the welfare consequences of small revenue-neutral reforms can be characterized using local welfare and revenue derivatives (Dixit 1975; Guesnerie 1977). The projection theorem of Tirole and Guesnerie (1981) shows that projecting the welfare gradient onto the revenue-neutral policy space identifies the best small reform under a specified policy metric.[1] I build on that projection theorem—and the geometric approach to tax reform—to show that the angle between the gradients is sufficient to identify the proximity of the tax system to the planner's local optimality condition and the welfare gains from reforming in its direction. Hence, the cosine and sine of that angle are sufficient statistics for assessing tax systems as a whole. That remains true across many economies, not just in the static variants occasionally used by the classic literature to study commodity taxation (Dixit 1975). I also extend the representation to nonlinear tax instruments. In that setting, the planner can change the broad burden imposed by an instrument, like the income tax, as well as the distribution of the burden within it. In short, I show how taking the insights of the early literature one step further permits an aggregate diagnostic across a variety of tax systems and settings.

My approach to estimating the sufficient statistics builds on the marginal value of public funds literature's insight that fiscal externalities can be estimated from causal evidence and that the welfare gradient is observable via the envelope theorem (Hendren 2016; Finkelstein and Hendren 2020). I estimate these fiscal externalities using narrative identification, constructing present-

1. Diewert (1978) and Dixit (1979) provide related many-consumer treatments.

value revenue gradients from impulse responses to tax shocks.[2] In seeking apply the reform literature empirically, I build most closely on Ahmad and Stern (1984), which applied the commodity tax reform framework to Indian commodity taxes with a static demand system. Whereas their estimates required estimates of the full matrix of own- and cross-price demand responses, I estimate macro revenue semi-elasticities. This builds on McKay and Wolf (2023) and Caravello, McKay, and Wolf (2024), who show that impulse responses are sufficient to construct Lucas-critique-robust counterfactuals under alternative policy rules. My application is narrower, evaluating whether actual tax changes moved toward or away from the locally optimal direction rather than simulating counterfactual policies. The geometric approach—testing whether welfare and revenue gradients are aligned—also parallels Barnichon and Mesters (2023), who evaluate monetary policy by testing whether the gradient of the loss function with respect to policy shocks is zero; both frameworks detect suboptimal policies from sufficient statistics without requiring a fully specified structural model.

Finally, this paper brings together the sufficient statistics literature in public finance and the more parametric literature in macroeconomics. Existing approaches to evaluating tax reform face a trade-off between adequately capturing general equilibrium forces and avoiding parametric disagreement. Structural general equilibrium models capture both cross-base spillovers and dynamic accumulation, but require parametric commitments that can lead to fundamentally different conclusions. For example, a subclass of neoclassical models would conclude the postwar shift toward personal taxation was optimal (Chari, Nicolini, and Teles 2020), while overlapping generations models reach the opposite conclusion (Conesa, Kitao, and Krueger 2009). Even within the neoclassical framework, the optimal tax on capital—and hence the optimal direction of reform—depends sensitively on preference specifications and heterogeneity (Straub and Werning 2020). Sufficient statistics approaches avoid much of this model dependence by working directly with causal elasticities (Chetty 2009; Saez and Stantcheva 2018; Hendren 2016; Kleven 2021). Many empirical applications, however, rely on static own-base elasticities estimated from micro data (Saez 2001; Hendren and Sprung-Keyser 2020). Such estimates do not by themselves recover the cross-base and dynamic spillovers that may be important for evaluating tax reform in certain contexts.[3] Both approaches are generally used to characterize optimal tax rules or the direction of a particular reform rather than to summarize how much scope for local reform remains across an entire tax system.[4] Rather than infer cross-base and dynamic responses from a structural model or approx-

2. I build on the narrative approach to identification by modestly extending Mertens and Montiel Olea (2018) through the Tax Cuts and Jobs Act, as well as refining the Besten and Känzig (2026) tariff instruments at the quarterly level.

3. Saez, Slemrod, and Giertz (2012) likewise emphasize that tax responses may spill across instruments, in their case between capital gains and labor income, and Kleven et al. (2025) directly estimates dynamic elasticities.

4. Vergara and Swonder (2025), for example, incorporates cross-base fiscal externalities from corporate taxes on the labor base in a static optimal corporate tax formula.

imate them with micro elasticities, I estimate them directly from causal variation in aggregate tax policy. The resulting revenue gradient, together with an incidence-adjusted welfare gradient, yields an aggregate diagnostic of local Ramsey alignment in a specified reform space without requiring a complete model of preferences or production.

2 The Geometry of Reform

When a tax system is not optimal, two questions follow: which direction to reform it, and how far it stands from the optimum. This section answers both with two sufficient statistics, the welfare gradient and the revenue gradient, which can be estimated from reduced-form evidence without specifying preferences, production, or market structure. The answer is geometric: the optimal direction of revenue-neutral reform is the orthogonal projection of the welfare gradient onto the revenue-neutral hyperplane, and the system's alignment with the Ramsey optimum is the cosine of the angle between the two gradients.

2.1 Setup: Taxes, Bases, and Gradients

Consider a government evaluating small changes to an existing tax system. The system may contain any number of taxes, and a tax change at one date may affect tax bases and revenue at other dates. Let $\{\tau_{i,t}\}_{i=1,\dots,n}^{t=0,\dots,T}$ denote n tax rates over $T+1$ periods. The index i may refer either to a broad tax or to an individual rate within a nonlinear tax schedule. Stack these rates into the policy vector

$$\boldsymbol{\tau} := (\tau_{1,0}, \dots, \tau_{n,0}, \tau_{1,1}, \dots, \tau_{n,1}, \dots, \tau_{1,T}, \dots, \tau_{n,T})^\top \in \mathbb{R}^{n(T+1)}.$$

Let $B_{i,t}(\boldsymbol{\tau})$ denote the tax base associated with rate $\tau_{i,t}$. If instrument i indexes a tax rate within a nonlinear tax schedule, then $B_{i,t}$ is the amount of taxable income mechanically exposed to that particular rate. Importantly, I allow for tax bases to potentially depend on the entire policy sequence, reflecting the fact that tax reforms have spillovers across bases and over time. Consequently, a change to $\tau_{i,t}$ may affect not only base i at date t, but also other bases at other dates. For example, a corporate tax cut today may raise investment, increasing the capital stock and expanding future labor tax bases through production complementarity. If that cut was anticipated, then it would also affect tax bases before the reform even happens. Aggregating over bases and tax rates, it is convenient to define the present value of consolidated revenue as

$$R(\boldsymbol{\tau}) := \sum_{t=0}^{T} \beta^t \sum_{i=1}^{n} \tau_{i,t} B_{i,t}(\boldsymbol{\tau}), \tag{1}$$

where $0 < \beta \leq 1$ is the discount factor. I assume that tax bases respond smoothly to small changes in policy.

Assumption 1. *Each tax base $B_{i,t}(\boldsymbol{\tau})$ is continuously differentiable in $\boldsymbol{\tau}$ in a neighborhood of $\bar{\boldsymbol{\tau}}$.*

This assumption is satisfied in standard models. As long as the equilibrium is unique and fairly well behaved in the neighborhood of policy, it will hold. For example, standard neoclassical and overlapping generations models typically satisfy this condition. The key exclusion is an economy with lumpy behavior in the aggregate, such as a model with multiple equilibria in which a small tax change triggers a discrete jump. Because present value revenue in equation (1) is constructed from these tax bases, the assumption also implies that $R(\boldsymbol{\tau})$ is continuously differentiable. Under that assumption, we can characterize how present value revenue changes directly from the response of each tax base.

The revenue gradient. Given differentiability, define the revenue gradient $r := \nabla_{\boldsymbol{\tau}} R \in \mathbb{R}^{n(T+1)}$, which measures how present value revenue responds to marginal tax changes. Conceptually, it is useful to consider two separate decompositions of this response because they illustrate different dimensions of general equilibrium. The first decomposes the revenue response into three components. Suppose $\tau_{i,t}$ rises. It has a mechanical effect from taxing the existing base, a behavioral effect through the same tax base, and spillovers to other tax bases. Differentiating equation (1) gives Differentiating equation (1) gives

$$r_{i,t} = \underbrace{\beta^t B_{i,t}}_{\text{Mechanical effect}} + \underbrace{\sum_{s=0}^{T} \beta^s \tau_{i,s} \frac{\partial B_{i,s}}{\partial \tau_{i,t}}}_{\text{Own-base responses}} + \underbrace{\sum_{s=0}^{T} \beta^s \sum_{k \neq i} \tau_{k,s} \frac{\partial B_{k,s}}{\partial \tau_{i,t}}}_{\text{Cross-base spillovers}}. \tag{2}$$

The mechanical effect is the static revenue gain, while the second term includes all dynamic own-base responses, and the final term includes spillovers to other tax bases across dates. Second, the same revenue response can instead be decomposed according to when it occurs. Let $R_s(\boldsymbol{\tau}) := \sum_{k=1}^{n} \tau_{k,s} B_{k,s}(\boldsymbol{\tau})$ denote revenue collected at date s. Then

$$r_{i,t} = \underbrace{\beta^t \frac{\partial R_t}{\partial \tau_{i,t}}}_{\text{Contemporaneous}} + \underbrace{\sum_{s>t} \beta^s \frac{\partial R_s}{\partial \tau_{i,t}}}_{\text{Persistence}} + \underbrace{\sum_{s<t} \beta^s \frac{\partial R_s}{\partial \tau_{i,t}}}_{\text{Anticipation}}. \tag{3}$$

Equations (2) and (3) provide two complementary decompositions of the same revenue gradient. The first distinguishes how revenue changes across tax bases, while the second distinguishes when those effects occur. Existing empirical work often focuses on a particular own-base response, usually in a static partial equilibrium setting, rather than the full revenue gradient. The following two examples illustrate what the additional terms capture.

Example 1: Human capital accumulation (persistence, own base). As a first example, consider an increase in the labor tax at $t = 0$. In standard models, that may reduce human capital investment because it would lower the return to skills, lowering workers' future productivity and taxable earnings. As a result, the tax base would decline persistently beyond the reform:

$$\frac{\partial B_{L,t}}{\partial \tau_{L,0}} < 0 \quad \text{for all } t > 0. \tag{4}$$

These effects enter the own-base term in equation (2) and the persistence term in equation (3). In present value, revenue declines, and the gradient $r_{L,0}$ is smaller than the contemporaneous revenue response alone would suggest. Indeed, theoretical and quantitative work from Badel, Huggett, and Luo (2020) and empirical work from Kleven et al. (2025) suggest that accounting for this channel is particularly important, so the revenue gradient must incorporate dynamic tax base responses rather than only contemporaneous responses.

Example 2: Capital accumulation (persistence, cross base). As a second example, consider an increase in the corporate tax rate. In most models—and in canonical models for scoring tax reform (Hall and Jorgenson 1967)—that will reduce investment and therefore shrink the capital stock over time. If capital and labor are complements in production ($F_{KL} > 0$), the lower capital stock reduces the marginal product of labor and may thereby erode the future labor tax base:

$$\sum_{t=1}^{T} \beta^t \tau_{L,t} \frac{\partial B_{L,t}}{\partial \tau_{K,0}} < 0.$$

As long as future labor tax rates are positive, the channel contributes to the present-value revenue gradient $r_{K,0}$ the quantity

$$\sum_{t=1}^{T} \beta^t \tau_{L,t} \frac{\partial B_{L,t}}{\partial \tau_{K,0}}. < 0$$

Once more, this contribution appears in the cross-base term of equation (2) and in the persistence term of equation (3). It is distinct from both the mechanical revenue effect of the corporate tax change and its effects on the corporate tax base itself. Indeed, production complementarity of this kind provides a central supply-side rationale for reductions in capital and corporate income taxes (Lucas 1990).[5]

5. Anticipation provides another intertemporal channel. If agents learn at date 0 that the capital tax will change at date 5, they may adjust beforehand, so that $\partial B_{K,t}/\partial \tau_{K,5} \neq 0$ for $t < 5$. These responses enter the anticipation term in equation (3) and can make the present-value revenue effect differ from its contemporaneous counterpart. See Mertens and Ravn (2012) and Ramey (2016), and Leeper, Richter, and Walker (2012) for the relevant theoretical

The revenue gradient describes what happens to revenue when taxes change on the margin. But the purpose of reform is also to change welfare. Thus, the next piece of the setup is what happens to welfare when taxes change on the margin—the welfare gradient.

The welfare gradient. Let $W(\boldsymbol{\tau})$ denote the money-metric social welfare cost of the tax system, measured in present-value dollars, with higher values indicating lower welfare. We construct W by aggregating present-value compensating variation across groups using social welfare weights $\{\omega_h\}$, where groups are a potentially arbitrary population. For example, they could be domestic residents versus foreigners, or the top-1% versus the bottom 99% of the income distribution. Regardless, we will eventually want to incorporate how welfare changes for groups locally, which requires an assumption on the differentiability of welfare.

Assumption 2 (Welfare)**.** *$W(\boldsymbol{\tau})$ is continuously differentiable in $\boldsymbol{\tau}$ in a neighborhood of $\bar{\boldsymbol{\tau}}$.*

Mirroring the revenue gradient, the welfare gradient captures how welfare changes when taxes change on the margin:

$$g := \nabla_{\boldsymbol{\tau}} W \in \mathbb{R}^{n(T+1)}.$$

Consider a small increase $d\tau_{i,t}$. Holding the tax base fixed, the resulting increase in tax payments at date t is $B_{i,t}d\tau_{i,t}$, or $\beta^t B_{i,t}d\tau_{i,t}$ in present value dollars. Locally, agents are already choosing across all margins optimally given the tax system. The envelope theorem therefore implies that their behavioral adjustments do not generate a separate first-order term in welfare. Those adjustments can still change tax bases and government revenue, which is why they enter the revenue gradient (Finkelstein and Hendren 2020).

The remaining issue is how the mechanical burden is distributed across groups and how the planner values those groups. Let $s_{h,i,t}$ denote group h's present value burden per dollar of mechanical burden. That burden may operate through direct tax payments or through changes in wages, prices, and profits. A negative value of $s_{h,i,t}$ means that the group benefits from the tax change. Given social welfare weights $\{\omega_h\}$, define

$$\tilde{\gamma}_{i,t} := \sum_h \omega_h s_{h,i,t}.$$

Aggregating the weighted burdens across groups gives

$$g_{i,t}(\bar{\boldsymbol{\tau}}) = \beta^t \tilde{\gamma}_{i,t} B_{i,t}(\bar{\boldsymbol{\tau}}). \tag{5}$$

Thus, $\beta^t B_{i,t}$ determines the size of the mechanical burden, while $\tilde{\gamma}_{i,t}$ captures how that burden is

discussion and empirics.

distributed and socially valued. When the groups represented in W collectively bear the entire burden, $\sum_h s_{h,i,t} = 1$. That need not be the case when, for example, part of a tariff is borne abroad and foreign welfare is excluded from W. A useful benchmark, and one which I will rely on later in the empirics, assigns equal social welfare weights, $\omega_h = 1$, to all groups whose welfare is included. If those groups bear the full marginal burden of tax $\tau_{i,t}$, then

$$\sum_h s_{h,i,t} = 1,$$

so that $\tilde{\gamma}_{i,t} = 1$ and the welfare gradient simplifies to

$$g_{i,t} = \beta^t B_{i,t}. \tag{6}$$

This representation is important for the empirical implementation in this paper, which I discuss later.

The welfare representation here comes with a major caveat. Like other derivations of the marginal value of public funds, it requires all first-order consequences of a tax change to be included either in the measured burdens or in consolidated government revenue (Hendren 2016). However, if tax changes interact with an externality, market power, or another distortion that creates additional welfare consequences, those consequences must be incorporated into the welfare gradient.[6] In the empirical application, I will maintain that these additional welfare effects are either absent or small enough to ignore. This is an important restriction, but it is the same one maintained in the MVPF literature.

Under the maintained assumptions, the distinction between the welfare gradient and the revenue gradient is exact to first order: g measures the socially weighted burden on the groups represented in W, while r measures the change in government resources. Together, they suffice to measure the welfare cost of a given tax instrument through the marginal value of public funds (Mayshar 1990; Bastani 2025).

The marginal value of public funds. For any tax rate with a nonzero marginal revenue effect, the welfare and revenue gradients define a welfare-weighted marginal value of public funds.

$$\text{MVPF}_{i,t} := \frac{g_{i,t}}{r_{i,t}} = \frac{\beta^t \tilde{\gamma}_{i,t} B_{i,t}}{r_{i,t}}, \qquad r_{i,t} \neq 0. \tag{7}$$

6. For example, the effects of federal tax changes on state and local budgets should be included in consolidated revenue; pollution caused by an expansion in production creates an additional welfare effect; and changes in foreign prices following a tariff reform affect domestic welfare through the terms of trade.

When $r_{i,t} > 0$, this ratio measures the marginal social burden created per dollar of present-value revenue raised by increasing $\tau_{i,t}$. Because the social welfare weights are already incorporated into $g_{i,t}$ through $\tilde{\gamma}_{i,t}$, no additional welfare weight is applied to the ratio.

At an interior Ramsey optimum, the welfare and revenue gradients are positively proportional. Consequently, the MVPFs in equation (7) are equal across all tax rates for which they are defined (Appendix A.3 derives this condition formally). When the gradients are not proportional, there is scope for a welfare-improving revenue-neutral reform. The next subsection states this condition directly in terms of the two gradients.

2.2 Ramsey Alignment

Having characterized the welfare and revenue effects of marginal tax changes, I now proceed in two steps. First, I ask how those changes should be combined to generate an optimal local reform, which is an old question in public finance (Feldstein 1976). Second, I use the solution to that problem to provide a natural way to summarize how closely the existing system satisfies the Ramsey condition.

I start by considering local reforms. Let $d\boldsymbol{\tau}$ denote a small reform around the baseline policy vector $\bar{\boldsymbol{\tau}}$. To first order, the welfare gain associated with that change is $-g(\bar{\boldsymbol{\tau}})^\top d\boldsymbol{\tau}$, while its effect on present-value revenue is $r(\bar{\boldsymbol{\tau}})^\top d\boldsymbol{\tau}$. The reform problem holds revenue fixed:

$$r(\bar{\boldsymbol{\tau}})^\top d\boldsymbol{\tau} = 0.$$

We are primarily interested in identifying the direction of reform, but will not be able to pin down how far the government should move in that direction. Consequently, I compare reforms of a common maximum size $\alpha > 0$ and restrict attention to reforms satisfying $\|d\boldsymbol{\tau}\| \leq \alpha$, where each included reform direction is measured in percentage point units and $\|\cdot\|$ denotes theEuclidean norm. For a change in a single tax rate, this is simply the percentage point change in that rate. The next subsection defines the corresponding measure when a reform changes several rates within the same instrument. The planner's local reform problem is

$$\max_{d\boldsymbol{\tau}} \quad -g(\bar{\boldsymbol{\tau}})^\top d\boldsymbol{\tau} \quad \text{subject to} \quad r(\bar{\boldsymbol{\tau}})^\top d\boldsymbol{\tau} = 0, \quad \|d\boldsymbol{\tau}\| \leq \alpha. \tag{8}$$

Equation (8) separates the two parts of the reform problem. The revenue gradient determines which combinations of tax changes hold revenue fixed, while the welfare gradient ranks those combinations by their welfare effects. The following lemma characterizes the best one.

Lemma 1 (Best local revenue-neutral reform)**.** *Let $\bar{\boldsymbol{\tau}}$ be an interior baseline, and write $g := g(\bar{\boldsymbol{\tau}})$*

and $r := r(\bar{\boldsymbol{\tau}})$*, with* $r \neq 0$*. Define*

$$P_{\perp r} := I - r(r^\top r)^{-1} r^\top.$$

If $P_{\perp r} g \neq 0$*, the unique solution to equation* (8) *is*

$$d\boldsymbol{\tau}^* = -\alpha \frac{P_{\perp r} g}{\|P_{\perp r} g\|}. \tag{9}$$

The resulting first-order welfare gain is

$$-g^\top d\boldsymbol{\tau}^* = \alpha \|P_{\perp r} g\|. \tag{10}$$

If $P_{\perp r} g = 0$*, every first-order revenue-neutral reform has zero first-order welfare effect. In that case,* $d\boldsymbol{\tau}^* = 0$ *may be selected.*

To see the result, begin with the unrestricted reform direction $-g$, which produces the largest first-order reduction in welfare cost. In general, that reform also changes revenue, which eliminates it from the reform set. However, if we project g onto the hyperplane perpendicular to r, then we remove the component associated with that change in revenue. That leaves $P_{\perp r} g$, the component of the welfare gradient that can be changed while holding revenue fixed to first order. Reversing and normalizing this component gives the reform in the lemma, while its length determines the available welfare gain for a given reform size α. Appendix A.4 provides the proof.

The solution to this problem is not new. Variations on it exist in Dixit (1979), which studies a similar reform problem in a many-consumer economy, while Tirole and Guesnerie (1981) find a similar result in developing a gradient-projection process for revenue-neutral reform of linear commodity taxes. However, one difference, which comes down to labeling, is that the object projected here is the gradient of present-value consolidated revenue, which includes behavioral effects across tax bases and across dates. The resulting reform direction therefore incorporates intertemporal and cross-base revenue effects, something which is not apparent in the early reform literature.

One of the theoretical contributions of this paper is using the projection result to show how to assess the tax system itself. At an interior Ramsey optimum, the welfare and revenue gradients point in the same direction. That is akin to saying that welfare and revenue are perfectly aligned, which requires that the revenue-neutral projection is zero. Geometrically, however, it means that the angle between the gradients summarizes how aligned they are, and that is a measurable quantity.

Proposition 1 (Ramsey alignment)**.** *Suppose* $g \neq 0$ *and* $r \neq 0$*, and let* $\theta \in [0, \pi]$ *denote the angle* *between them. Then*

$$\cos\theta = \frac{g^\top r}{\|g\|\,\|r\|}, \qquad \sin\theta = \frac{\|P_{\perp r} g\|}{\|g\|}. \tag{11}$$

For the best revenue-neutral reform satisfying $\|d\boldsymbol{\tau}\| \le \alpha$*,*

$$-g^\top d\boldsymbol{\tau}^* = \alpha \|g\| \sin\theta. \tag{12}$$

Moreover, $\cos\theta = 1$ *if and only if* g *and* r *are positively proportional.*

The proof is straightforward.

Proof. The expression for $\cos\theta$ is the inner-product definition of the angle. Because $P_{\perp r} g$ is orthogonal to the projection of g onto r,

$$\|P_{\perp r} g\|^2 = \|g\|^2 - \frac{(g^\top r)^2}{\|r\|^2} = \|g\|^2(1 - \cos^2\theta) = \|g\|^2 \sin^2\theta.$$

Equation (12) then follows from Lemma 1. The final statement follows from the equality condition for the Cauchy–Schwarz inequality. □

Crucially, as I discuss later in more detail, θ is a measurable object and does not require a complete structural model. The welfare gradient can be constructed from baseline tax bases and the weights placed on their burdens, while the revenue gradient can be constructed from estimated responses of tax bases. Once both gradients have been constructed for the same tax changes, substituting them into equation (11) gives a direct measure of local Ramsey alignment. Beyond measurement, however, there is a clear economic interpretation. Consider $\sin\theta$. If revenue were allowed to change, then the best reform of size α would produce a first-order welfare gain of $\alpha\|g\|$. Requiring the reform to be revenue neutral reduces that gain to $\alpha\|g\|\sin\theta$. Thus, $\sin\theta$ measures how much of the best local welfare improvement remains available when revenue must be held fixed.

On the other hand, the cosine of that angle is required to interpret cases in which there may be no revenue-neutral gain. To see that, note that $\cos\theta = 1$ and $\cos\theta = -1$ produce the same measured $\sin\theta$. If $\cos\theta = 1$, then the revenue and welfare gradients point in the same direction and the tax system satisfies the local Ramsey first-order condition. For example, Chamley (1986) shows that the Ramsey allocation sets the capital tax to zero in the long run because a permanent capital tax compounds distortions to saving over time. In the gradient representation, the welfare and revenue gradients are aligned at that allocation.

On the other hand, if $\cos\theta = -1$, then the welfare and revenue gradients point in opposite directions and there is no revenue-neutral gain from reform, so $\sin\theta = 0$. Under the equal-incidence benchmark, that corresponds to being on the wrong side of the Laffer curve: reducing tax rates raises both revenue and welfare.[7] In the intermediate case, in which $\cos\theta = 0$, the welfare and revenue gradients are orthogonal. That means that the unconstrained welfare-improving direction $-g$ is already revenue neutral.

Next, I use the same geometric approach to distinguish reforms across broad tax instruments from reforms that change rates within an instrument.

2.3 Alignment Across and Within Tax Instruments

Broadly, policymakers may be interested in two types of reform. The first concerns the burden imposed through a broad tax instrument. The second concerns the distribution of that burden within the instrument. In the first type of reform, the policymaker would be interested in changing, for example, how much revenue is raised via income taxation. The second type of reform would hold fixed the total income tax payments that would be collected before behavioral responses and alter the progressivity of the income tax. Many important policy questions concern the second type of reform. Beyond the progressivity of income taxes, one may worry about the composition of tariffs across product groups or the tariffs levied against certain countries, as well as the distribution of business taxes across different legal entities, industries, and so on. Importantly, although the preceding subsection did not distinguish between these two sets of questions, the degree of alignment with the local Ramsey optimum can be meaningfully measured for both using the same types of inputs.

The alignment criterion from the previous subsection applies whether a reform changes broad tax rates or rates within an instrument. What remains is to separate these two types of change and measure them together. Throughout this subsection, I treat reform as a bundled problem. Rather than require revenue-neutrality within a change in broad tax rates or within a distributional reform, I require neutrality across the reform as a whole. Thus, for example, the optimal reform may cut corporate taxes relative to personal taxes but finance the cut with greater progressivity. Let $d\boldsymbol{\tau}_A$ denote the broad component of a reform and $d\boldsymbol{\tau}_S$ the within-instrument component, and let dR_A and dR_S denote their respective first-order revenue effects. Overall revenue neutrality requires

$$dR_A + dR_S = 0. \tag{13}$$

Neither component must be revenue-neutral separately; only their combined revenue effect must be zero.

Within each instrument, the distinction between the two components turns on whether a reform changes total tax payments at the existing tax bases or changes the distribution of those

7. This is one of the key advantages over using raw MVPFs; they become undefined on the wrong side of the Laffer curve, whereas tax instruments remain meaningful even on the wrong side of the Laffer curve.

payments across rates. Consider an instrument containing J rates at date t. Any set of rate changes can be separated into a common change and changes in each rate relative to that common change. Choose the common change to reproduce the same change in total tax payments at the existing tax bases:

$$d\tau_{A,t} = \frac{\sum_{j=1}^{J} B_{j,t}\, d\tau_{j,t}}{\sum_{j=1}^{J} B_{j,t}}, \qquad d\tau_{S,j,t} = d\tau_{j,t} - d\tau_{A,t}. \tag{14}$$

The first expression defines the broad component, while the second assigns the remaining changes to the within-instrument component. Multiplying the second expression by the existing tax bases and summing across rates gives

$$\sum_{j=1}^{J} \beta^t B_{j,t}\, d\tau_{S,j,t} = 0. \tag{15}$$

Thus, the within-instrument component changes the distribution of tax payments across rates without changing their total at the existing tax bases. This condition does not require the within-instrument component to be revenue-neutral. Once tax bases respond, it may raise or lower revenue, and that effect enters the revenue-neutrality constraint for the reform as a whole.

The base weights in equation (15) also enter the welfare gradient. Applying the definition from the setup, the first-order welfare effect of a within-instrument change is

$$dW_S = \sum_{j=1}^{J} \beta^t \tilde{\gamma}_{j,t} B_{j,t}\, d\tau_{S,j,t}. \tag{16}$$

When the incidence-adjusted welfare weights $\tilde{\gamma}_{j,t}$ differ across rates, changes that offset in dollars need not offset in socially weighted welfare. When the planner applies equal weights and the groups represented in W collectively bear the full marginal burden, however, $\tilde{\gamma}_{j,t} = 1$ for every rate. In that case, equation (15) implies $dW_S = 0$ for every within-instrument direction. This special case is an important benchmark because it implies that reforming toward progressivity may be optimal even without distributional motives. A distributional reform that raises revenue can finance a tax cut elsewhere along a margin with a positive welfare cost. For example, a planner may place no direct value on progressivity but still prefer a more progressive schedule because it allows taxes on another base to be reduced. Nevertheless, this benchmark is only an illustrative case.

To combine the two types of reform in the same alignment metric, I need to measure their size in the same units. I measure the size of a within-instrument reform by the square root of the average squared percentage-point changes in its rates, weighting each rate by its share of the existing tax base. This preserves the percentage-point interpretation from the previous subsection: if all the rates move together, the measure is simply their common change. It also ensures that the measured size of a reform does not change simply because the same instrument is divided into more rates. Equation (15) says that the weighted average of the within-instrument changes is zero. Because the broad component changes every rate by the same amount, the two components are orthogonal under this measure.

Appendix A.6 proves this result and defines the corresponding coordinates formally. In those coordinates, write the complete welfare and revenue gradients as

$$g = (g_A, g_S), \qquad r = (r_A, r_S).$$

The equal incidence result above then implies $g_S = 0$. Suppose $g \neq 0$ and $r \neq 0$. At a regular interior Ramsey optimum, provided that a larger revenue requirement increases welfare cost, there must exist some $\mu > 0$ such that

$$g_A = \mu r_A, \qquad g_S = \mu r_S.$$

Following the notation of the previous subsection, let θ denote the angle between the complete welfare and revenue gradients. When g_A and r_A are nonzero, let θ_A denote the angle between their broad components.

Proposition 2 (Alignment with within-instrument reforms)**.** *The alignment metric over the complete reform space is*

$$\cos\theta = \frac{g_A^\top r_A + g_S^\top r_S}{\sqrt{\|g_A\|^2 + \|g_S\|^2}\sqrt{\|r_A\|^2 + \|r_S\|^2}}. \tag{17}$$

Under the equal incidence benchmark, $g_S = 0$. If g_A and r_A are nonzero, this specializes to

$$\cos\theta = \cos\theta_A \frac{\|r_A\|}{\sqrt{\|r_A\|^2 + \|r_S\|^2}}. \tag{18}$$

Consequently, under this benchmark, $\cos\theta = 1$ if and only if $\cos\theta_A = 1$ and $r_S = 0$.

Appendix A.6 provides the proof and derives the corresponding expression for the scope of revenue-neutral reform below.

Compared to the broad notion of alignment in the previous subsection, adding within-instrument reforms can change alignment through both gradients. With general welfare weights, a distributional change may have a direct welfare effect through g_S as well as a revenue effect through r_S. The result is sharpest under the equal incidence benchmark, when $g_S = 0$. In that case, a nonzero r_S adds a component to the revenue gradient without adding one to the welfare gradient. Consequently, when broad alignment is positive, full alignment equals broad alignment multiplied by the fraction of the revenue gradient's magnitude that comes from the broad tax directions. Intuitively, even though the planner does not value progressivity directly, it may still prefer a more progressive schedule if shifting the burden toward higher rates raises revenue because rates at different points in the schedule generate different behavioral responses and fiscal spillovers. This creates a revenue margin that is absent from the broad tax calculation $\cos\theta_A$. Even if $\cos\theta_A = 1$, the system may be misaligned with the Ramsey optimum because there are gains from changing the burden within an instrument.

The same logic applies to the scope for revenue-neutral reform. Under the equal incidence benchmark, adding within-instrument reforms cannot reduce that scope:

$$\sin^2\theta = \sin^2\theta_A + \cos^2\theta_A \frac{\|r_S\|^2}{\|r_A\|^2 + \|r_S\|^2}. \tag{19}$$

The additional term is positive whenever $r_S \neq 0$ and $\cos\theta_A \neq 0$. This term is larger when within-instrument revenue effects are large relative to broad revenue effects and when the broad reform space leaves less scope for revenue-neutral reform. Economically, a revenue-raising within-instrument change can then finance a welfare-improving change in broad tax rates without adding a direct first-order welfare cost of its own.

Thus, the framework provides a local criterion for assessing alignment with the Ramsey first-order condition in the enlarged policy space. The next section shows how to map the broad and within-instrument gradients into empirical objects.

3 Estimation the Revenue Gradient

As an application of the geometric framework, I evaluate the alignment of the U.S. federal tax system with the Ramsey optimum. Throughout the postwar era, personal and corporate income taxes accounted for over 90 percent of federal tax revenue, with tariffs contributing a negligible share. Figure 1 shows that until quite recently, most reform happened on the income tax mix, with corporate rates and revenue share falling dramatically and personal taxes rising modestly to pick up the burden. At the same time, tax policy appears to have become less progressive over time, with the marginal tax rate of the top-1% falling relative to the top. That motivates a retrospective look at whether the federal tax system became more or less aligned over time, both across broad tax instruments, as well as within the personal tax mix.

3.1 From Theory to Empirics

I consider a version of the planner's problem in which he has access to four instruments at each time t: a broad corporate tax rate $\tau_{C,t}$, a personal income tax rate $\tau_{P,t}$, a tariff $\tau_{T,t}$, and a progressivity instrument d_t. Each of the first three instruments is an average tax rate, while the progressivity instrument allows the planner to alter the tax rates paid by the the top-1% relative to the lower 99% of the income distribution, holding total personal tax payments fixed when evaluated at the existing tax bases. There are then two empirical objects: the welfare gradient $g_t = (g_{C,t}, g_{P,t}, g_{T,t}, g_{d,t})^\top$ and the revenue gradient $r_t = (r_{C,t}, r_{P,t}, r_{T,t}, r_{d,t})^\top$. Following the decomposition in Section 2.3, write $g_t = (g_{A,t}, g_{S,t})$ and $r_t = (r_{A,t}, r_{S,t})$, where $g_{A,t} = (g_{C,t}, g_{P,t}, g_{T,t})^\top$, $r_{A,t} = (r_{C,t}, r_{P,t}, r_{T,t})^\top$, and $g_{S,t} = g_{d,t}$ and $r_{S,t} = r_{d,t}$.

As a baseline, I take advantage of the envelope theorem and apply equal incidence weights across the groups bearing each of the three broad tax burdens. That means the mechanical welfare cost of raising each tax rate is proportional to its observable tax base in the national accounts data, i.e., $g_{i,t} = \tilde{B}_{i,t}$ for $i = C, P, T$.[8] Under equal incidence, the progressivity instrument has no direct first-order welfare effect, so $g_{d,t} = 0$. I discuss these assumptions later in the paper, as well as in the appendix. Later, I vary the equal incidence assumption to allow for preferences over progressivity.

The more difficult object is the revenue gradient. Its construction requires present-value elasticities of the tax bases with respect to each instrument, accounting for dynamic adjustments and spillovers across bases. In this context, that is particularly important for three reasons. First, the instruments themselves have economically important effects which unfold over time. For example, a corporate tax cut should increase investment, which raises the corporate tax base dynamically (Hall and Jorgenson 1967). Second, the instruments probably have large cross-base effects. For instance, corporate tax cuts may affect the personal tax base by causing wages to rise over time, as in the canonical model, or they may cause the personal tax base to fall due to organizational form shifting (Barro and Wheaton 2019). Third, I am studying a national tax system, and that system naturally has important aggregate effects. Thus, the revenue gradient should account for all of these spillovers. As an example, the corporate tax's entry in the revenue gradient is, after suppressing time subscripts,

$$r_C = \tilde{B}_C \left[1 - \frac{\tau_C}{1-\tau_C} \varepsilon_{CC} - \frac{\tau_P}{1-\tau_C} \frac{\tilde{B}_P}{\tilde{B}_C} \varepsilon_{PC} - \frac{\tau_T}{1-\tau_C} \frac{\tilde{B}_T}{\tilde{B}_C} \varepsilon_{TC} \right]. \tag{20}$$

Here, ε_{kC} is the present-value elasticity of base k with respect to the corporate net-of-tax rate.

8. Equal incidence is a potentially important assumption for tariffs and corporate taxes. For the former, the burden should be split in principle between the incidence is split between domestic consumers and foreign exporters, with the division depending on terms-of-trade effects. Recent evidence on the 2018-19 trade war with China indicates that the United States bore around 60-80 percent of the incidence on the low end (Ganapati and Hottman 2026), while a significant number of studies show full passthrough (Amiti, Redding, and Weinstein 2019, 2020; Cavallo et al. 2021; Fajgelbaum et al. 2020). At baseline, I assume that domestic consumers bear the full burden of tariffs and show throughout the effect of that assumption. For corporate taxes, many corporations are owned in part by foreigners and under a nationalist planner, they should be excluded from the baseline under equal incidence.

The first term in brackets is the mechanical revenue gain. The second captures the present-value response of the corporate tax base. The remaining terms are the cross-elasticities of the personal and tariff bases, scaled by the tax rate applying to each base and its size relative to the corporate base. That matters since tariff revenue is small, which means that the tariff cross-term receives little weight even if the elasticity is large. The same basic formula holds for the entries of tariffs and personal taxes in the revenue gradient, with the tariff elasticity defined relative to the gross-of-tax rate. The broad entries hold d fixed by definition, since the reform space separates the level of each instrument from the distribution of rates within it.[9]

While the broad entries in $r_{A,t}$ evaluate changes in the overall level of each tax instrument while holding fixed the distribution of rates within the personal tax schedule, the distributive entry $r_{d,t}$ asks whether a change in that distribution affects revenue. I define d to raise the tax rate paid by the top 1% relative to the lower 99%, while reducing rates elsewhere so that total personal tax payments would be unchanged if taxable incomes did not respond. I implement this single direction using four income groups—the bottom 90%, the 90th to 95th percentiles, the 95th to 99th percentiles, and the top 1%—which enter separately because their tax rates, tax bases, and behavioral responses can differ. This yields

$$r_d = \tilde{B}_P \left[\sum_{\ell=1}^{4} \tau_\ell \frac{\tilde{B}_\ell}{\tilde{B}_P} e_{\ell d} + \tau_C \frac{\tilde{B}_C}{\tilde{B}_P} e_{Cd} + \tau_T \frac{\tilde{B}_T}{\tilde{B}_P} e_{Td} \right]. \tag{21}$$

Here, τ_ℓ and $\tilde{B}_\ell$ are the tax rate and discounted base for income group ℓ, and e_{kd} denotes the present-value semi-elasticity of base k with respect to d. There is no mechanical revenue term because the change in tax payments offsets across groups when evaluated at the existing tax bases. Every term in equation (21) therefore reflects a behavioral response: the first sum captures responses within the personal tax base, while the final two terms capture spillovers to the corporate and tariff bases.[10]

Given (20) and (21), the difficult empirical task is obtaining the relevant elasticities. The rest of this section does exactly that in two pieces. First, I obtain the elasticities for the broad tax terms

9. The empirical counterpart of that separation is a stronger requirement. The quarterly system contains the three average rates and the three bases and carries no progressivity coordinate, which means that nothing in the estimator holds d fixed. I am therefore assuming that the narrative reforms identifying the broad entries carry no systematic component along the progressivity direction, and that assumption is an exclusion restriction on the identifying variation. The restriction does not hold exactly, since the level and distributive components of the narrative record move together. Figure G.4 re-estimates the broad block on the part of the personal narrative orthogonal to the progressivity direction, and alignment rises at every date, so the restriction biases the reported series downward rather than upward. The raw estimate credits a pure change in the average personal rate with the base expansion that came from the accompanying change in the top rate, and taxable income at the top responds more elastically.

10. I work in semi-elasticities rather than elasticities because the policy reform is with respect to a percentage-point change in progressivity.

using inputs from Mertens and Ravn (2013) and Besten and Känzig (2026). Second, I use Mertens and Montiel Olea (2018) to obtain the inputs for the progressivity reform.

3.2 Broad Tax Instruments

I start by estimating the inputs to $r_{A,t}$ using time series methods over the medium run. This largely captures the dynamic general equilibrium effects which are critical for properly assessing tax system alignment in this setting.

Data and Specification

I start with a baseline specification of a six-variable quarterly VAR:

$$Z_t = \Big[\log\left(1 - \tau_{P,t}\right), \log\left(1 - \tau_{C,t}\right), -\log(1 + \tau_{T,t}), \log B_{P,t}, \log B_{C,t}, \log B_{T,t}\Big],$$

where $\tau_{i,t}$ is the average tax rate and $B_{i,t}$ is the corresponding tax base. All data are constructed using the National Income and Product Accounts as described in Appendix B. The personal and corporate tax variables exactly correspond to those used in Mertens and Ravn (2013) and Cloyne et al. (2022), while the tariff rate is customs duties divided by total imports of goods, where the latter is the base. Although these definitions are standard, there is one important shortcoming given the nature of the exercise. In the national accounts, corporate profits include both C-corporations and S-corporations, and they are not typically taxed the same. As a result, some cross-base income shifting, which is a natural response to tax changes, may not show up if S- and C-corps are treated as being in the same base. I discuss the implications of this choice in more detail later in the section. As a robustness check, I also later use the Statistics of Income to reallocate income from the corporate tax base to the personal tax base, with corresponding details about construction in Appendix B.1. The sample covers 1947Q1–2019Q4 with four lags.[11]

I estimate impulse responses to the system using the two-stage Bayesian local projection framework developed by Cloyne et al. (2022). The first stage estimates a Bayesian VAR (BVAR) with hierarchical priors following Giannone, Lenza, and Primiceri (2015). Conditional on identification described later in the section, I recover a draw of the impact matrix A_0 from the BVAR posterior, which delivers the causal impact of each tax shock on all variables in the system at horizon zero.[12]

The second stage estimates a separate local projection for each variable and up to k horizons

11. Note that the average tax rate is the object required by theory since revenue from each base is $R_i = \tau_i B_i$.
12. Hyperparameters governing overall shrinkage are treated as random and sampled via Metropolis-Hastings.

from the original shock:

$$Z_{t+k} = c^{(k)} + \sum_{\ell=1}^{p} B_\ell^{(k)} Z_{t-\ell} + e_{t+k}^{(k)}, \tag{22}$$

where $p = 4$, $c^{(k)}$ is the vector of constants, $B_\ell^{(k)}$ collects the coefficients on the ℓth lag of the six variables, and $e_{t+k}^{(k)}$ is an error term. For horizons $h \geq 1$, the impulse response is

$$IRF_h = B_1^{(h-1)} A_0.$$

The index differs by one because A_0 supplies the impact response, while the projection with lead zero supplies the response one quarter later. I estimate each equation using a Minnesota prior and Student-t errors to accommodate fat tails. Appendix C provides complete details on the prior specification and MCMC algorithm, as well as all estimation details. I estimate the responses of all six variables to the three tax shocks from impact through $h = 19$. These five-year response paths are converted into present-value elasticities after the shocks are identified below.

Bayesian estimation offers three advantages: it propagates identification uncertainty from A_0 into the impulse response posterior, it improves precision at long horizons where LP variance is high (Li, Plagborg-Møller, and Wolf 2024), and it produces posterior draws for straightforward inference on the alignment metrics. I use local projections rather than propagating shocks through a VAR because the present-value gradient requires reliable estimates over extended horizons, where VARs can suffer from misspecification bias due to lag truncation. Later, however, I also show that the main results are robust to a variety of alternative estimation methods including standard local projections, penalized local projections, smooth local projections, Bayesian VARs, proxy SVARs, and LP-IV. I also show robustness to this approach using a larger system including GDP, debt, and government spending across estimators.

Identification

Identification comes from the narrative approach. To recover the own- and cross-elasticities of each tax base with respect to each tax rate, we need exogenous variation in each instrument. I rely on two separate sources to supply the narrative shocks.

First, I use the tax shock series for personal and corporate shocks which was initially developed by Mertens and Ravn (2013) and later extended by Cloyne et al. (2022) through the 2017 Tax Cuts and Jobs Act. Those shocks, initially from Romer and Romer (2010), come from reading the Congressional Record and Treasury documents to classify major federal tax legislation from 1945 to 2019 according to the motivations policymakers articulated at the time. Reforms are exogenous when the legislative record indicates policymakers pursued objectives unrelated to the contemporaneous business cycle—such as long-run deficit reduction, ideological commitments, or tax simplification—rather than short-run stabilization. To avoid anticipation effects, I exclude reforms implemented more than 90 days after announcement.[13] Each reform receives a quantitative measure equal to the projected revenue impact scaled by the lagged tax base, as announced at the time of enactment. This scaling delivers an approximation to the change in the average tax rate on that instrument. The resulting series are denoted m_t^P for personal income taxes and m_t^C for corporate income taxes. Cloyne et al. (2022) argues that these instruments are strong and I provide additional diagnostics in the appendix in Table F.1. I find that the personal instrument is strong, but both the corporate and the tariff instrument are borderline relative to the critical threshold.

To get exogenous variation in tariffs, I rely on the narrative series developed by Besten and Känzig (2026). Mirroring the Romer and Romer (2010) approach, their series classifies postwar tariff changes as exogenous when the legislative record indicates motivations unrelated to contemporaneous economic conditions, primarily GATT negotiating rounds and the 2018–2019 tariff increases. I make two modifications to improve instrument strength. First, I assign each shock to the implementation quarter rather than distributing it across adjacent years. Second, I replace their sign-only coding with magnitudes, using tariff rate changes from Bown and Irwin (2015) for the pre-Trump period and from Amiti, Redding, and Weinstein (2020) for the Trump-era tariffs. Appendix Figure B.1 plots the resulting shock series alongside the Mertens-Ravn shocks. Denote this instrument as m_t^T.

In the baseline BLP method, the narrative measures serve as instruments for latent structural tax shocks rather than entering as regressors directly. Let Z_t denote the vector of endogenous variables. A reduced-form VAR produces residuals u_t, which relate to structural shocks ε_t via $u_t = A_0\varepsilon_t$, where ε_t are mutually orthogonal with unit variance and A_0 is the contemporaneous impact matrix. Following Mertens and Ravn (2013) and Cloyne et al. (2022), the narrative measures identify the columns of A_0 corresponding to tax shocks through the moment conditions:

$$E[m_t \epsilon'_{tax,t}] = \Phi \neq 0, \quad E[m_t \epsilon'_{other,t}] = 0, \quad \text{rank}(\Phi) = 3$$

where $m_t = (m_t^P, m_t^C, m_t^T)'$, $\epsilon_{tax,t}$ denotes the three structural tax shocks, and $\epsilon_{other,t}$ denotes all other structural disturbances. The first condition requires the instruments to correlate with the

13. This mitigates anticipation effects that operate between enactment and implementation, but does not address anticipation that precedes enactment itself. For example, it was widely expected that President Reagan would pursue tax reform well before specific legislation was introduced, and similarly that President Trump would push for a large corporate tax cut in 2017. If agents adjusted behavior in advance of these reforms, the narrative shock—which is dated at enactment—would be preceded by movements in the tax bases. Figure F.1 provides a direct test by estimating impulse responses at estimating impulse responses at negative horizons $h = -8, \dots, -1$, which measure whether narrative tax shocks predict *prior* movements in tax bases. Under valid identification, these responses should be zero. Pre-period responses are uniformly small and statistically indistinguishable from zero.

tax innovations; the second requires them to be uncorrelated with non-tax shocks.

Because Congress often adjusts several taxes in the same legislation, the narrative measures may be contemporaneously correlated. Together, they identify three independent sources of exogenous tax variation, but they do not require a reform associated with one tax to leave the other two tax rates unchanged. For example, the 2017 Tax Cuts and Jobs Act altered both the personal tax rate and the corporate tax rate, while tariff reform was nearly contemporaneous. Consequently, For each BVAR draw, I follow Mertens and Ravn (2013) and apply a Cholesky decomposition ordering the shock of interest first. Thus, I place personal taxes first when estimating responses to a personal tax shock, corporate taxes first when estimating responses to a corporate tax shock, and tariffs first when estimating responses to a tariff shock. In each case, I retain the responses of all three tax rates and all three tax bases.

Repeating this procedure for all three shocks produces the complete set of rate and base responses, which is sufficient for translating the responses of tax bases into the required elasticities. There is one more step, which is required because each shock may move more than one tax rate, so the raw impulse responses do not isolate the effect of moving a single tax rate. Thus, the final step purifies that response to give an elasticity. Toward that end, let D_h denote the 3×3 matrix of transformed tax rate responses at horizon h, where each column comes from the Cholesky ordering associated with that column's shock. Let Y_h denote the corresponding matrix of tax base responses. I construct $D = \sum_{h=0}^{19} \beta^h D_h$ and $Y = \sum_{h=0}^{19} \beta^h Y_h$. Then the matrix of present-value elasticities is simply

$$E = YD^{-1}.$$

For example, the personal tax column of E gives the tax base responses to a change in the personal tax rate while setting the discounted five-year changes in the corporate and tariff rates equal to zero. The same calculation applies to the other two rates. Appendix C provides the complete construction.

Present-Value Elasticities

The revenue gradient for each instrument depends on the full matrix of present-value elasticities. Table 1 reports the corresponding 3×3 matrix under the Bayesian local projection estimator. Each column gives the response to a pure cut in the named tax rate, holding the other two rates fixed. The diagonal tells a familiar story: the median own-elasticity of the personal base is small, at roughly 0.3, though imprecisely estimated; the corporate base rises by roughly 2.5 percent following a corporate tax cut; and the tariff base rises by roughly 15 percent following a tariff cut. The most directly comparable estimate comes from Besten and Känzig (2026). In their postwar sample, a one percentage point tariff increase reduces real imports by approximately 10 percent on impact, with the response persisting at 5–6 percent through year five. Converting their impulse responses into a permanent-equivalent elasticity comparable to ε_{TT} yields a value of approximately 10.[14]

However, the off-diagonal pattern is striking. The corporate base is highly responsive to both personal tax cuts and tariff cuts; the former expands the corporate base by roughly nine percent, while a tariff cut expands it by roughly five percent. By contrast, a corporate tax cut reduces the personal tax base by roughly 0.3 percent, and the three remaining cross-elasticities are imprecisely estimated. The clearest cross-base pattern is therefore that the corporate base responds to changes elsewhere in the tax system. Whether those responses create large fiscal externalities depends on the tax rate and size of the responding base, which is what the revenue gradient incorporates.

Table 1: Present-Value Elasticities of Tax Bases

	PIT Cut	CIT Cut	Tariff Cut
$\varepsilon_{P\cdot}$ (Personal Base)	0.31	-0.30	-0.90
	(-0.20, 0.80)	(-0.52, -0.11)	(-1.83, 0.09)
$\varepsilon_{C\cdot}$ (Corporate Base)	8.89	2.50	5.27
	(6.96, 11.05)	(1.76, 3.38)	(1.91, 9.23)
$\varepsilon_{T\cdot}$ (Tariff Base)	-0.09	0.41	14.85
	(-1.48, 1.63)	(-0.16, 1.06)	(12.24, 18.06)

Note: Each elasticity ε_{ij} is the cumulative discounted percent change in base i from a one-percent cut in rate coordinate j, holding the other two coordinates fixed, computed draw by draw as an element of $E = YD^{-1}$. Parentheses report 90% intervals. The baseline is estimated by Bayesian local projection; Table D.1 reports alternative methods.

The large cross-elasticity from personal tax shocks to corporate income survives across three different robustness checks. First, Appendix Table D.1 shows that when we maintain the same six-variable system, the large cross-elasticity from personal tax shocks to corporate income remains positive in the point estimates across methods, although its magnitude varies. Some methods, including the proxy SVAR, feature wide confidence intervals. Second, Appendix Table D.3 shows

14. The trade elasticity literature typically estimates substitution elasticities—either across foreign varieties of a product or between domestic and foreign goods—rather than the response of the aggregate import base to an aggregate tariff change. The former captures reallocation across suppliers, while ε_{TT} additionally reflects income effects, general equilibrium feedbacks, and tariff avoidance, compounded over the five-year horizon. See Boehm, Levchenko, and Pandalai-Nayar (2023) for a comprehensive discussion of how concept, horizon, and identification shape trade elasticity estimates.

that this large cross-elasticity is largely unchanged when reallocating S-corporate profits to the personal tax base. That is surprising. S-corporations are passthrough entities taxed at personal rates, so when personal taxes fall, S-corporation activity may expand and appear in the measured corporate base even though it does not generate corporate tax revenue. In principle, this channel should be quantitatively important because S-corporations grew from a negligible share of business income in the 1970s to roughly half by the 2010s (Dyrda and Pugsley 2024; Smith et al. 2022), with growth accelerating after the Tax Reform Act of 1986 lowered the top personal rate below the corporate rate for the first time. Nevertheless, this particular source of reclassification appears unimportant for the large cross-elasticity. Next, Appendix Table D.2 shows that the large cross-elasticity also survives when the system expands to nine variables including government spending, government debt, and GDP. Finally, Appendix Figure D.2 shows that few reforms are influential for the magnitudes in the resulting elasticity matrix, but the Tax Reform Act of 1986 is particularly pivotal for corporate taxes.

Rationalizing the Cross-Elasticity. Given the survival of the large cross-elasticity from personal taxes to the corporate base across various robustness checks, it is important to rationalize. One potential explanation, which I explore in greater detail in Appendix D.1, originates in operating leverage. Corporate profits are a residual claim on revenue minus costs. Some of these costs, including capital payments, debt service, leases, and overhead, adjust slowly or are entirely predetermined. When the cost structure of firms is slow to adjust to shocks, profits absorb more of the change in revenue. Consider a firm with \$100 in revenue, \$85 in costs, and \$15 in profits. If revenue rises 1 percent to \$101 but costs do not immediately adjust, profits rise to \$16. In other words, with sluggish adjustment, profits rise by $1/0.15 \approx 6.7$ percent in response to a small shock.

The same arithmetic applies to the aggregate corporate sector. In the national accounts, corporate value added decomposes into wages wL, profits π, and other costs F that are plausibly predetermined. When output rises by dY and wages absorb their proportional share $\gamma\, dY$, where γ is the labor share, the remainder $(1-\gamma)\, dY$ accrues entirely to profits. Because profits are only a share s_π of output, this translates into an elasticity of $(1-\gamma)/s_\pi$. This ratio is directly observable from the National Income and Product Accounts and ranges from approximately 1.9 to 3.4 over the postwar period (Appendix Figure D.3). Theoretically, assuming a frictionless labor market, the cross-elasticity is then the product of this leverage and the output response to personal tax cuts: $\varepsilon_{CL} \approx [(1-\gamma)/s_\pi] \times \varepsilon_{YL}$. When we add GDP to the specification, the output elasticity is $\varepsilon_{YL} \approx 0.9$. Combining a postwar average leverage of 2.59 with this output elasticity predicts $\varepsilon_{CL} \approx 2.5$, roughly a third of the baseline estimate of 8.82. This prediction assumes frictionless labor, and it is a lower bound if wages adjust more slowly than output. When wages are also slow to adjust, a larger share of output gains accrues to profits, raising effective leverage above the frictionless-labor benchmark. Table 2 extends the decomposition to the PIT–CIT elasticity matrix. The frictionless-labor benchmark tends to underpredict corporate entries and overpredict personal entries, which is consistent with what one would expect when wages are slow to adjust.

Table 2: Frictionless-Labor Predictions vs. Data

	Frictionless Labor	Preferred BLP	Macro Controls
ε_{CC}	$2.59 \times 0.9 \approx 2.2$	2.50	4.10
ε_{CP}	$2.59 \times 1.0 \approx 2.5$	8.89	8.91
ε_{PC}	$1 \times 0.9 \approx 0.9$	−0.30	0.51
ε_{PP}	$1 \times 1.0 \approx 1.0$	0.31	0.11

Note: The first column applies operating leverage $(1-\gamma)/s_\pi$, whose 1948–2023 NIPA mean is 2.59, to the draw-level GDP elasticities of the Macro Controls specification; personal-base pass-through is one under frictionless labor. The two data columns are posterior medians of five-year present-value pure-rate responses from BLPs that jointly estimate the personal, corporate, and tariff rates and bases. Macro Controls adds GDP, government spending, and federal debt.

3.3 The Personal Tax Progressivity Gradient

The previous subsection estimates the broad entries of the revenue gradient by treating the personal income tax as a single instrument. I now estimate the remaining entry of the revenue gradient, which captures how tax revenue responds to a change in the progressivity of personal taxes. In particular, I consider a reform that raises the marginal tax rate paid by the top one percent of the income distribution by one percentage point relative to the lower 99 percent. This approach to nonlinearity and progressivity is intentionally stylized.

Data and Specification

I start with a baseline specification of a six-variable annual VAR:

$$Z_t^A = \left[\log(1-\tau_{L,t}), \log(1-\tau_{H,t}), \log B_{L,t}, \log B_{H,t}, \log B_{C,t}, \log B_{T,t}\right],$$

where L denotes the lower 99 percent and H denotes the top one percent. The rates $\tau_{L,t}$ and $\tau_{H,t}$ are the marginal tax rate for each group from labor income (including income and payroll taxes). $B_{L,t}$ and $B_{H,t}$ are the corresponding tax bases excluding capital income. The corporate and tariff bases are the same as in the quarterly analysis in the previous section, but aggregated to an annual frequency. Including those bases allows us to eventually trace forward the effects of a "progressivity shock" on changes in corporate profits and revenue, which is fundamental to recover the semi-elasticities necessary to estimate the revenue gradient.

The corporate and tariff bases are the same as in the previous section. However, the group-specific personal tax rates and income bases come from Mertens and Montiel Olea (2018). Their original series end in 2012, which I extend through 2023 using Statistics of Income data. Appendix B.2 describes the extension and the construction of the lower-99 and top-1 series. The annual sample covers 1947–2023.

I estimate responses using the same two-stage Bayesian local projection framework as in the previous subsection. The first stage estimates an annual BVAR with one lag and, conditional on the identification described in the next subsection, recovers the two personal tax columns of the contemporaneous impact matrix A_0. Those columns supply the impact responses that the second-stage local projections trace forward through year four, giving the same five-year window as the quarterly analysis. I include annualized corporate and tariff narrative shocks as contemporaneous controls in both stages. Appendix C provides the complete estimation details.

Identification

Once more, identification comes from an existing set of narrative instruments, this time from Mertens and Montiel Olea (2018). For each tax reform they classify as exogenous, they calculate the change in the average marginal tax rate faced by each income group. I use their measures for the lower 99 percent and the top 1 percent, which Figure B.4 displays together with my extension to the 2017 Tax Cuts and Jobs Act.

The simultaneity problem is even more pronounced than in the broad tax analysis because reforms almost always change both personal tax rates at once; a reform associated with the top 1 percent will generally also change the rate faced by the lower 99 percent, and vice versa. The two narrative measures therefore cannot be treated as separate experiments in which one rate changes while the other remains fixed. Instead, they must be used jointly to recover the two sources of exogenous variation in the personal tax schedule. To do that, I first purge from each narrative measure the part associated with contemporaneous corporate and tariff narrative shocks. I then use the remaining variation as instruments for the two personal tax rate innovations in the annual BVAR. Then, for each posterior draw, the relation between these instruments and the BVAR innovations, together with the covariance matrix of those innovations, recovers the two personal tax columns of the impact matrix A_0 up to a rotation. As in the broad tax analysis, I resolve that rotation using a Cholesky decomposition, ordering the lower-99 rate before the top-1 rate. Thus, the identification approach is essentially the same as in the previous section.

With that identification, feeding each column of A_0 gives the impact response to one of the two personal tax shocks and, propagating forward, gives complete matrices of tax rate and tax base responses to the two identified shocks. Next, I convert those responses into the necessary semi-elasticities for the revenue gradient in two steps. First, let D_h^A denote the 2×2 matrix containing the responses of the lower-99 and top-1 rates at horizon h, after converting the log retention-rate responses into ordinary percentage-point changes, with its discounted sum represented as D^A. Let Y_h^A denote the corresponding 4×2 matrix containing the responses of the lower-99, top-1, corporate, and tariff bases, and its discounted sum as Y^A. Then the matrix of present-value semi-elasticities is

$$E^A = Y^A (D^A)^{-1}. \tag{23}$$

The first column of E^A gives the response of every tax base to a one-percentage-point increase in the lower-99 rate while holding the top-1 rate fixed. The second gives the corresponding response to the top-1 rate. Equation (23) therefore separates the effects of the two rates using their joint variation across the nine reforms. This is the same role played by the 3×3 inversion in the broad tax analysis.[15]

However, the first step does not isolate a pure progressivity shock because changing one rate also changes the aggregate personal tax burden. I therefore combine the two rate changes such that the top-1-minus-lower-99 rate gap rises by one percentage point while mechanical personal tax payments remain fixed. To do that, let $m_{L,t}$ and $m_{H,t}$ denote the amount of income exposed to a marginal change in each tax rate. I construct these exposures from group income and the official ex ante liability scores in the MMO reform data, as described in Appendix B.2. Define the top-1 share of total exposure as

$$w_{H,t} = \frac{m_{H,t}}{m_{L,t} + m_{H,t}}.$$

However, the first step does not isolate a pure progressivity shock because changing one rate also changes the aggregate personal tax burden. I therefore combine the two rate changes such that the top-1-minus-lower-99 rate gap rises by one percentage point while mechanical personal tax payments remain fixed. To do that, let $m_{L,t}$ and $m_{H,t}$ denote the amount of income exposed to a marginal change in each tax rate. I construct these exposures from group income and the official ex ante liability scores in the MMO reform data, as described in Appendix B.2. Define the top-1 share of total exposure as

$$w_{H,t} = \frac{m_{H,t}}{m_{L,t} + m_{H,t}}.$$

In that case, a one-percentage-point increase in progressivity changes the two rates in the direction $\mathbf{h}_t = \begin{pmatrix} -w_{H,t} & 1 - w_{H,t} \end{pmatrix}^\top$. The group with the larger initial exposure receives the smaller rate

15. Any common nonsingular rotation of the columns of D^A and Y^A cancels from equation (23). Appendix C provides the exact construction.

change. The rate gap therefore rises by one percentage point. The construction also satisfies

$$m_{L,t}h_{L,t} + m_{H,t}h_{H,t} = 0,$$

so initial personal tax payments remain fixed. Applying this reform to the two pure-rate responses gives

$$e_{d,t} = E^A \mathbf{h}_t = \begin{pmatrix} e_{Ld,t} & e_{Hd,t} & e_{Cd,t} & e_{Td,t} \end{pmatrix}^T. \tag{24}$$

These are the present-value semi-elasticities of the four tax bases with respect to a one-percentage-point increase in progressivity.

Present-Value Semi-Elasticities

Table 3 reports the present-value semi-elasticities of each tax base with respect to progressivity. While each semi-elasticity is computed by date-by-date, I report its 1948-2023 average within each posterior draw before reporting the posterior distribution across draws. I find that a one-percentage-point increase in progressivity reduces the tax base of the top income percentile by 2.2 percent, with credible interval well clear of zero. The response of the bottom-99 percent, by contrast is imprecisely estimated. However, the refor does increase the corporate tax base by roughly 3.7%, likely reflecting income shifting, and it also increases the tariff base by roughly 7.6%. Both of those cross-base responses are significantly different from zero.

Table 3: Five-Year Present-Value Semi-Elasticities with Respect to Personal-Tax Progressivity

	Bottom 99	Top 1	Corporate base	Tariff base
BLP	-0.22	-2.22	4.03	7.58
	(-0.69, 0.36)	(-3.69, -1.02)	(1.98, 10.47)	(4.63, 17.46)

Note: Posterior medians with 90-percent credible intervals in parentheses. Entries are the percentage change in the discounted five-year response of each tax base to a one-percentage-point increase in the top-1-minus-bottom-99 average marginal tax-rate gap, holding initial personal tax payments fixed, evaluated draw by draw at each year's observed rates and exposure weights and averaged over 1948–2023. The personal block has two groups, identified by two narrative instruments over nine reform years, conditioning on the annual corporate and tariff narrative measures.

The appendix addresses three concrete concerns about this interpretation. First, the estimated responses could depend on the BLP propagation procedure or on the recovery of tax shocks from reduced-form innovations. Appendix Table E.1 shows that standard and smooth local projections produce a similar pattern of point estimates: the top-1 response remains negative, while the corporate and tariff responses remain positive. The bias-corrected local projection has the same broad pattern but is much less precise. The more restrictive BVAR changes the decomposition, producing a corporate response near zero and a substantially smaller tariff response, while the SVAR–IV estimates are too imprecise to adjudicate between these possibilities. Direct LP–IV does not rely on the same invertibility condition, but its intervals are likewise too wide to independently confirm the baseline pattern (Appendix Table E.2). Second, the corporate and tariff spillovers could reflect omitted aggregate conditions or the accounting treatment of passthrough business income. Appendix Table E.3 shows that adding GDP, government spending, and debt leaves both spillovers positive, although smaller than in the baseline. Appendix Table E.4 shows that reallocating S-corporation income strengthens the estimated corporate response and leaves the tariff response similar, while changing the estimated responses within the personal tax base. Third, the annual exercise is identified from a small number of narrative reforms, including the Tax Cuts and Jobs Act as the only post-2012 source of variation. Appendix Figure E.1 re-estimates the semi-elasticities after omitting each reform in turn. The top-1 response remains negative and the corporate and tariff responses remain positive throughout, although the lower-99 response and the magnitudes of the spillovers vary across omissions.

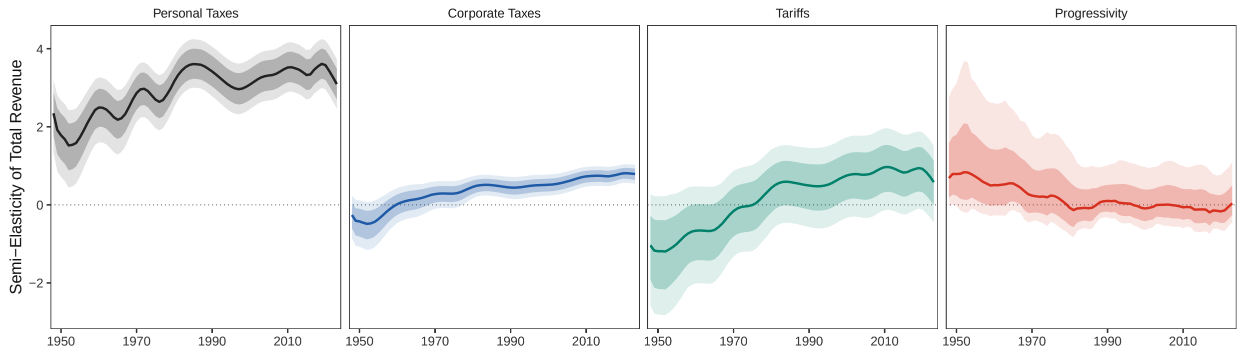
Thus, the evidence supports a revenue margin from changing progressivity that is not explained by the baseline macro specification, passthrough accounting, or a single reform episode. Its magnitude and the composition of the cross-base effects nevertheless remain method-sensitive.

4 The Revenue Gradient over the Postwar Era

The previous section estimated the inputs to the components of the revenue gradient $r_t = (r_{A,t}, r_{d,t})^\top$. This subsection illustrates their combination in Figure 2, which shows the semi-elasticity of total tax revenue with respect to each instrument. Each panel shows the consolidated revenue semi-elasticity along with 68% and 90% credible intervals.

Positive values mean that increasing the tax rate—or increasing progressivity—raises revenue, while a negative value means that the instrument is on the wrong side of the Laffer curve. The personal tax revenue semi-elasticity is the highest of any instrument by a considerable degree and it remains significantly positive throughout the sample. By contrast, the corporate tax and tariff instruments initially have negative point estimates, indicating that in the early postwar era higher taxes of either kind would have reduced revenue, though not significantly so. However, by the 1970s, the corporate tax instrument would have raised revenue. That is similarly true for the tariff instrument, but only for the 68% credible interval. In contrast to the broad instruments, which all saw their semi-elasticities rise over time, the progressivity semi-elasticity of consolidated revenue declined over time and the point estimate has been roughly zero for several decades. In other words, raising the top-1% tax rate relative to the bottom-99 by one percentage point would not raise revenue.

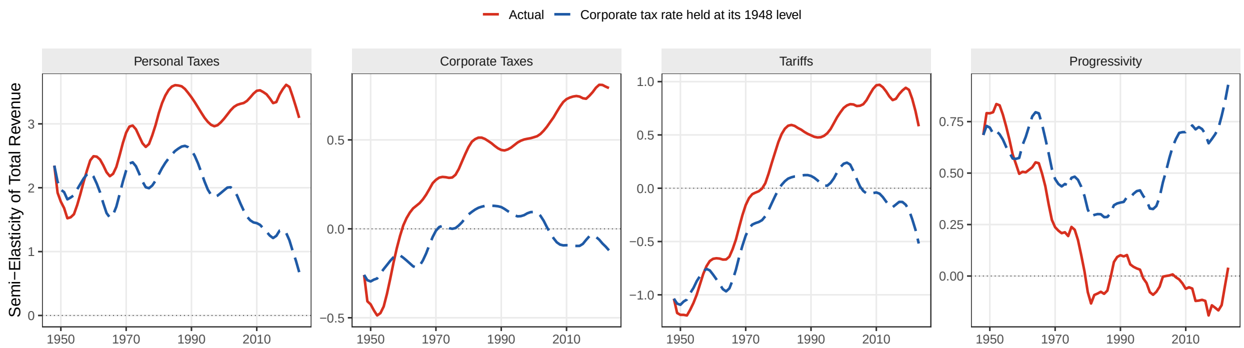
Figure 2: Revenue Semi-Elasticities of the Four Reform Margins



Note: Each panel plots the semi-elasticity of consolidated revenue with respect to a one percentage point reform. Broad-rate changes hold the other broad rates fixed. The progressivity coordinate widens the top-1-minus-lower-99 rate gap while holding initial personal tax payments fixed. Solid lines are posterior medians; dark and light shaded regions are 68-percent and 90-percent credible intervals.

All four trends in Figure 2 are largely explained by the decline in the corporate tax rate. Figure 3 shows that if the corporate tax rate were held at its 1948 level—roughly 40%—then the personal tax semi-elasticity would have been stagnant until the 1990s and then declined, the increases in the corporate tax and tariff semi-elasticities would have been considerably less steep, and the progressivity semi-elasticity would have hardly budged. Hence, much of the movement in the components of the revenue gradient are explained by the large cross-partials between each of the instruments with the corporate tax base, which the previous subsections have shown is robust.. Indeed, they suggest that raising revenue via broad tax instruments is considerably easier now on the margin because corporate tax rates have declined so much since the 1940s.

Figure 3: The Corporate Tax Rate Revalues the Revenue Margins



Note: Solid lines show posterior-median revenue semi-elasticities. Dashed lines recompute each margin after holding the average corporate tax rate at its 1948 level. All other tax rates, tax bases, and estimated behavioral responses follow their historical values. Each series is expressed relative to actual total revenue. Panels use different vertical scales.

While this section can sign the revenue gradient, it has nothing to say about alignment with the local Ramsey optimum. The next section combines these objects to evaluate Ramsey alignment in the postwar era for progressively larger reform spaces.

5 Ramsey Alignment and Local Reform

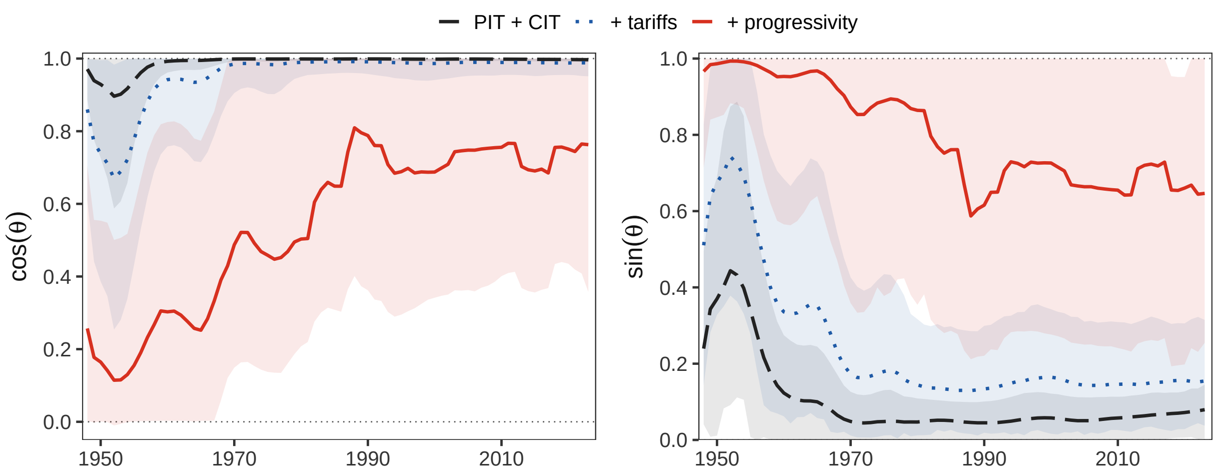
The previous section constructed four components of the revenue gradient for the federal tax system. I now combine them with the welfare gradient to evaluate the proximity of the system to the local Ramsey optimum. Since the annual and quarterly blocks are estimated separately, I treat their posteriors as independent. Consequently, I evaluate every alignment statistic on the full Cartesian product of the two sets of draws, so its posterior reflects uncertainty in both. It is likely that this is conservative because the two blocks share data and are similar policy experiments, so the errors are likely positively correlated.[16]

Figure 4 shows how Ramsey alignment (captured by $\cos\theta$) and the gains to reform (captured by $\sin\theta$) change as more ingredients are added to the tax system. Under the equal incidence benchmark, which I use throughout the main results, the personal–corporate tax system is closely aligned with the Ramsey optimum throughout the sample, showing only modest misalignment before the 1980s. On its own, that would suggest little room for reform post-1980. However, adding in tariffs substantially raises the degree of misalignment early in the period, and the right

16. Because the cosine rises in the broad revenue gradient and falls in the progressivity gradient, positively dependent errors would narrow this posterior rather than widen it, so the reported intervals are conservative under the plausible violation; re-coupling the two posteriors at the extreme admissible dependence moves the 90 percent width by less than ten percent after 1970 and by eighteen percent in 1948.

panel indicates that there are significant gains to reform left when only evaluating the broad tax mix. Finally, adding in progressivity, which is the red line and associated 90% credible interval, significantly raises both the degree of misalignment and the uncertainty around it throughout the postwar era. Between 1950 and 1990, the median posterior draw increases of $\cos\theta$ goes from 0.2 to roughly 0.6, but stays around that point from 1990 onward.[17] That contrasts strongly with the broad instrument alignment, which is close to one with a tight credible interval. Importantly, the contrast also emerges despite equal incidence, which assigns zero weight to progressivity, so the entirety of the effect comes through the effect of progressivity on revenue.

Figure 4: Ramsey Alignment in Nested Reform Spaces



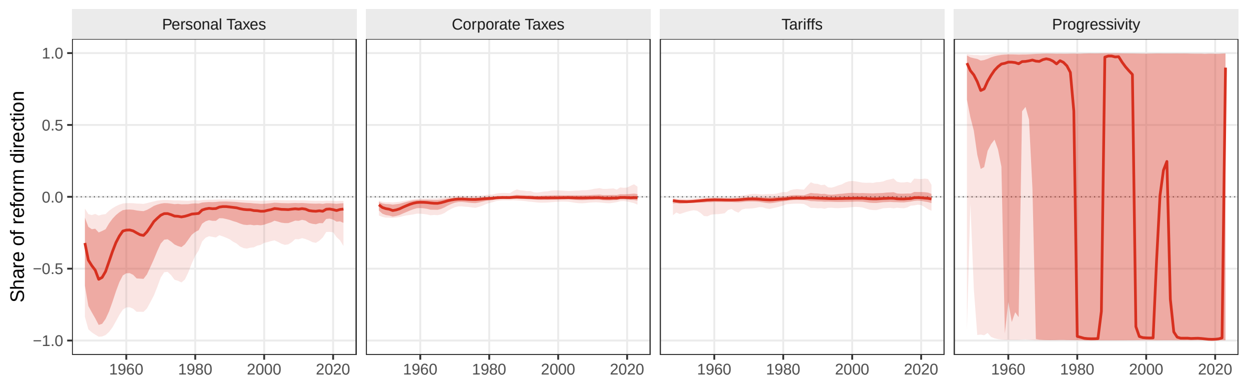
Note: The left panel plots $\cos\theta$ and the right plots $\sin\theta$. The black series permits changes in personal and corporate tax rates; the blue series additionally permits changes in tariffs; and the red series additionally admits progressivity. The blue series is the broad tax space. Shaded regions are 90-percent posterior credible intervals.

Figure 4 establishes that revenue-neutral reforms to the tax system existed throughout the postwar era, although the degree to which they would matter declines over time. Next, I apply the projection result from equation (9) to show which ones would have been most welfare-improving over time in Figure 5. Since the magnitude of reform is normalized to the unit vector, each panel should be interpreted as its contribution to that along with associated posterior uncertainty. Importantly, the uncertainty in each panel should be understood as increasing as alignment goes up because the space for obvious reforms goes away. From left, the planner would optimally have significantly cut taxes on personal income throughout the postwar era. That cut was largest in the early period and would become more mild by the 1980s. That was similarly true for both tariffs and corporate taxes until the 1980s, after which both become indistinguishable from zero.

17. Since the interval is wide, Appendix Figure G.1 plots the probability that alignment has improved compared to the beginning of the period, finding that it exceeds 0.9 by 1990 and remains stable.

By contrast, the rightmost panel indicates that, within personal taxes, the tax system should have become more progressive. Pointwise, that held true until the 1980s. Since the planner places no weight on progressivity under equal incidence, the progressivity coordinate of the optimal reform is proportional to the progressivity semi-elasticity of consolidated revenue. That coordinate therefore collapses toward zero, and its sign stops being identified, once the semi-elasticity becomes indistinguishable from zero. Following the discussion in the previous section, all of the trends in Figure 5 are largely explicable with the falling corporate tax rate, which is the most elastic of all of the tax bases.

Figure 5: Coordinates of the Locally Improving Revenue-Neutral Reform



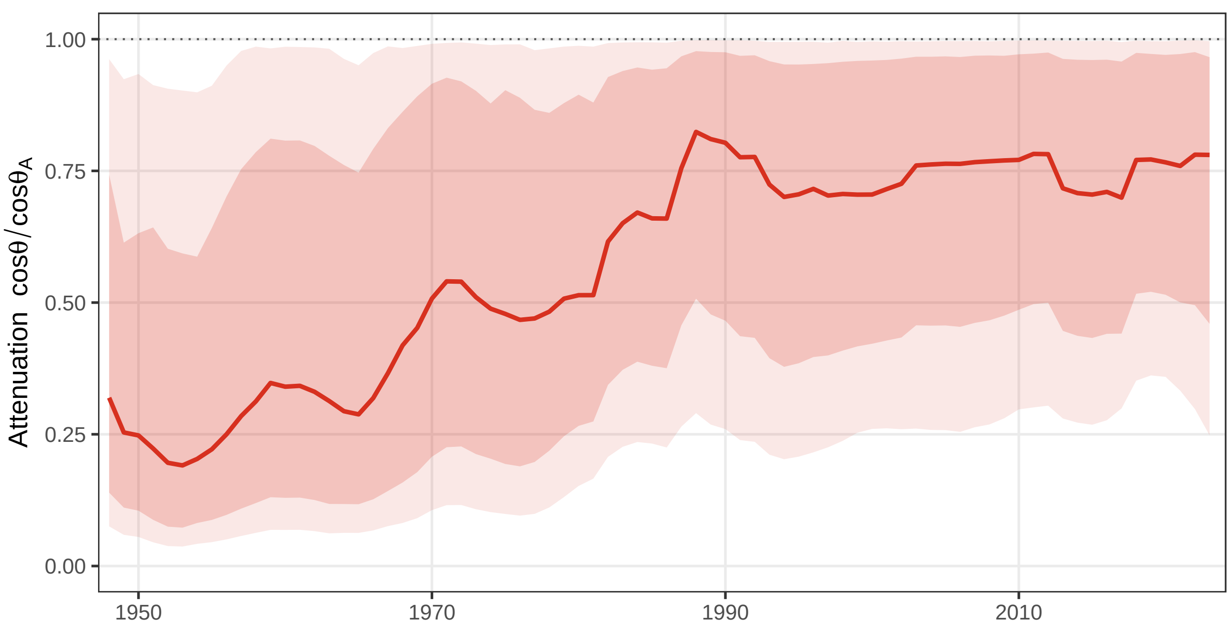
Note: The figure plots the coordinates of the best local revenue-neutral reform, $-P_{\perp r}g/||P_{\perp r}g||$, in the stated policy metric. A negative broad-rate coordinate denotes a tax cut; a positive progressivity coordinate raises the top-1-minus-lower-99 rate gap. At each posterior draw, the four coordinates have unit Euclidean norm. Solid lines are coordinatewise posterior medians; dark and light shaded regions are 68-percent and 90-percent credible intervals. Consequently, the four median series need not themselves form a unit-norm reform vector.

Figures 4 and 5 do not establish that every feature of the tax schedule is misaligned. Rather, the takeaway, particularly since the approach to progressivity is stylized, is that alignment among broad tax rates is not sufficient to establish alignment in a policy space that also allows for progressivity. The remaining question is why admitting progressivity changes the alignment statistic even when the broad tax rates are closely aligned. Under equal incidence, and whenever $\cos\theta_A \neq 0$, Proposition 2 implies that the ratio of alignments between the full system and the broad tax system is

$$\frac{\cos\theta}{\cos\theta_A} = \frac{||r_A||}{\sqrt{||r_A||^2 + ||r_S||^2}},$$

where $||r_S|| = |r_d|/\sqrt{m_d}$ is the progressivity component of the gradient in our application, expressed in the same broad personal-rate units as the entries of r_A. The metric m_d is the squared length of the progressivity direction, given in equation (A.34). The attenuation factor on the right is well defined whenever $r_A \neq 0$, so it is the object plotted below. The ratio between the two is the share of broad alignment that remains after admitting the within-personal revenue margin. Moreover, as long as r_S is nonzero, then the revenue gradient moves away from the revenue gradient without adding any tilt to the welfare gradient. Since, in practice, r_S is distinguishable from zero before the 1980s, that substantially increased the degree of misalignment relative to the broad tax system. Figure 6 reports the degree of attenuation draw by draw. The ratio is low in the early postwar period, meaning that relatively little of broad alignment survives once progressivity is admitted. It rises later in the sample, although the interval remains wide, because the progressivity semi-elasticity of revenue goes to zero. A more progressive personal tax schedule raises the corporate base in the median, and that response is more valuable when corporate income faces a high tax rate. However, holding the estimated behavioral responses fixed, the counterfactual in the previous section shows that this revaluation accounts for much of the median decline in the progressivity margin.

Figure 6: The Attenuation of Broad Alignment from Progressivity



Note: The figure plots $\cos\theta/\cos\theta_A$, where θ_A is alignment in the three-instrument personal, corporate, and tariff reform space and θ additionally admits progressivity. The solid line is the posterior median; dark and light shaded regions are 68-percent and 90-percent credible intervals.

Taken together, the results illustrate the central point of the framework. The broad tax mix can be close to its local Ramsey condition while the tax system remains farther from that condition in the declared policy space once the distribution of personal tax payments is admitted as a reform margin. Thus, an assessment based only on broad tax rates cannot determine whether the complete first-order condition is satisfied. The next section examines the robustness of this conclusion.

5.1 Robustness

The alignment estimates combine two empirical blocks. The broad revenue gradient uses medium-run responses estimated from quarterly data, while the progressivity component uses annual variation in the personal tax schedule. The relevant robustness questions are therefore different. Some concern the dynamic response matrix itself; some concern the construction of the annual progressivity experiment; and some concern the accounting and welfare assumptions used to translate these responses into gradients. Although the estimation section shows that the components of the revenue gradients are fairly stable across methods and specifications, their resulting composition in the alignment metric is sensitive.

I first vary how the revenue gradient is estimated methodologically. Appendix Figures G.2, G.3, and G.6 compare the baseline Bayesian local projection with alternative estimators, direct IV methods, and different horizons for the present-value elasticities. The direct-IV exercises address whether the results depend on recovering tax shocks from the reduced-form innovations, while the matched-horizon exercise changes the annual and quarterly present-value windows together. The latter is necessary because full alignment combines the broad and progressivity blocks in the same revenue gradient. These exercises do not produce identical levels of alignment. In particular, the full-system cosine is sensitive to both the method used to estimate the response matrix and the horizon over which those responses are valued. While the LP methods and the proxy SVAR agree on the rise in alignment, the BVAR and LP-IV methods are considerably flatter and lie beneath the posterior median BLP.

I also investigate whether my stylized approach to measuring productivity matters. Mertens and Montiel Olea (2018) include shocks for four income groups in their analysis, so I can split the tax schedule into up to three kinks. I therefore redo the analysis for two, three, and four income groups. For each partition, I reconstruct r_d from the corresponding semi-elasticities and recompute alignment in the resulting complete reform space. Appendix Figure G.8 shows that moving from two to three and then four income groups progressively lowers the cosine and raises the scope for revenue-neutral reform. A finer partition admits additional changes in the shape of the personal tax schedule, and those directions reveal revenue effects that are hidden by the top-1-versus-lower-99 comparison. However, splitting the tax schedule up into beyond one kink substantially reduces the strength of the first stage, which makes it difficult to interpret the estimates for additional partitions.

Next, I check how important it is that the corporate income measured by the BEA includes S-corporations. Appendix Figure G.7 shows the path of alignment when moving S-corp income to the personal bucket, and I find that the path of alignment is unchanged, but the level is shifted down.

Finally, I investigate whether the assumed equal incidence welfare weights in the bencharm alter the resulting conclusions about the path of alignment. Appendix Figure G.9 varies the social value assigned to burdens borne by the top 1 percent relative to the lower 99 percent and, separately, the value assigned to the corporate tax burden relative to the personal tax burden. Alignment is especially sensitive to the within-personal comparison: changing the relative weight on the top 1 percent materially changes both the level and the historical path of the cosine. However, this sensitivity is generally only relevant if the planner values the 1-percent above the 99-percent, which may not be empirically relevant given what is typically estimated in inverse welfare exercises (Hendren 2020). If personal income is, on average, valued 10 times more as corporate income, then the alignment path does not change. However, if corporate income is valued 10 times more than personal income, then the measured alignment path does not change, but the level shifts down moderately. In similar exercises, I check whether altering assumptions on foreign ownership of corporations (Figure H.2) and the incidence of tariffs (Figure H.3) matter. They do not.

The robustness exercises show that the baseline is sensitive to a number of assumptions. That is, in part, a consequence of estimating macro elasticities. As a result, the baseline should be interpreted as a five-year, medium-run benchmark, rather than as a specification-free estimate of the distance between the tax system and the Ramsey optimum. Nevertheless, taken together, the exercises support treating the baseline as a transparent empirical benchmark for how to implement Ramsey alignment and evaluate the tax system as a whole across both linear and nonlinear dimensions.

6 Conclusion

This paper uses tools from the literature on the marginal value of public funds to answer old questions about the direction of optimal reform and develops new sufficient statistics for the alignment of a tax system with the local Ramsey optimum. The projection of the welfare gradient onto the revenue-neutral reform space gives the best local reform direction, while the angle between the welfare and revenue gradients measures alignment. These statistics are valid in many economic settings, including in general equilibrium. I show how to apply the framework both to alignment across broad tax instruments, which are relevant in the macro literature on optimal taxation (Straub and Werning 2020), and to nonlinear instruments. Under equal incidence, a change in the distribution of payments within an instrument may have no direct first-order welfare effect and still matter through its effect on revenue. The necessary ingredients are a welfare gradient, constructed from tax bases and incidence-adjusted welfare weights, and a revenue gradient.

Empirically, I construct the revenue gradient for four margins of the federal tax system in the United States. Because it is appropriate in this context to account for spillovers across bases and over time, I estimate it using time series methods with established instruments from the literature on tax and tariff shocks (Besten and Känzig 2026; Mertens and Ravn 2013; Mertens and Montiel Olea 2018). The resulting estimates yield what can be interpreted as general equilibrium semi-elasticities of consolidated revenue for corporate taxes, personal taxes, tariffs, and a top-1-versus-lower-99 measure of progressivity. There are large cross-base effects, particularly from the other instruments to corporate profits. Those cross-base effects drive much of the variation in both the time series of revenue gradients and the resulting measures of alignment. That's because the corporate tax rate has declined considerably in the postwar era. Combining these objects, I find that the broad tax instruments are now close to alignment under the baseline specification. Adding progressivity lowers measured alignment and introduces considerably more uncertainty, although the posterior-median revenue effect of progressivity itself has been close to zero in recent decades.

While the application is specific to a particular problem, the framework is more general. The same gradients could be constructed for other settings and larger sets of taxes. For example, one could estimate the revenue gradient within tariffs using a full matrix of product-specific elasticities, including spillovers to other tax bases. Together with the corresponding welfare gradient, this would yield a first-order notion of the alignment of the tariff schedule with the local Ramsey condition. One could also incorporate state and local taxes into the system, which would likely change my own estimates of alignment. The framework is therefore broadly applicable provided that the welfare and revenue gradients can be reliably constructed for the same policy space.

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Appendices

Online Appendix

A Theoretical Derivations

This appendix provides the formal derivations underlying Section 2. Unless otherwise stated, all derivatives are evaluated at the baseline policy vector $\bar{\boldsymbol{\tau}}$. Bars are retained where they prevent ambiguity and otherwise suppressed. The main text writes $g = \nabla_{\boldsymbol{\tau}} W$ and $r = \nabla_{\boldsymbol{\tau}} R$. When the coordinate system matters, I write these same derivatives as

$$g^{\tau} := \nabla_{\boldsymbol{\tau}} W, \qquad r^{\tau} := \nabla_{\boldsymbol{\tau}} R.$$

The superscript distinguishes derivatives in the original rate coordinates from their representation after a change in the policy metric. It does not denote a different welfare or revenue object.

A.1 The Revenue Gradient

Recall that present value revenue is

$$R(\boldsymbol{\tau}) = \sum_{s=0}^{T} \beta^s \sum_{k=1}^{n} \tau_{k,s} B_{k,s}(\boldsymbol{\tau}).$$

Assumption 1 permits differentiation term by term. For any rate $\tau_{i,t}$,

$$\begin{aligned} r_{i,t} := \frac{\partial R}{\partial \tau_{i,t}} &= \beta^t B_{i,t} + \sum_{s=0}^{T} \beta^s \sum_{k=1}^{n} \tau_{k,s} \frac{\partial B_{k,s}}{\partial \tau_{i,t}} \\ &= \beta^t B_{i,t} + \sum_{s=0}^{T} \beta^s \tau_{i,s} \frac{\partial B_{i,s}}{\partial \tau_{i,t}} + \sum_{s=0}^{T} \beta^s \sum_{k \neq i} \tau_{k,s} \frac{\partial B_{k,s}}{\partial \tau_{i,t}}. \end{aligned} \tag{A.1}$$

The first equality follows from the product rule. The second merely separates responses of the same tax base from responses of other tax bases. No sign restriction is imposed on either set of behavioral responses.

The elasticity representation in Section 3.1 follows directly from equation (A.1). Suppose that $B_{k,s} > 0$ and $\bar{\tau}_{i,t} < 1$, and define

$$\varepsilon_{k,s;i,t} := \left. \frac{\partial \ln B_{k,s}(\boldsymbol{\tau})}{\partial \ln(1 - \tau_{i,t})} \right|_{\bar{\boldsymbol{\tau}}}.$$

By the chain rule,

$$\varepsilon_{k,s;i,t} = \frac{1 - \bar{\tau}_{i,t}}{B_{k,s}} \frac{\partial B_{k,s}}{\partial (1 - \tau_{i,t})} = -\frac{1 - \bar{\tau}_{i,t}}{B_{k,s}} \frac{\partial B_{k,s}}{\partial \tau_{i,t}}, \tag{A.2}$$

and hence

$$\frac{\partial B_{k,s}}{\partial \tau_{i,t}} = -\frac{B_{k,s}}{1 - \bar{\tau}_{i,t}} \varepsilon_{k,s;i,t}. \tag{A.3}$$

A positive net-of-tax elasticity therefore means that the base contracts when the tax rate rises. Substituting equation (A.3) into equation (A.1) gives the exact local formula

$$r_{i,t} = \beta^t B_{i,t} - \frac{1}{1 - \bar{\tau}_{i,t}} \sum_{s=0}^{T} \beta^s \sum_{k=1}^{n} \bar{\tau}_{k,s} B_{k,s} \varepsilon_{k,s;i,t}. \tag{A.4}$$

Separating the term $(k,s)=(i,t)$, the remaining responses of base i, and the responses of all other bases yields

$$r_{i,t}=\beta^t B_{i,t}-\frac{1}{1-\bar{\tau}_{i,t}}\Bigg[\beta^t\bar{\tau}_{i,t}B_{i,t}\varepsilon_{i,t;i,t} + \sum_{s\neq t}\beta^s\bar{\tau}_{i,s}B_{i,s}\varepsilon_{i,s;i,t}+\sum_{s=0}^{T}\beta^s\sum_{k\neq i}\bar{\tau}_{k,s}B_{k,s}\varepsilon_{k,s;i,t}\Bigg]. \tag{A.5}$$

Thus, each elasticity is weighted by the baseline revenue collected from the responding base, $\beta^s\bar{\tau}_{k,s}B_{k,s}$, and by the conversion from a change in the tax rate to a proportional change in its net-of-tax rate. An unweighted sum of elasticities would not recover the revenue gradient.

The same derivatives also recover the response to a reform that changes several rates at once. For any infinitesimal reform $d\boldsymbol{\tau}$, the chain rule and equation (A.3) give

$$dB_{k,s}=-B_{k,s}\sum_{t=0}^{T}\sum_{i=1}^{n}\frac{\varepsilon_{k,s;i,t}}{1-\bar{\tau}_{i,t}}d\tau_{i,t}. \tag{A.6}$$

Thus, rate-specific elasticities can be combined using the rate changes that define the reform. Alternatively, an empirical design may estimate the left side of equation (A.6) directly for the combined reform. In either case, the reform's first-order fiscal effect is $dR=r^\top d\boldsymbol{\tau}$.

The decomposition by date is the same derivative regrouped in a different way. Defining $R_s(\boldsymbol{\tau})=\sum_k\tau_{k,s}B_{k,s}(\boldsymbol{\tau})$ gives

$$r_{i,t}=\sum_{s=0}^{T}\beta^s\frac{\partial R_s}{\partial\tau_{i,t}}.$$

Partitioning the sum into $s=t$, $s>t$, and $s<t$ gives equation (3). This temporal decomposition and the base decomposition in equation (A.1) are therefore two partitions of the same derivative, not separate assumptions about the revenue response.

A.2 The Welfare Gradient

The formula for the welfare gradient requires a distinction between the envelope argument and the incidence accounting that follows it. Let $C_h(\boldsymbol{\tau})$ denote group h's present value money-metric welfare cost after its members have optimized, and write

$$W(\boldsymbol{\tau})=\sum_h\omega_h C_h(\boldsymbol{\tau}).$$

For an interior private choice, the envelope theorem removes the terms that operate solely through the induced change in that choice. For example, if $x_h^*(\boldsymbol{\tau})$ denotes the vector of choices made by members of group h, the derivative of optimized welfare contains no term proportional to the private first-order condition times $\partial x_h^*/\partial\tau_{i,t}$. This does not remove changes in wages, prices, profits, transfers, or other equilibrium objects from welfare. Those changes affect optimized welfare directly and must be included in $\partial C_h/\partial\tau_{i,t}$.

When $\beta^t B_{i,t}>0$, define the incidence share

$$s_{h,i,t}:=\frac{1}{\beta^t B_{i,t}}\frac{\partial C_h}{\partial\tau_{i,t}}. \tag{A.7}$$

This definition assigns to group h every first-order effect on its money-metric welfare, including effects operating through equilibrium wages, prices, and profits. A negative share means that the group benefits from the tax increase. Combining equation (A.7) with the definition of W gives

$$\begin{aligned} g_{i,t} &:= \frac{\partial W}{\partial \tau_{i,t}} = \sum_h \omega_h \frac{\partial C_h}{\partial \tau_{i,t}} \\ &= \beta^t B_{i,t} \sum_h \omega_h s_{h,i,t} = \beta^t B_{i,t} \tilde{\gamma}_{i,t}, \end{aligned} \tag{A.8}$$

where $\tilde{\gamma}_{i,t} := \sum_h \omega_h s_{h,i,t}$. Once the incidence shares are defined to include the complete money-metric burden, the final equality is an accounting identity. The envelope theorem explains why the agents' induced optimizing choices do not need to be added to that burden as separate first-order terms; it does not, by itself, determine the incidence shares.

As in the main text, equation (A.8) requires every first-order consequence of the reform to enter either household money-metric burdens or consolidated government revenue. A fiscal effect on a government omitted from R must be added to consolidated revenue. A nonfiscal externality or a policy interaction that changes social resources must be added to the appropriate C_h or entered as a separate term in W. Otherwise, equation (A.8) omits a first-order consequence of the reform.

The further condition

$$\sum_h s_{h,i,t} = 1 \tag{A.9}$$

is not implied by the envelope theorem. It instead requires the groups represented in W to bear the entire marginal mechanical burden. When equation (A.9) holds and $\omega_h = 1$ for every included group, $\tilde{\gamma}_{i,t} = 1$ and $g_{i,t} = \beta^t B_{i,t}$. When some burden falls on groups excluded from W, or when social welfare weights differ, the general expression in equation (A.8) applies.

A.3 The Ramsey First-Order Condition

Consider the local Ramsey problem

$$\min_{\boldsymbol{\tau}} W(\boldsymbol{\tau}) \qquad \text{subject to} \qquad R(\boldsymbol{\tau}) = G. \tag{A.10}$$

Proposition 3 (Ramsey first-order condition)**.** *Suppose that $\boldsymbol{\tau}^*$ is an interior solution to equation (A.10), that W and R are continuously differentiable in a neighborhood of $\boldsymbol{\tau}^*$, and that the revenue constraint is regular, so $r(\boldsymbol{\tau}^*) \neq 0$. Then there exists a scalar μ such that*

$$g(\boldsymbol{\tau}^*) = \mu r(\boldsymbol{\tau}^*). \tag{A.11}$$

If the minimized welfare cost is nondecreasing in the revenue requirement and $g(\boldsymbol{\tau}^*) \neq 0$, then $\mu > 0$. For every component with $r_{i,t}(\boldsymbol{\tau}^*) \neq 0$,

$$\frac{g_{i,t}(\boldsymbol{\tau}^*)}{r_{i,t}(\boldsymbol{\tau}^*)} = \mu. \tag{A.12}$$

Proof. Form the Lagrangian

$$\mathcal{L}(\boldsymbol{\tau}, \mu) = W(\boldsymbol{\tau}) - \mu\big(R(\boldsymbol{\tau}) - G\big).$$

Interior first-order conditions give $g(\boldsymbol{\tau}^*) - \mu r(\boldsymbol{\tau}^*) = 0$. Let $V(G)$ denote the minimized value in equation (A.10). The envelope theorem gives $V'(G) = \mu$ wherever the value function is differentiable, so the economically standard condition that an additional revenue requirement does not lower the minimized welfare cost implies $\mu \geq 0$. Monotonicity alone does not deliver a strict inequality, since a strictly increasing function can have a zero derivative at a point. The sign is pinned down instead by equation (A.11): if μ were zero then $g(\boldsymbol{\tau}^*)$ would be zero as well, so $g(\boldsymbol{\tau}^*) \neq 0$ gives $\mu \neq 0$ and hence $\mu > 0$. Dividing the vector condition by any nonzero component of r gives equation (A.12). □

The vector condition in equation (A.11) is more general than the ratio formulation. If $r_{i,t} = 0$, the corresponding MVPF is undefined, but the first-order condition remains well defined and requires $g_{i,t} = 0$ at an interior optimum. Positive μ is also what distinguishes Ramsey alignment from negative proportionality. These are necessary first-order conditions at a regular interior optimum. They do not by themselves establish that a candidate is a local or global optimum.

A.4 Proof of the Projection Lemma

Proof of Lemma 1. Let

$$\mathcal{H}_r := \{d\boldsymbol{\tau} : r^\top d\boldsymbol{\tau} = 0\}$$

denote the revenue-neutral hyperplane. Since $r \neq 0$, $P_{\perp r} = I - r(r^\top r)^{-1} r^\top$ is the orthogonal projector onto $\mathcal{H}_r$. For every $d\boldsymbol{\tau} \in \mathcal{H}_r$,

$$g^\top d\boldsymbol{\tau} = (P_{\perp r} g)^\top d\boldsymbol{\tau},$$

because the component of g parallel to r is orthogonal to every feasible reform. The Cauchy–Schwarz inequality therefore gives

$$-g^\top d\boldsymbol{\tau} = -(P_{\perp r} g)^\top d\boldsymbol{\tau} \leq \|P_{\perp r} g\| \, \|d\boldsymbol{\tau}\| \leq \alpha \|P_{\perp r} g\|.$$

If $P_{\perp r} g \neq 0$, both inequalities bind only when

$$d\boldsymbol{\tau} = -\alpha \frac{P_{\perp r} g}{\|P_{\perp r} g\|}.$$

This vector belongs to $\mathcal{H}_r$, has norm α, and is therefore the unique solution. Its objective value is $\alpha \|P_{\perp r} g\|$. If $P_{\perp r} g = 0$, then $g^\top d\boldsymbol{\tau} = 0$ for every feasible reform. Every feasible reform is then a solution to the first-order problem, and the zero reform may be selected. □

A.5 Proof of the Alignment Identities

Proof of Proposition 1. For nonzero g and r, the inner-product definition of the angle gives

$$\cos\theta = \frac{g^\top r}{\|g\| \, \|r\|}.$$

The decomposition of g into its components parallel and perpendicular to r is orthogonal, so

$$\begin{aligned} \|P_{\perp r} g\|^2 &= \|g\|^2 - \left\| r \frac{r^\top g}{r^\top r} \right\|^2 = \|g\|^2 - \frac{(g^\top r)^2}{\|r\|^2} \\ &= \|g\|^2 (1 - \cos^2\theta) = \|g\|^2 \sin^2\theta. \end{aligned}$$

Because $\theta \in [0, \pi]$, $\sin\theta \geq 0$, which gives $\sin\theta = \|P_{\perp r} g\| / \|g\|$. Combining this identity with Lemma 1 yields

$$-g^\top d\boldsymbol{\tau}^* = \alpha \|g\| \sin\theta.$$

Finally, equality in the Cauchy–Schwarz inequality implies $|\cos\theta| = 1$ if and only if the two gradients are proportional. The sign of their inner product distinguishes the two cases: $\cos\theta = 1$ if and only if $g = \mu r$ for some $\mu > 0$. □

A.6 Broad and Within-Instrument Reforms

The projection lemma already applies to a policy vector containing every individual rate. This subsection does not introduce a new geometry. It decomposes that same policy space into broad directions and their within-instrument complements under a normalization that nests the broad reform problem. Restricting attention to broad reforms sets the second block to zero; allowing within-instrument reforms relaxes that restriction. The two components are therefore parts of one complete reform, and revenue neutrality applies only after they have been combined. This subsection makes the construction exact and proves Proposition 2.

Let $f \in \mathcal{F}$ index instrument-date families. Family f contains J_f rates, and t_f denotes its date. A broad instrument represented by a single rate is included by setting $J_f = 1$. Let $\mathbf{1}_f$ denote the J_f-vector of ones. For each family, write

$$B_{f+} := \sum_{j=1}^{J_f} B_{fj}, \qquad w_{fj} := \frac{B_{fj}}{B_{f+}}, \qquad \Omega_f := \operatorname{diag}(w_{f1}, \ldots, w_{fJ_f}),$$

and assume that every rate retained as a separate coordinate has $B_{fj} > 0$. Given a vector of rate changes $d\boldsymbol{\tau}_f$, define

$$da_f := \sum_{j=1}^{J_f} w_{fj} d\tau_{fj}, \qquad d\boldsymbol{\tau}_{S,f} := d\boldsymbol{\tau}_f - da_f \mathbf{1}_f. \tag{A.13}$$

This gives a unique decomposition of every change in family f. It also implies

$$\sum_{j=1}^{J_f} \beta^{t_f} B_{fj} d\tau_{S,fj} = \beta^{t_f} B_{f+} \sum_{j=1}^{J_f} w_{fj} d\tau_{S,fj} = 0. \tag{A.14}$$

The common component therefore reproduces the change in tax payments at the existing bases, while the remaining component redistributes those payments across the rates. Equation (A.14) defines the within-instrument component. It does not require that component to be revenue-neutral once tax bases respond.

The base weights also provide a common measure of reform size. Stack the family-specific weight matrices as

$$\Omega := \bigoplus_{f \in \mathcal{F}} \Omega_f$$

and define

$$\|d\boldsymbol{\tau}\|_\Omega^2 := d\boldsymbol{\tau}^\top \Omega d\boldsymbol{\tau} = \sum_{f \in \mathcal{F}} \sum_{j=1}^{J_f} w_{fj} (d\tau_{fj})^2.$$

Within each family, equation (A.14) implies

$$\left\langle da_f \mathbf{1}_f, d\boldsymbol{\tau}_{S,f} \right\rangle_{\Omega_f} = da_f \sum_{j=1}^{J_f} w_{fj} d\tau_{S,fj} = 0.$$

Adding these identities across families gives

$$\|d\boldsymbol{\tau}\|_{\Omega}^2 = \sum_{f \in \mathcal{F}} (da_f)^2 + \sum_{f \in \mathcal{F}} \|d\boldsymbol{\tau}_{S,f}\|_{\Omega_f}^2. \tag{A.15}$$

Thus, changes across broad instruments and changes within instruments are orthogonal blocks of the same reform space. A one-percentage-point common change in one family has size one. Dividing that family into additional rate brackets does not change the size of the common change. The main text takes a common movement in all rates as the broad change. Another definition of a broad change can be used in the same way after its direction has been specified and normalized. The personal instrument in the application is such a case. Its broad coordinate is the average personal tax rate rather than the common movement in the two group marginal rates, and the two are worth different amounts at the existing bases, so the within-family direction is converted to the units of the broad coordinate before its length is taken. Equations (A.33) and (A.34) carry out that conversion.

To express the blocks in ordinary Euclidean coordinates, define

$$d\boldsymbol{z} := \Omega^{1/2} d\boldsymbol{\tau}.$$

For each family, let

$$e_f := \Omega_f^{1/2} \mathbf{1}_f = (\sqrt{w_{f1}}, \ldots, \sqrt{w_{fJ_f}})^\top,$$

so $e_f^\top e_f = 1$. The orthogonal projectors for that family are

$$P_{A,f} := e_f e_f^\top, \qquad P_{S,f} := I_{J_f} - e_f e_f^\top.$$

Thus, if $d\boldsymbol{z}_f = \Omega_f^{1/2} d\boldsymbol{\tau}_f$, its components are

$$d\boldsymbol{z}_{A,f} := P_{A,f} d\boldsymbol{z}_f = e_f da_f, \qquad d\boldsymbol{z}_{S,f} := P_{S,f} d\boldsymbol{z}_f.$$

Embed each e_f in the complete standardized policy space and collect the embedded vectors as the columns of E. Because different families have disjoint coordinates, $E^\top E = I$. For each family, choose a matrix U_f whose columns form an orthonormal basis for the range of $P_{S,f}$, and set

$$U := \bigoplus_{f \in \mathcal{F}} U_f.$$

The columns of E and U together form an orthonormal basis. Every complete reform therefore has the unique representation

$$d\boldsymbol{z} = E\, d\boldsymbol{a} + U\, d\boldsymbol{s}, \qquad d\boldsymbol{a} := E^\top d\boldsymbol{z}, \qquad d\boldsymbol{s} := U^\top d\boldsymbol{z}. \tag{A.16}$$

The particular matrices U_f used to represent the within-instrument subspaces do not affect the result. A different orthonormal basis rotates the coordinates in $d\boldsymbol{s}$, g_S, and r_S together and therefore leaves their inner products, norms, and implied reforms unchanged. The entry da_f is the common change in equation (A.13) and corresponds to $d\tau_{A,t_f}$

in the main text. The vector $\boldsymbol{ds}$ is an orthonormal, standardized representation of the rate changes $\{d\boldsymbol{\tau}_{S,f}\}_{f\in\mathcal{F}}$; it is not the unstandardized rate vector itself. It follows directly that

$$\|d\boldsymbol{\tau}\|_{\Omega}^{2} = \|d\boldsymbol{a}\|^{2} + \|d\boldsymbol{s}\|^{2}.$$

Return now to the original rate-coordinate derivatives g^{τ} and r^{τ}. Their gradients in the standardized coordinates are

$$\widetilde{g} := \Omega^{-1/2} g^{\tau}, \qquad \widetilde{r} := \Omega^{-1/2} r^{\tau}, \tag{A.17}$$

because $(g^{\tau})^{\top} d\boldsymbol{\tau} = \widetilde{g}^{\top} d\boldsymbol{z}$ and similarly for revenue. Define

$$g_A := E^{\top}\widetilde{g}, \qquad g_S := U^{\top}\widetilde{g}, \qquad r_A := E^{\top}\widetilde{r}, \qquad r_S := U^{\top}\widetilde{r}. \tag{A.18}$$

Then

$$dW = g_A^{\top} d\boldsymbol{a} + g_S^{\top} d\boldsymbol{s}, \qquad dR = r_A^{\top} d\boldsymbol{a} + r_S^{\top} d\boldsymbol{s}.$$

Thus, g_S and r_S are the welfare and revenue derivatives with respect to the standardized within-instrument coordinates. Their ordinary Euclidean norms are the norms used for the within-instrument block in the main text. For the remainder of this subsection, and in Proposition 2, write

$$g := \begin{pmatrix} g_A \\ g_S \end{pmatrix}, \qquad r := \begin{pmatrix} r_A \\ r_S \end{pmatrix} \tag{A.19}$$

for the complete gradients in these standardized block coordinates. This is an explicit change of coordinates from g^{τ} and r^{τ}, not a change in the underlying welfare and revenue derivatives. When every family has one rate, $\Omega = I$ and the within-instrument block is empty, so this notation reduces to the broad reform problem.

In these coordinates, that same reform problem becomes

$$\begin{aligned} \max_{d\boldsymbol{a}, d\boldsymbol{s}} \quad & -g_A^{\top} d\boldsymbol{a} - g_S^{\top} d\boldsymbol{s} \\ \text{subject to} \quad & r_A^{\top} d\boldsymbol{a} + r_S^{\top} d\boldsymbol{s} = 0, \\ & \|d\boldsymbol{a}\|^{2} + \|d\boldsymbol{s}\|^{2} \le \alpha^{2}. \end{aligned} \tag{A.20}$$

Setting $d\boldsymbol{s} = 0$ recovers the broad reform problem and the corresponding angle θ_A. Allowing $d\boldsymbol{s}$ to vary enlarges that same reform space; it does not introduce a separate alignment criterion. Neither block must be revenue-neutral on its own. Only their combined revenue effect must be zero. The projection and alignment results therefore apply without modification to the complete gradients in equation (A.19).

This construction also proves the equal incidence result. For rate j in family f, equal incidence gives

$$g_{fj}^{\tau} = \beta^{tf} B_{fj} = \beta^{tf} B_{f+} w_{fj}.$$

The standardized welfare gradient for that family is therefore

$$\widetilde{g}_f = \Omega_f^{-1/2} g_f^{\tau} = \beta^{tf} B_{f+} e_f.$$

It lies entirely in the broad direction, so $U_f^{\top}\widetilde{g}_f = 0$ for every family and $g_S = 0$. The same result holds whenever $\tilde{\gamma}_{fj}$ is constant within each family, although the constant may differ across families. The revenue gradient need not lie entirely in the broad block because rates within a family can generate different behavioral responses. The mechanical component of the revenue gradient is proportional to $(B_{f1}, \ldots, B_{fJ_f})$ and therefore lies in the broad direction as well. Consequently, r_S is generated by behavioral effects on the taxed base, other tax bases, or revenue at other dates.

Because the broad and within-instrument blocks are orthogonal in the standardized coordinates, the alignment result applies to their complete gradients without modification. Their inner product and squared norms are

$$g^\top r = g_A^\top r_A + g_S^\top r_S, \qquad \|g\|^2 = \|g_A\|^2 + \|g_S\|^2, \qquad \|r\|^2 = \|r_A\|^2 + \|r_S\|^2.$$

Substitution into the definition of the cosine gives

$$\cos\theta = \frac{g_A^\top r_A + g_S^\top r_S}{\sqrt{\|g_A\|^2 + \|g_S\|^2}\sqrt{\|r_A\|^2 + \|r_S\|^2}}. \tag{A.21}$$

At a regular interior Ramsey optimum, Proposition 3 requires a common $\mu > 0$ such that

$$g_A = \mu r_A, \qquad g_S = \mu r_S.$$

Thus, alignment in only the broad block is not sufficient for alignment in the complete reform space.

Under equal incidence, $g_S = 0$. If g_A and r_A are nonzero and θ_A is the angle between them, equation (A.21) becomes

$$\begin{aligned} \cos\theta &= \frac{g_A^\top r_A}{\|g_A\|\sqrt{\|r_A\|^2 + \|r_S\|^2}} \\ &= \cos\theta_A \frac{\|r_A\|}{\sqrt{\|r_A\|^2 + \|r_S\|^2}}. \end{aligned} \tag{A.22}$$

The multiplicative term is at most one and is strictly positive. It equals one if and only if $r_S = 0$. Consequently, $\cos\theta = 1$ if and only if $\cos\theta_A = 1$ and $r_S = 0$.

The bundled reform behind this result can also be constructed directly. Suppose the broad block is positively aligned, $g_A = \mu r_A$ for some $\mu > 0$, with $r_A \neq 0$, while $g_S = 0$ and $r_S \neq 0$. Choose $d\boldsymbol{s}$ such that $r_S^\top d\boldsymbol{s} > 0$; reversing any direction with a nonzero revenue effect produces this sign. Because $r_A \neq 0$, choose

$$d\boldsymbol{a} = -\frac{r_S^\top d\boldsymbol{s}}{r_A^\top r_A} r_A.$$

The complete reform is revenue neutral, since $r_A^\top d\boldsymbol{a} + r_S^\top d\boldsymbol{s} = 0$, while its first-order effect on welfare cost is

$$\begin{aligned} g_A^\top d\boldsymbol{a} + g_S^\top d\boldsymbol{s} &= \mu r_A^\top d\boldsymbol{a} \\ &= -\mu r_S^\top d\boldsymbol{s} < 0. \end{aligned}$$

Both components can be scaled down together to satisfy any prescribed local size bound. A nonzero within-instrument revenue gradient therefore leaves a revenue-neutral welfare improvement even when the broad block by itself satisfies the Ramsey condition.

Finally, squaring equation (A.22) and using $\sin^2 x = 1 - \cos^2 x$ gives

$$\begin{aligned}\sin^2\theta &= 1 - \cos^2\theta_A \frac{\|r_A\|^2}{\|r_A\|^2 + \|r_S\|^2} \\ &= \sin^2\theta_A + \cos^2\theta_A \frac{\|r_S\|^2}{\|r_A\|^2 + \|r_S\|^2},\end{aligned} \tag{A.23}$$

which is equation (19). The additional term is nonnegative. Under equal incidence, enlarging the reform space to include within-instrument directions therefore cannot reduce the first-order scope for revenue-neutral reform under the stated normalization.

B Data

The empirical implementation combines quarterly data on broad personal, corporate, and tariff rates with annual data on four personal marginal rates and group incomes, together with annual corporate and tariff variables. The quarterly BLP is estimated over 1947Q1–2019Q4. The underlying quarterly rate and base series extend beyond that estimation endpoint and supply annual corporate and tariff states and accounting histories through 2023. The annual BLP spans 1947–2023, and the accounting series used to construct the alignment measures begin in 1948.

B.1 Quarterly Tax Data

The personal and corporate series extend the data in Mertens and Ravn (2013) and Cloyne et al. (2022). The tariff series are constructed over the same quarterly sample.

- **Personal Income Tax Base.** The personal income tax base is NIPA personal income (Table 2.1 Line 1) plus contributions for government social insurance (Table 3.2 Line 11) less transfers (Table 2.1 Line 17).

- **Personal Income Tax Revenue and Rate.** Personal tax revenue is the sum of NIPA Table 3.2 Lines 3 and 11. The average personal income tax rate is revenue divided by the contemporaneous base.

- **Corporate Income Tax Base.** The corporate tax base is corporate profits with inventory valuation and capital consumption adjustments (NIPA Table 1.12 Line 13), less Federal Reserve Bank profits from Historical Tables 6.16B–D. However, corporations are not domestically owned entirely, and the share of foreign ownership has changed overtime. In some robustness exercises, I remove the foreign owners from the tax base using the annual rest-of-world share of U.S. corporate equity in the Flow of Funds. I measure it as

$$\text{Foreign ownership share}_t = \frac{\text{BOGZ1LM263064105A}_t}{\text{BOGZ1LM893064105A}_t},$$

  where both series are from FRED. I match the ratio to the accounting year.

- **Corporate Income Tax Revenue and Rate.** Corporate tax revenue is federal taxes on corporate income excluding Federal Reserve Banks (NIPA Table 3.2 Line 8). The average corporate income tax rate is revenue divided by the contemporaneous base.

- **Tariff Base, Revenue, and Rate.** The tariff base is imports of goods (NIPA Table 4.1 Line 19). Tariff revenue is federal customs duties (NIPA Table 3.2 Line 6), and the average tariff rate is customs duties divided by imports of goods. However, the quarterly NIPA customs-duty series begins in 1959Q1. Over 1947–1958, I combine the annual NIPA totals with monthly Treasury customs receipts from the NBER Macrohistory Database (FRED Series M15001USM144NNBR). I seasonally adjust the monthly series using X-13ARIMA-SEATS, aggregate it to quarters, and use proportional Denton–Cholette benchmarking to distribute each annual NIPA total across the four quarters. The quarterly values are expressed at seasonally adjusted annual rates and spliced to the NIPA series in 1959Q1

All base series are deflated by the GDP deflator (NIPA Table 1.1.9 Line 1), divided by the population over age 16, and log-transformed. Population is the Bureau of Labor Statistics series CNP16OV. The series begins in 1948, so I linearly extrapolate it over the four quarters of 1947. The three policy variables enter the quarterly system as

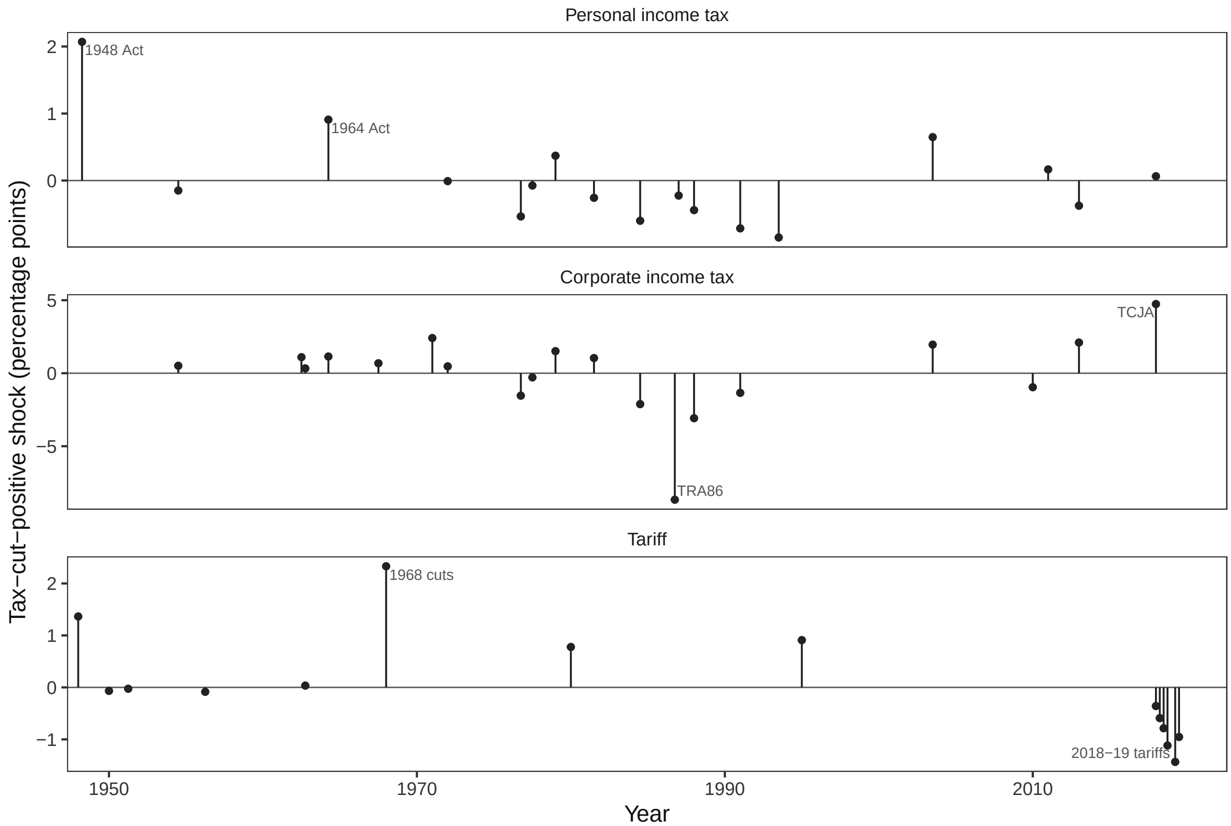
$$\log(1-\tau_{P,t}), \qquad \log(1-\tau_{C,t}), \qquad -\log(1+\tau_{T,t}),$$

where P, C, and T denote the broad personal, corporate, and tariff coordinates. The last sign convention puts a tariff cut in the same direction as an income-tax cut.

The unfiltered transformed rates and log bases enter the quarterly BLP over its 1947Q1–2019Q4 estimation sample. The same underlying series, extended beyond that endpoint, provide the annual corporate and tariff states described below. For the accounting histories used to evaluate the alignment measures, I instead apply the symmetric filter of Christiano and Fitzgerald (2003) to the personal and corporate rates, the tariff rate, and the log bases, removing components with periods between 2 and 40 quarters. The resulting trends supply the broad quarterly accounting histories. For the annual nonlinear accounting, I average them within each year and denote the annual bases by $B_{P,t}$, $B_{C,t}$, and $B_{T,t}$ for personal income, corporate income, and imports, respectively. Thus the annual corporate and tariff states and the smooth accounting histories are constructed from the same underlying quarterly data but serve different purposes.

Narrative Instruments. The personal and corporate instruments are the narrative series in Mertens and Ravn (2013), extended through 2019 by Cloyne et al. (2022). I construct the tariff instrument from the narrative episodes in Besten and Känzig (2026), whose annual series records the sign of each change. I assign a change to its implementation quarter and set the other quarters to zero. Before the Trump tariffs, I scale each change by the prevailing tariff rate and the tariff reduction reported by Bown and Irwin (2015); for the Trump years, I use the timing and magnitudes in Amiti, Redding, and Weinstein (2020). The three instruments enter jointly, and I impose no exclusion that a narrative measure affect only its named tax rate. For each series, I subtract its mean across nonzero reform observations and leave non-reform quarters at zero. These recentered series are the instruments used in the quarterly BLP.

Figure B.1: Narrative Tax Reforms



Note: The top panel reports the quarterly personal income tax, the middle the corporate income tax, and the bottom the tariff narrative measures. Only nonzero observations are shown.

Other Macro Variables. The expanded-information-set specification adds real federal government consumption and gross investment (NIPA Table 1.1.3 Line 23), real GDP (NIPA Table 1.1.3 Line 1), and federal debt held by the public. The debt series splices the measure in Favero and Giavazzi (2012) to FRED Series FYGFDPUN. These variables are auxiliary controls; they are not part of the baseline revenue gradient. Government spending and GDP are divided by the population over age 16 and logged. I deflate nominal debt with the GDP deflator, divide it by the same population series, and log the resulting real per-capita quantity.

SOI Construction

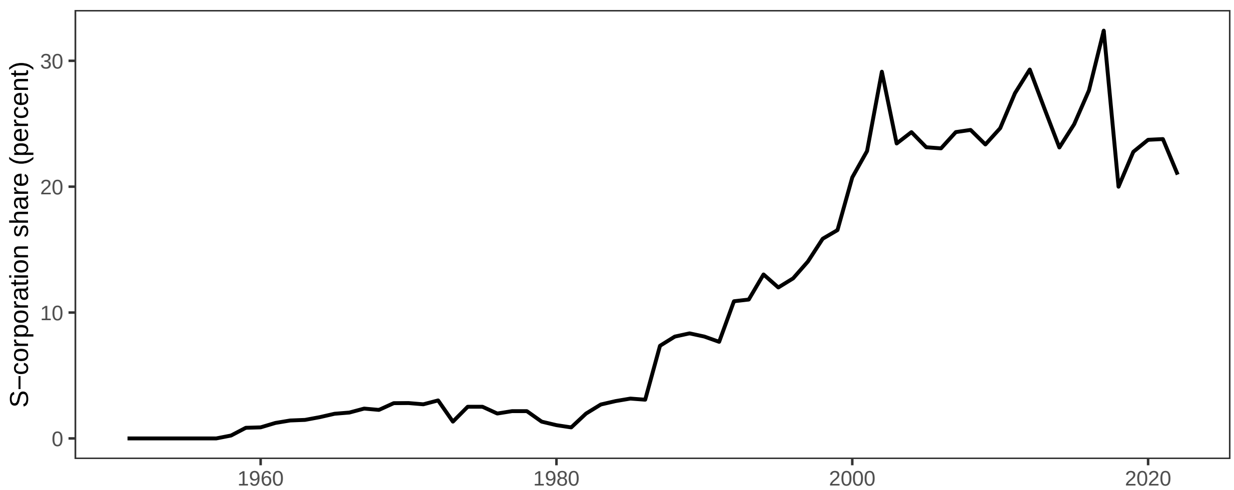
The baseline specification uses NIPA corporate profits as the corporate tax base, which includes both C-corporation and S-corporation income. Since S-corporations are passthrough entities whose income is taxed at the shareholder level rather than at the entity, one might be concerned that the results are driven by organizational form arbitrage rather than traditional general equilibrium channels.

To address this concern, we construct adjusted tax bases that reallocate S-corporation income from the corporate to the personal base. I use the Annual Corporate Reports from the Statistics of Income (SOI) to compile a time series of the profits of C-corporations and S-corporations. I measure profits for the former using the SOI's measure of "Income Subject to Tax" for form 1120 filers (excluding 1120-REIT, 1120-RIC, and 1120-S filers) and profits for the latter as "Net Income (Less Deficit)" for form 1120-S filers. Since Subchapter S was enacted in 1958, allowing qualifying corporations to elect passthrough treatment, I do this for every year from 1958–2022 and carry the 2022 log adjustment forward for the 2023 robustness observation. Toward reallocating NIPA corporate income for S-corps, I compute the S-corporation share of total corporate income as:

$$s_t = \frac{\text{S-corp net income}_t}{\text{S-corp net income}_t + \text{C-corp net income}_t} \tag{A.24}$$

using SOI data for both numerator and denominator to maintain consistency. The resulting series is plotted in Figure B.2. As is well-documented elsewhere, s_t is negligible until after the Tax Reform Act of 1986; since then, S-corps have been large relative to C-corps (Dyrda and Pugsley 2024).

Figure B.2: S-Corporate Share of Corporate Profits



Note: The S-corporate share of corporate profits is computed from the Statistics of Income's Annual Corporate Reports.

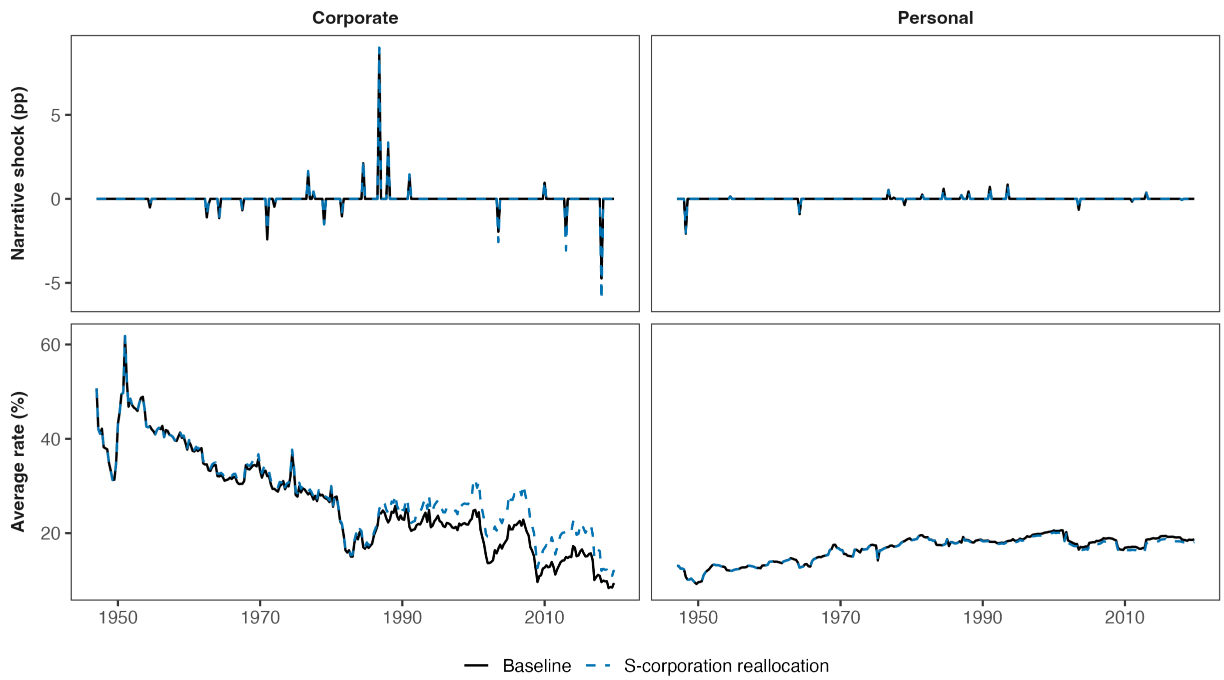
NIPA applies a number of conceptual adjustments to corporate profits which are sensible economically, but are not present in the SOI data. To maintain consistency, I assume that these adjustments are proportional, and get quarterly tax bases by applying s_t in year t to NIPA data in every quarter q of that year as:

$$\text{Corporate base}_{q,t}^{adj} = \text{NIPA corporate profits}_{q,t} \times (1 - s_t) \tag{A.25}$$

$$\text{Personal base}_{q,t}^{adj} = \text{NIPA personal income}_{q,t} + \text{NIPA corporate profits}_{q,t} \times s_t \tag{A.26}$$

The reallocation also changes the measured average rates and associated quarterly first-stage inputs. Figure B.3 records those source series before they enter the robustness estimation.

Figure B.3: Comparing Mertens-Ravn and Reallocated S-Corp Tax Rates and Shocks



Note: After reallocating S-corps to the personal tax base, I recompute average tax rates and tax shocks (as shown above). There is a somewhat sizable difference for the average corporate tax rate, but the results are similar otherwise.

B.2 Progressivity Data and Construction

The annual progressivity exercise uses the lower 99 percent and top 1 percent series constructed by Mertens and Montiel Olea (2018). For each group, the tax rate is the average marginal individual income tax rate plus the average marginal payroll tax rate. The outcome is log real average market income excluding capital gains. The narrative measures are selected federal reforms scored as changes in the marginal tax rate faced by each group. I use $\log(1-\tau_{g,t})$ as the policy variable for group $g \in \{L, H\}$, where L denotes the lower 99 percent and H denotes the top 1 percent. The corresponding narrative measure is the negative of the MMO rate change divided by 100, so that a positive value denotes a tax cut.

The annual dataset also includes the corporate and tariff tax bases. I construct these series from their quarterly counterparts. Annualized corporate and tariff narrative measures enter the annual system as contemporaneous controls. The corporate and tariff rates do not enter the annual state vector.

Extension after 2012

The original MMO data end in 2012. I preserve their observations through that year and extend the tax rate and income series through 2023 using Statistics of Income data. Beginning in 2013, I apply year-over-year changes from SOI Tables 3.5, 4.1, and 1.4 to the original 2012 levels. The individual income tax and payroll tax components are constructed separately and add to the combined marginal rate.

The post-2012 top-1 series is extended directly. I construct the lower-99 series by aggregating the bottom 90 percent, 90th–95th percentiles, and 95th–99th percentiles. Rates are aggregated using AGI-dollar band weights; average income is aggregated as total band income excluding capital gains divided by total band returns. Both series are anchored to their MMO values in 2012. The post-2012 income extension uses returns rather than tax units. The replication package provides the complete construction.

I add one post-2012 narrative reform: the Tax Cuts and Jobs Act. I construct an MMO-style direct-schedule score using published SOI cells and the fixed 2017 income distribution. The underlying rate changes for the bottom 90 percent, 90th–95th percentiles, 95th–99th percentiles, and top 1 percent are −2.407, −3.036, −4.996, and −2.819 percentage points, respectively. Aggregating the first three changes with the accounting schedule's group income shares in the reform year gives a lower-99 narrative measure of 0.03033, while the top-1 measure is 0.02819. These values are added in code to the eight original MMO reform years.

The TCJA was enacted in December 2017 and applied in 2018, so I classify it as a long-run reform under MMO's timing criterion. I also report estimates that omit it. I do not include the 2013 extension of the American Taxpayer Relief Act because its near-term stabilization motivation does not satisfy the same narrative criterion. All other post-2012 observations of the narrative measures are zero.

Mechanical Tax Exposure and the Progressivity Direction

The annual estimates describe how the two group incomes and the corporate and tariff bases respond to changes in the lower-99 and top-1 marginal tax rates. To define the progressivity reform, I use the MMO liability scores to determine how much each rate must change for initial personal tax payments to remain fixed.

Let $Y_{L,t}$ and $Y_{H,t}$ denote current-dollar income, in millions, assigned to the lower 99 percent and top 1 percent. Through 2012, these series are constructed from MMO market income excluding capital gains. Thereafter, I extend the 2012 total using aggregate SOI AGI excluding capital gains and allocate it between the two groups using the extended average-income series and their fixed group masses.

I use the official ex ante liability scores from the eight original MMO reforms to map group income into mechanical tax exposure. These scores include the effects of individual income tax and payroll tax marginal-rate provisions rather than the total revenue effects of every provision in each law. Let $s_{H,e-1}$ denote the lagged top-1 income share in reform episode e, let $\Delta\tau_{H,e}$ and $\Delta\tau_{L,e}$ denote the scored tax rate changes for the top 1 percent and lower 99 percent, and let $\Delta L_e/Y_{e-1}$ denote the associated liability change relative to lagged market income. I estimate

$$\frac{\Delta L_e}{Y_{e-1}} = b_H s_{H,e-1}\Delta\tau_{H,e} + b_L(1 - s_{H,e-1})\Delta\tau_{L,e} + u_e. \tag{A.27}$$

The estimates are $b_H = 0.191$ and $b_L = 0.599$.

The resulting mechanical exposures are

$$m_{L,t} = b_L Y_{L,t}, \qquad m_{H,t} = b_H Y_{H,t}.$$

Define the corresponding exposure shares as

$$w_{L,t} = \frac{m_{L,t}}{m_{L,t} + m_{H,t}}, \qquad w_{H,t} = \frac{m_{H,t}}{m_{L,t} + m_{H,t}}. \tag{A.28}$$

Let $B^{\text{nom}}_{P,t}$ denote the nominal NIPA personal tax base, in billions. The scale of total group exposure relative to the broad personal base is

$$q_t = \frac{(m_{L,t} + m_{H,t})/1000}{B^{\text{nom}}_{P,t}}. \tag{A.29}$$

The accounting bases used to value the resulting income responses are

$$\widehat{B}_{g,t} = B_{P,t} \frac{Y_{g,t}/1000}{B^{\text{nom}}_{P,t}}, \qquad g \in \{L, H\}. \tag{A.30}$$

Because the group masses are fixed, the estimated response of log average income is also the response of log total group income.

A change dd_t in the progressivity instrument changes the two marginal rates by $\mathbf{h}_t dd_t$, where

$$\mathbf{h}_t = \begin{pmatrix} -w_{H,t} \\ 1 - w_{H,t} \end{pmatrix}. \tag{A.31}$$

The lower-99 rate therefore falls by $w_{H,t}$ percentage points and the top-1 rate rises by $1 - w_{H,t}$ percentage points. This direction satisfies

$$w_{L,t} h_{L,t} + w_{H,t} h_{H,t} = 0, \qquad h_{H,t} - h_{L,t} = 1. \tag{A.32}$$

The first condition holds mapped mechanical personal tax liability fixed to first order. The second makes dd_t a one-percentage-point increase in the gap between the top-1 and lower-99 marginal rates.

I measure its length with the metric of Appendix A.6, which assigns family f the weights $w_{fj} = B_{fj}/B_{f+}$ and gives a within-family direction the squared length $\sum_j w_{fj}(d\tau_{S,fj})^2$. The mechanical-neutrality condition in (A.32) is $w_{L,t} h_{L,t} + w_{H,t} h_{H,t} = 0$, so the bases that define this family are the two exposures $m_{L,t}$ and $m_{H,t}$ and the family total is $B_{f+} = m_{L,t} + m_{H,t}$. Applying the metric to $\mathbf{h}_t$ directly gives

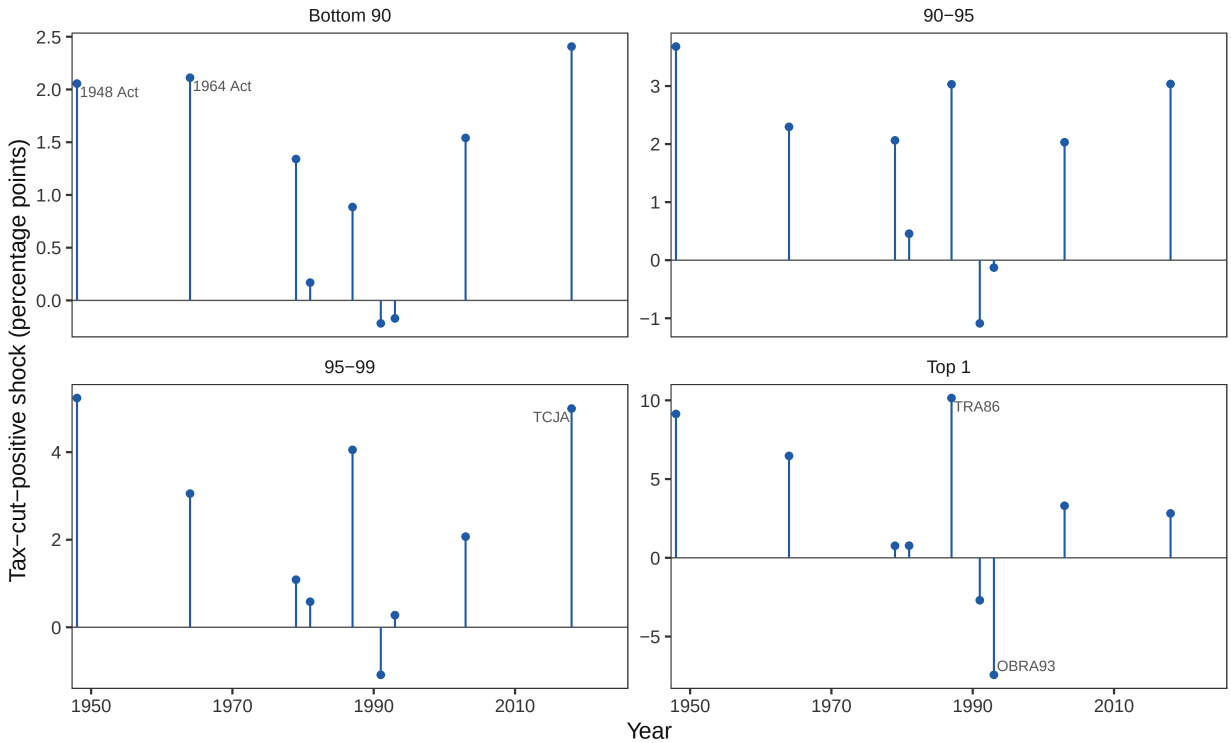
$$w_{L,t} h^2_{L,t} + w_{H,t} h^2_{H,t} = w_{L,t} w^2_{H,t} + w_{H,t} w^2_{L,t} = w_{L,t} w_{H,t}. \tag{A.33}$$

That length is stated in the units of the family's own common change, which raises both group marginal rates by one percentage point and moves mechanical liability by $m_{L,t} + m_{H,t}$. The broad personal coordinate in this paper is not that reform. It is a one-percentage-point change in the average personal tax rate, which moves mechanical liability by $B^{\text{nom}}_{P,t}$. By (A.29) the two differ by the factor q_t, so one percentage point on the group marginal rates is worth q_t percentage points of the broad personal rate at the existing bases. Putting the progressivity direction in broad personal-rate units therefore replaces $\mathbf{h}_t$ by $q_t \mathbf{h}_t$ before the length is taken, and its squared length is

$$m_{d,t} = q^2_t \left(w_{L,t} h^2_{L,t} + w_{H,t} h^2_{H,t} \right) = q^2_t w_{L,t} w_{H,t}. \tag{A.34}$$

The conversion preserves the normalization of Appendix A.6. The family's common change has size one in group marginal-rate units and size q_t in broad personal-rate units, which is what it is worth at the existing bases, and the four revenue coordinates are then lengths of reforms measured in one common unit.

Figure B.4: Narrative Personal Tax Reform Measures by Income Group



Note: The panels report the narrative personal tax measures for the lower 99 percent and top 1 percent. A positive value denotes a tax cut. Only the nine nonzero reform years are shown, including the recovered 2018 Tax Cuts and Jobs Act measures.

C Estimation and Posterior Construction

This appendix describes how the two empirical designs are mapped into the revenue gradients used in the paper. The quarterly block estimates the responses of the personal, corporate, and tariff bases to broad tax-rate changes. The annual block estimates the additional response to a change in the distribution of personal tax rates. The former produces the three broad entries of the revenue gradient; the latter produces its progressivity entry. Appendix B describes the underlying series and Appendix B.2 describes the personal tax data, which are constructed for four income brackets and aggregated to two groups.

C.1 Estimator and Posterior Draws

Both blocks use the two-stage Bayesian local-projection procedure of Cloyne et al. (2022). Let Z_t denote the relevant state vector. The first stage estimates

$$Z_t = c + \sum_{\ell=1}^{p} A_\ell Z_{t-\ell} + F_C m_t^C + F_T m_t^T + u_t, \qquad u_t \sim N(0, \Sigma), \tag{A.35}$$

where $p = 4$ in the quarterly block and $p = 1$ in the annual block. The terms containing m_t^C and m_t^T are absent from the quarterly specification. In the annual specification, m_t^C and m_t^T are the annualized corporate and tariff narrative shocks, which enter as contemporaneous controls.

For each outcome v and projection lead $k \geq 0$, the second stage estimates

$$Z_{v,t+k} = c_v^{(k)} + b_{v,1}^{(k)\prime} Z_{t-1} + \sum_{\ell=2}^{p} b_{v,\ell}^{(k)\prime} Z_{t-\ell} + f_{C,v}^{(k)} m_t^C + f_{T,v}^{(k)} m_t^T + \eta_{v,t+k}^{(k)}. \tag{A.36}$$

The two control terms are again omitted from the quarterly specification. Stacking the first-lag coefficient rows across outcomes gives $B_1^{(k)}$. For a given posterior draw and impact vector a_s, the response is

$$IRF_{0,s} = a_s, \qquad IRF_{h,s} = B_{1,s}^{(h-1)} a_s, \quad h \geq 1. \tag{A.37}$$

Thus, the impact response comes from the first stage and later responses come from the corresponding local projections. The quarterly block uses $h = 0, \ldots, 19$ and the annual block uses $h = 0, \ldots, 4$. These windows contain twenty quarterly observations and five annual observations, respectively. The quarterly discount factor is $\beta = 0.9926$, and the annual discount factor is $\beta_A = \beta^4$. All response matrices below are constructed draw by draw.

Bayesian estimation and computation. I use the hierarchical Minnesota specification of Giannone, Lenza, and Primiceri (2015) in the first-stage BVAR. The prior centers each first own lag at one and all remaining lag coefficients at zero. Lag variances decay quadratically, the prior variance of the constant is 10^7, and the lag-decay parameter is fixed at two. The gamma hyperprior for overall tightness has mode 0.2, standard deviation 0.4, and support $[10^{-4}, 5]$. The variable-specific innovation scales are calibrated from univariate autoregressions. I also use the sum-of-coefficients prior of Doan, Litterman, and Sims (1984) and the single-unit-root dummy-observation prior of Sims (1993). Their tightness hyperpriors each have mode and standard deviation equal to one and support $[10^{-4}, 50]$. Overall tightness and the two dummy-prior tightness parameters are sampled by Metropolis–Hastings; conditional on them, the VAR coefficients and Σ are drawn from their conjugate posterior. In the annual block, the coefficients on the two contemporaneous narrative controls have zero prior mean and variance 10^7.

Each first-stage chain runs for 6,700 iterations. I discard the first 200 iterations and retain every fifth draw, producing 1,300 reduced-form posterior draws. After constructing the relevant impact basis, I retain 1,000 valid draws without replacement.

The local projections are estimated separately for every outcome and lead. For a given outcome, the Minnesota prior mean places its preliminary univariate AR(1) coefficient on the first own lag and zero on all other lag coefficients. If s_y and s_j are the sample standard deviations of the outcome and regressor j, respectively, the prior variance on lag ℓ is

$$\left(\frac{s_y}{s_j}\right)^2 \frac{10^2}{\ell^2},$$

and the prior variance on the constant is 100. In the annual block, the coefficients on the two contemporaneous narrative controls have zero prior mean and variance 10^7. I allow for heavy-tailed projection errors through the scale-mixture representation

$$\eta_{v,t+k}^{(k)} \mid v_{v,t,k} \sim N\left(0, \frac{\sigma_{v,k}^2}{v_{v,t,k}}\right), \qquad v_{v,t,k} \sim \text{Gamma}\left(\frac{\nu_{v,k}}{2}, \frac{\nu_{v,k}}{2}\right).$$

Integrating out $v_{v,t,k}$ gives a Student-t error with $\nu_{v,k}$ degrees of freedom. I use $p(\nu_{v,k}) \propto \exp(-\nu_{v,k}/10)$ for $\nu_{v,k} > 2$.

Conditional on the remaining parameters, the latent precisions, regression coefficients, and error variance have standard Gamma, Normal, and inverse-Gamma updates. I update $\nu_{v,k}$ using the random-walk proposal $\nu'_{v,k} = \nu_{v,k} + \xi$, with $\xi \sim N(0,1)$, rejecting proposals at or below two.

Each second-stage chain runs for 5,200 iterations. I discard the first 200 iterations and retain every fifth draw, leaving 1,000 draws. Within each block, every retained local-projection draw is paired with one distinct first-stage impact-basis draw. This preserves the dependence among the response columns used to construct a given gradient.

C.2 Quarterly Broad-Tax Block

The quarterly state vector is

$$Z_t^Q = \left(\log(1-\tau_{P,t}), \log(1-\tau_{C,t}), -\log(1+\tau_{T,t}), \log B_{C,t}, \log B_{P,t}, \log B_{T,t}\right)^\top .$$

The sample covers 1947Q1–2019Q4. Let z_t^Q collect the three quarterly narrative measures. Writing $\varepsilon_{\text{tax},t}$ for the three tax innovations and $\varepsilon_{\text{other},t}$ for all remaining innovations, the proxy restrictions are

$$\operatorname{rank}\left(\mathbb{E}[z_t^Q \varepsilon_{\text{tax},t}^\top]\right) = 3, \qquad \mathbb{E}[z_t^Q \varepsilon_{\text{other},t}^\top] = 0. \tag{A.38}$$

For each BVAR draw, these restrictions and the reduced-form covariance matrix recover three independent tax innovations up to a 3×3 rotation. I resolve that rotation with a single Cholesky factorization in one fixed ordering—personal, corporate, tariff—and take all three columns of the impact matrix from that same factorization, scaling each column only by the impact response of its own transformed rate. This normalization does not imply that a named tax shock moves only its corresponding tax rate: the corporate and tariff columns carry nonzero impacts on the rates ordered before them. I retain the response of every broad rate and every tax base to each impact direction. Because equation (A.40) inverts the complete rate-response matrix, the semi-elasticities it returns do not depend on this ordering.

Let $D_{h,s}^Q$ denote the matrix of responses of the three transformed tax rates at horizon h in draw s, and let $Y_{h,s}^Q$ denote the corresponding matrix of log-base responses. Their discounted sums are

$$D_s^Q = \sum_{h=0}^{19} \beta^h D_{h,s}^Q, \qquad Y_s^Q = \sum_{h=0}^{19} \beta^h Y_{h,s}^Q.$$

At accounting date t, the corresponding response of ordinary tax rates is

$$P_{s,t}^Q = J_t^Q D_s^Q, \qquad J_t^Q = -\operatorname{diag}\left(1-\tau_{P,t}, 1-\tau_{C,t}, 1+\tau_{T,t}\right). \tag{A.39}$$

The matrix of present-value semi-elasticities with respect to the three ordinary tax rates is

$$E_{s,t}^Q = Y_s^Q \left(P_{s,t}^Q\right)^{-1}. \tag{A.40}$$

The jth column of equation (A.40) therefore gives the response to a change in broad rate j while setting the discounted changes in the other two broad rates equal to zero. This is why the calculation uses the complete response matrices rather than treating each narrative measure as a one-rate experiment.

For $i, j \in \{P, C, T\}$, the resulting revenue derivative is

$$r_{j,s,t} = B_{j,t} + \sum_{i \in \{P,C,T\}} \tau_{i,t} B_{i,t} \left(E^{Q}_{s,t} \right)_{ij} . \tag{A.41}$$

The bases and rates in this expression are evaluated at the accounting date. A common discounted factor is suppressed because it cancels from the alignment measures.

C.3 Annual Progressivity Block

The annual state vector is

$$Z^A_t = \left(\log(1 - \tau_{L,t}), \log(1 - \tau_{H,t}), \log B_{L,t}, \log B_{H,t}, \log B_{C,t}, \log B_{T,t}\right)^\top ,$$

where L denotes the lower 99 percent of the income distribution and H denotes the top 1 percent. The personal tax rates and income measures are those constructed by Mertens and Montiel Olea (2018) and extended through 2023 as described in Appendix B.2. The annualized corporate and tariff narrative shocks enter both estimation stages as contemporaneous controls.

Let $z^A_t = (z_{L,t}, z_{H,t})^\top$ collect the two corresponding MMO narrative measures. The series used in estimation include the recovered 2018 Tax Cuts and Jobs Act measures, so both the two-group and four-group constructions draw on the same nine reform years. Let $\widetilde{z}^A$ denote the matrix obtained after residualizing each measure on a constant and the annualized corporate and tariff narrative shocks. For BVAR draw s, let U^A_s stack the reduced-form innovations. I construct the annual impact basis as

$$G^A_s = \frac{1}{N_A} \left(U^A_s \right)^\top \widetilde{z}^A. \tag{A.42}$$

The two columns of G^A_s identify two independent directions of exogenous variation in the personal tax schedule. Each column is used as an impact vector in equation (A.37).

Let $D^A_{h,s}$ denote the 2×2 matrix of responses of the two log retention rates at horizon h, and let $Y^A_{h,s}$ denote the corresponding 4×2 matrix of responses of the lower-99 income base, the top-1 income base, the corporate base, and the tariff base. Their five-year discounted sums are

$$D^A_s = \sum_{h=0}^{4} \beta^h_A D^A_{h,s}, \qquad Y^A_s = \sum_{h=0}^{4} \beta^h_A Y^A_{h,s}. \tag{A.43}$$

At accounting date t, I convert the log retention-rate responses into ordinary percentage-point tax-rate responses:

$$P^A_{s,t} = J^A_t D^A_s, \qquad J^A_t = -\operatorname{diag}\left(1 - \tau_{L,t}, 1 - \tau_{H,t}\right) . \tag{A.44}$$

The rates in J^A_t are the same lower-99 and top-1 average marginal tax rates used to construct the state vector.

The matrix of present-value semi-elasticities with respect to the two ordinary tax rates is

$$E^A_{s,t} = Y^A_s \left(P^A_{s,t} \right)^{-1} . \tag{A.45}$$

Its first column gives the base responses to a change in the lower-99 rate holding the top-1 rate fixed, while its second column gives the responses to a change in the top-1 rate holding the lower-99 rate fixed. The calculation does not require the two columns of G_s^A to be interpreted as separate lower-99 and top-1 structural shocks. For any nonsingular transformation Q of the impact basis,

$$Y_s^A Q \left(P_{s,t}^A Q\right)^{-1} = Y_s^A \left(P_{s,t}^A\right)^{-1}.$$

The complete rate-response matrix therefore removes the particular normalization of the two identified directions.

Let $m_{L,t}$ and $m_{H,t}$ denote the mechanical tax exposures of the lower 99 percent and top 1 percent, as constructed in Appendix B.2, and define

$$w_{H,t} = \frac{m_{H,t}}{m_{L,t} + m_{H,t}}.$$

A one-percentage-point increase in progressivity changes the two rates in the direction

$$\mathbf{h}_t = \begin{pmatrix} -w_{H,t} \\ 1 - w_{H,t} \end{pmatrix}. \tag{A.46}$$

The top-1-minus-lower-99 rate gap rises by one percentage point, while initial personal tax payments remain unchanged:

$$h_{H,t} - h_{L,t} = 1, \qquad m_{L,t} h_{L,t} + m_{H,t} h_{H,t} = 0.$$

The present-value semi-elasticities with respect to this reform are

$$e_{d,s,t} = E_{s,t}^A \mathbf{h}_t = \begin{pmatrix} e_{Ld,s,t} \\ e_{Hd,s,t} \\ e_{Cd,s,t} \\ e_{Td,s,t} \end{pmatrix}. \tag{A.47}$$

Let $\widehat{B}_{L,t}$ and $\widehat{B}_{H,t}$ denote the corresponding accounting bases. Because the reform leaves initial personal tax payments unchanged, its revenue-gradient entry contains no mechanical term:

$$r_{d,s,t} = \tau_{L,t} \widehat{B}_{L,t} e_{Ld,s,t} + \tau_{H,t} \widehat{B}_{H,t} e_{Hd,s,t} + \tau_{C,t} B_{C,t} e_{Cd,s,t} + \tau_{T,t} B_{T,t} e_{Td,s,t}. \tag{A.48}$$

Let $m_{d,t}$ denote the squared length of the progressivity direction under the policy metric defined in Appendix B.2. Under equal incidence, the expanded gradients are

$$g_t = \left(B_{P,t}, B_{C,t}, B_{T,t}, 0\right)^\top, \qquad r_{s_A,s_Q,t} = \left(r_{P,s_Q,t}, r_{C,s_Q,t}, r_{T,s_Q,t}, \frac{r_{d,s_A,t}}{\sqrt{m_{d,t}}}\right)^\top. \tag{A.49}$$

C.4 Combining the Two Blocks

The quarterly and annual blocks are estimated separately. I retain 1,000 posterior draws from each and evaluate the expanded gradient, cosine, and sine for every annual–quarterly draw pair. The resulting intervals are therefore conditional on the maintained product of the two marginal posteriors: they preserve uncertainty within each block but do not estimate dependence between the two blocks.

D Robustness of the Broad Tax Elasticity Estimates

This appendix reports robustness of the present-value elasticities from for the broad estimates to alternative estimation methods, accounting for S-corporations, adding in macro controls, and accounting for if the VAR may not be invertible.

1. **Frequentist local projections.** Standard local projections using the Mertens-Ravn structural shocks identified from the BVAR as regressors. Uncertainty about A_0 is propagated using the wild bootstrap proxy SVAR of Mertens and Ravn (2013).

2. **Smooth local projections.** Frequentist local projections with a penalty on deviations across adjacent horizons following Barnichon and Brownlees (2019), which reduces the lumpiness of standard LP estimates at the cost of imposing smoothness.

3. **Bias-corrected local projections.** Local projections with the bias correction of Herbst and Johannsen (2024), which adjusts for the finite-sample bias that arises from projecting on persistent regressors.

4. **Bayesian VAR.** A six-variable BVAR with four lags and hierarchical Minnesota priors following Giannone, Lenza, and Primiceri (2015). Impulse responses are computed by propagating shocks through the VAR companion form rather than estimating separate regressions at each horizon.

5. **Proxy SVAR.** A frequentist structural VAR using the narrative measures as external instruments following Mertens and Ravn (2013), extended to 2019Q4. Confidence intervals are constructed using the wild bootstrap.

6. **Linear trends.** Each of the above methods (where applicable) is re-estimated with a linear trend added to the specification. Results are reported in the lower panel of Table D.1.

7. **LP-IV.** At each horizon h, the log retention rate is instrumented directly with the narrative measure, controlling for four lags of all variables. This specification is robust to non-invertibility of the VAR representation by construction. Three variants are reported in Table D.5: plain LP-IV, smoothed LP-IV following Barnichon and Brownlees (2019), and bias-corrected LP-IV following Herbst and Johannsen (2024). All produce elasticities quantitatively similar to the baseline.

8. **TVP-BVAR.** A four-variable time-varying parameter VAR with stochastic volatility following Primiceri (2005), which allows the full transmission mechanism to drift over time.

Table D.1: Present-Value Elasticities of Tax Bases under Alternative Methods

	PIT Cut			CIT Cut			Tariff Cut		
Method	ε_{PP}	ε_{CP}	ε_{TP}	ε_{PC}	ε_{CC}	ε_{TC}	ε_{PT}	ε_{CT}	ε_{TT}
BLP	0.31	8.89	-0.09	-0.30	2.50	0.41	-0.90	5.27	14.85
	(-0.20, 0.80)	(6.96, 11.05)	(-1.48, 1.63)	(-0.52, -0.11)	(1.76, 3.38)	(-0.16, 1.06)	(-1.83, 0.09)	(1.91, 9.23)	(12.24, 18.06)
LP	0.13	9.76	-0.26	-0.42	2.79	1.11	-1.57	5.91	18.82
	(-0.96, 1.41)	(5.73, 18.17)	(-2.51, 4.21)	(-0.95, 0.04)	(1.20, 5.54)	(0.24, 2.28)	(-3.69, 0.62)	(-2.34, 14.48)	(13.78, 23.95)
SLP	0.13	9.76	-0.26	-0.42	2.79	1.11	-1.57	5.92	18.82
	(-0.97, 1.41)	(5.74, 18.19)	(-2.51, 4.22)	(-0.95, 0.04)	(1.20, 5.54)	(0.24, 2.28)	(-3.68, 0.62)	(-2.32, 14.49)	(13.78, 23.95)
Bias-corrected LP	-0.07	8.69	-2.13	-0.65	2.25	0.60	-2.47	-3.27	21.10
	(-1.62, 1.79)	(2.92, 20.84)	(-4.36, 1.80)	(-1.80, 0.15)	(-0.47, 10.22)	(-1.53, 2.01)	(-10.68, 3.61)	(-51.72, 16.83)	(12.58, 33.90)
BVAR	0.36	4.01	0.76	-0.45	2.92	0.06	0.32	7.29	15.57
	(-0.25, 0.89)	(0.00, 9.77)	(-1.66, 4.59)	(-0.68, -0.27)	(1.71, 5.05)	(-0.83, 1.47)	(-1.47, 2.01)	(-0.97, 20.95)	(9.55, 26.73)
SVAR-IV	0.70	14.92	-0.00	-0.59	3.60	-0.71	-1.42	15.70	11.14
	(-4.60, 8.30)	(-3.99, 51.44)	(-11.45, 15.87)	(-7.07, 5.11)	(-13.89, 32.50)	(-12.90, 12.39)	(-32.22, 23.72)	(-64.12, 130.58)	(-41.96, 57.93)

Note: Each elasticity ε_{ij} is the cumulative discounted percent change in base i from a one-percent cut in rate coordinate j, holding the other two coordinates fixed, computed draw by draw as an element of $E = YD^{-1}$. Parentheses report 90% intervals. LP, SLP, and Bias-corrected LP retain the BLP contemporaneous rate map and alter only propagation; BVAR and SVAR-IV re-estimate both maps. Bayesian rows report credible intervals, the remaining rows hybrid or bootstrap intervals.

Table D.2: Present-Value Elasticities of Tax Bases under Macro Controls

	PIT cut	CIT cut	Tariff cut
$\varepsilon_{P\cdot}$ (Personal Base)	0.11	0.51	0.30
	(-0.70, 1.53)	(0.11, 1.35)	(-1.67, 3.80)
$\varepsilon_{C\cdot}$ (Corporate Base)	8.91	4.10	3.37
	(5.83, 13.79)	(2.65, 7.01)	(-4.07, 15.72)
$\varepsilon_{T\cdot}$ (Tariff Base)	1.00	1.14	14.59
	(-0.93, 4.27)	(0.26, 3.11)	(10.57, 22.68)

Note: Each elasticity ε_{ij} is the cumulative discounted percent change in base i per one-percent increase in the named tax-cut coordinate. Within each draw the rate-response matrix D and base-response matrix Y are mapped into $E = YD^{-1}$, so each column holds the other coordinates fixed. Macro Controls adds government spending, output, and debt to the three rates and bases. Estimated by Bayesian local projection; parentheses report 90% credible intervals.

Table D.3: Present-Value Elasticities under S-Corporation Reallocation

	PIT Cut	CIT Cut	Tariff Cut
$\varepsilon_{P\cdot}$ (Personal Base)	0.66	-0.41	-0.93
	(0.04, 1.29)	(-0.60, -0.23)	(-1.91, -0.05)
$\varepsilon_{C\cdot}$ (Corporate Base)	8.43	3.10	6.51
	(6.05, 11.78)	(2.39, 3.98)	(3.41, 10.14)
$\varepsilon_{T\cdot}$ (Tariff Base)	0.59	0.70	16.17
	(-1.29, 2.85)	(0.12, 1.29)	(13.74, 19.04)

Note: Each elasticity ε_{ij} is the cumulative discounted percent change in base i from a one-percent cut in rate coordinate j, holding the other two coordinates fixed, computed draw by draw as an element of $E = YD^{-1}$. Parentheses report 90% intervals. S-corporation income is reallocated between the personal and corporate bases, and the rates and narrative measures are reconstructed to match.

Table D.4: Present-Value Elasticities of Tax Bases under Alternative Methods (PIT–CIT System)

	CIT cut		PIT cut	
Method	ε_{CC}	ε_{PC}	ε_{CP}	ε_{PP}
BLP	2.41	-0.15	9.82	0.73
	(1.82, 3.30)	(-0.32, 0.03)	(8.00, 12.29)	(0.27, 1.28)
LP	2.69	-0.23	13.20	0.58
	(1.22, 6.43)	(-0.78, 0.24)	(8.95, 23.15)	(-0.74, 2.32)
SLP	2.70	-0.23	13.23	0.58
	(1.22, 6.44)	(-0.78, 0.24)	(8.96, 23.21)	(-0.74, 2.33)
Bias-corrected LP	2.40	-0.39	14.21	0.51
	(0.73, 8.23)	(-1.16, 0.30)	(9.47, 28.29)	(-1.55, 2.78)
BVAR	3.84	-0.47	10.68	0.54
	(2.32, 5.84)	(-0.72, -0.27)	(4.55, 17.71)	(-0.19, 1.33)
SVAR-IV	2.87	-0.48	17.88	0.84
	(-1.42, 17.98)	(-3.66, 1.56)	(9.26, 53.64)	(-3.36, 6.32)

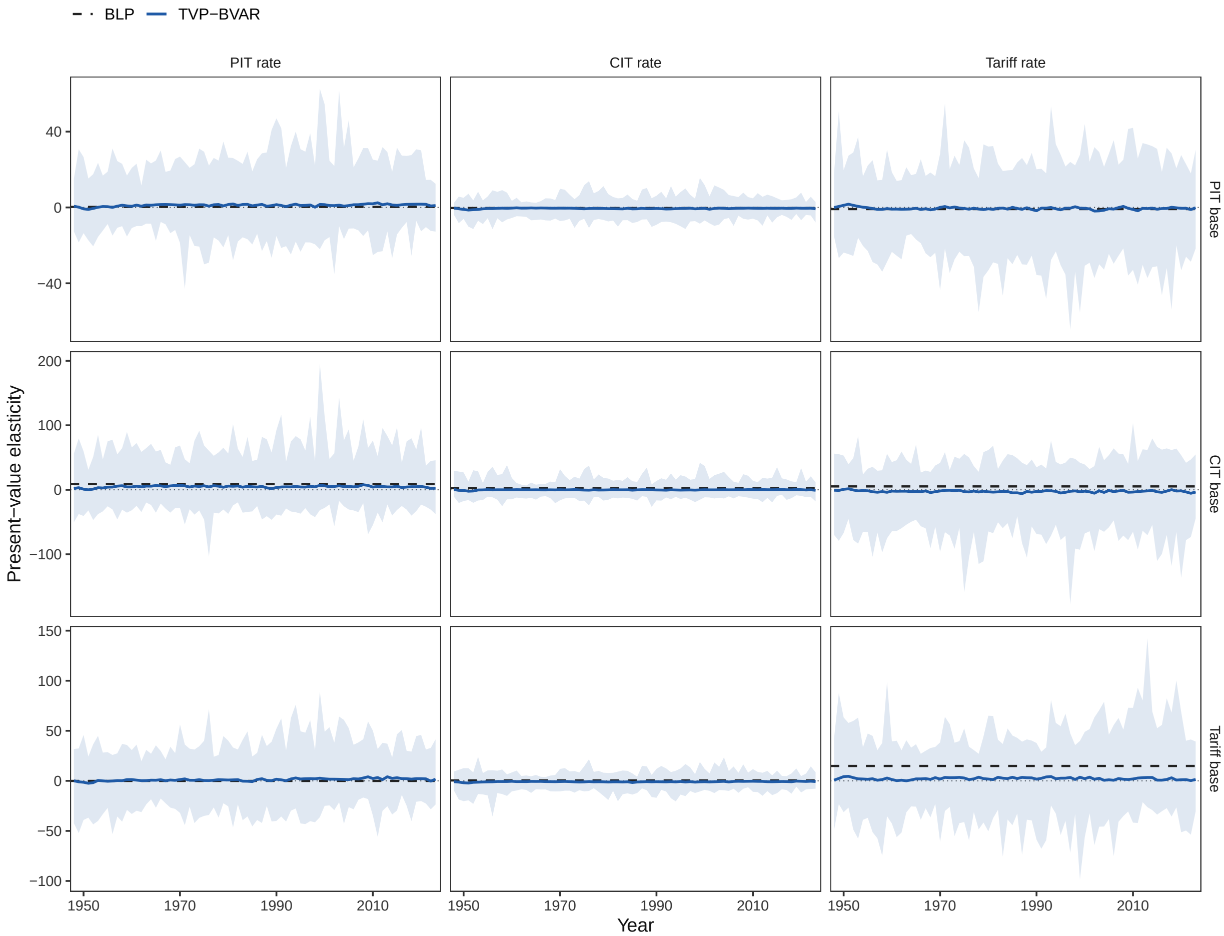
Note: Each elasticity is the cumulative discounted percent change in base i from a one-percent cut in the named log-retention-rate coordinate, holding the other broad rate fixed, computed within each draw as an element of $E = YD^{-1}$ for the PIT–CIT system. LP, SLP, and Bias-corrected LP retain the BLP contemporaneous impact basis and re-estimate only propagation; BVAR and SVAR-IV re-estimate both maps. Parentheses report 90% intervals.

Table D.5: Present-Value Elasticities under Direct LP-IV (Three-Instrument System)

	PIT Cut			CIT Cut			Tariff Cut		
Method	ε_{PP}	ε_{CP}	ε_{TP}	ε_{PC}	ε_{CC}	ε_{TC}	ε_{PT}	ε_{CT}	ε_{TT}
LP-IV	-0.76	6.51	1.77	-0.44	2.27	1.23	-2.34	10.34	8.09
	(-7.32, 4.13)	(-7.31, 27.34)	(-7.92, 18.75)	(-3.75, 1.71)	(-4.14, 12.23)	(-5.09, 8.17)	(-21.57, 11.98)	(-43.17, 74.12)	(-45.57, 51.97)
Smooth LP-IV	-0.76	6.51	1.77	-0.44	2.27	1.23	-2.34	10.34	8.09
	(-7.33, 4.13)	(-7.31, 27.34)	(-7.92, 18.75)	(-3.75, 1.71)	(-4.14, 12.22)	(-5.09, 8.17)	(-21.57, 11.98)	(-43.18, 74.10)	(-45.57, 51.97)
Bias-corrected LP-IV	-0.67	5.78	1.39	-0.45	2.18	1.05	-2.40	9.21	7.18
	(-6.29, 3.60)	(-6.42, 23.36)	(-7.33, 16.10)	(-3.46, 1.80)	(-3.71, 12.61)	(-5.12, 7.14)	(-22.28, 13.80)	(-42.17, 74.19)	(-50.58, 51.27)

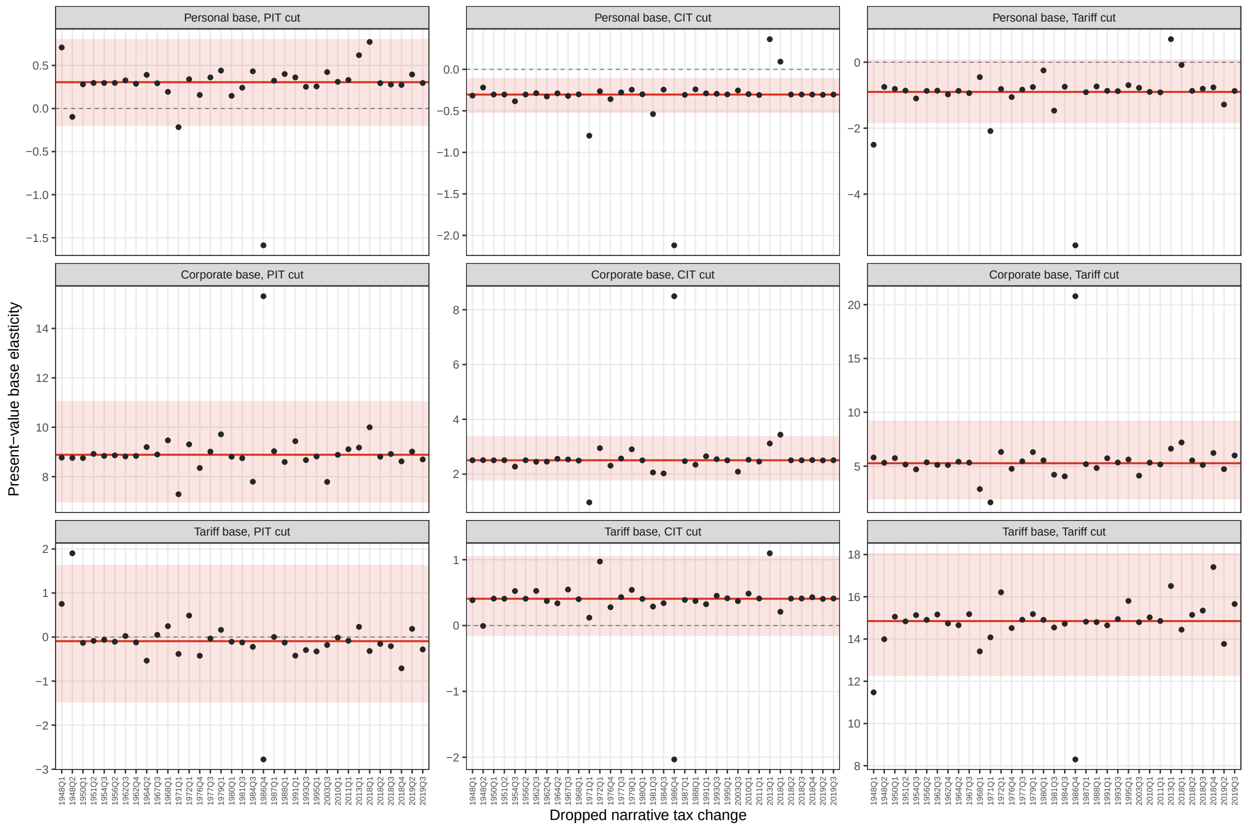
Note: Each elasticity ε_{ij} is the cumulative discounted percent change in base i from a one-percent cut in rate coordinate j, holding the other two coordinates fixed, computed draw by draw as an element of $E = YD^{-1}$. Parentheses report 90% intervals. The LP-IV rows identify the rate and base responses directly with the narrative measures before forming YD^{-1}, unlike the LP rows of Table D.1, which retain the BLP contemporaneous rate map. Intervals are bootstrap intervals.

Figure D.1: Time-Varying PIT–CIT–Tariff Elasticities



Note: The figure reports time-varying present-value elasticities in the PIT–CIT–tariff system.

Figure D.2: Tax Elasticity Matrix Leave-One-Out



Note: This figure shows the how the elasticity matrix changes leaving out one reform at a time.

D.1 Rationalizing the Magnitude of ε_{CL}

I find across a variety of methods that the elasticity of the corporate tax base with respect to personal taxes is the largest entry in the 2×2 elasticity matrix. This appendix shows that the magnitude is a predictable consequence of two well-established forces: operating leverage, which amplifies output movements into larger profit movements, and the fact that personal tax cuts generate larger output responses than corporate tax cuts. Together, these explain both the magnitude of the cross-elasicity and the full pattern of elasticities in Table D.1.

The intuition begins with a basic feature of national accounts: corporate profits are the residual left after a firm pays its workers and covers costs that are slow to adjust. Consider a firm with \$100 in revenue, \$85 in costs, and \$15 in profits. If revenue rises 1 percent to \$101 but costs do not immediately adjust, profits rise to roughly \$16, which is a 6.7% increase in the corporate tax base. The same 1 percent shock to revenue produced a 6.7 percent change in profits because profits are a thin margin between two large numbers. This is operating leverage: the smaller the profit margin, the larger the amplification.

Operating Leverage. Corporate output—or corporate gross value added—decomposes as

$$Y = wL + \pi + F,$$

where wL is the wage bill, π is pre-tax corporate profits, and F is the remainder of costs, including interest payments, taxes, depreciation, and so on. For convenience, define the labor share as $\gamma = wL/Y$ and the profit share as $s_\pi = \pi/Y$. When output changes by dY and F does not adjust, the change in profits is $d\pi = dY - d(wL)$. If wages are fully flexible and the labor share is roughly constant at the margin, then $d(wL) = \gamma\, dY$ and

$$d \log \pi = \frac{1-\gamma}{s_\pi}\, d \log Y.$$

The ratio $(1-\gamma)/s_\pi$ is the *frictionless* operating leverage. It is a lower bound on the true leverage because it assumes wages absorb their full share of output movements within the horizon. When wages are sticky, compensation adjusts by less than $\gamma\, dY$ in the short run, so a larger share of any output gain initially accrues to profits. This raises effective leverage above $(1-\gamma)/s_\pi$ and simultaneously attenuates the passthrough of output to the personal tax base below one. The same friction that makes the cross-elasticity larger than the frictionless prediction makes the personal base elasticities smaller. It exceeds one whenever $F > 0$ since non-labor income must cover both fixed costs F and profits π, so the percentage change in profits exceeds the percentage change in output by a factor of $(1-\gamma)/s_\pi$.[18] This is a lower bound on effective leverage because it assumes the labor bill freely adjusts. Unlike profits, wages have no leverage. When compensation adjusts freely, $d\log(wL) = d\log Y$, so a percent increase in output fully passes through to the personal tax base. When wages are sticky, the passthrough is attenuated further below one. In other words, the same friction that raises leverage on the corporate tax base lowers passthrough on the personal income tax base.

Tax Cuts and the Response of Output. The leverage result shows that the elasticity of the corporate base with respect to any shock ε_{CX} decomposes into the response of output to that shock, amplified by the factor $(1-\gamma)/s_\pi$:

$$\varepsilon_{CL} = \frac{1-\gamma}{s_\pi} \times \varepsilon_{YL} \qquad \text{and} \qquad \varepsilon_{CC} = \frac{1-\gamma}{s_\pi} \times \varepsilon_{YC}$$

where ε_{YL} is the elasticity of output with respect to the log personal retention rate. To evaluate the contribution of leverage to the response of the corporate base, we need to know both ε_{YL} and ε_{YC}. These are not directly estimated in the baseline specification, but I extend them to include GDP, government dept, and government spending. That allows direct estimation of both ε_{YL} and ε_{YC}. Across estimation methodologies, I find that both ε_{YL} and ε_{YC} are robustly positive, and that the personal response is the larger of the two. Under the Bayesian local projection, $\varepsilon_{YL} = 0.96$ and $\varepsilon_{YC} = 0.85$. The ordering is intuitive. Personal tax cuts raise labor supply through a higher after-tax wage, which immediately raises output. Corporate tax cuts lower the user cost of capital, but capital accumulation is slow and gradual, and the direct effect is confined to the corporate sector's investment margin. The output response to corporate tax cuts is therefore smaller and more back-loaded.

Since leverage applies equally to both shocks, the full elasticity matrix is determined by two objects: the leverage factor $(1-\gamma)/s_\pi$ and the output elasticities. For the corporate base, the predictions are given above. For the personal base, there is no leverage since wages should scale appropriately with output, so

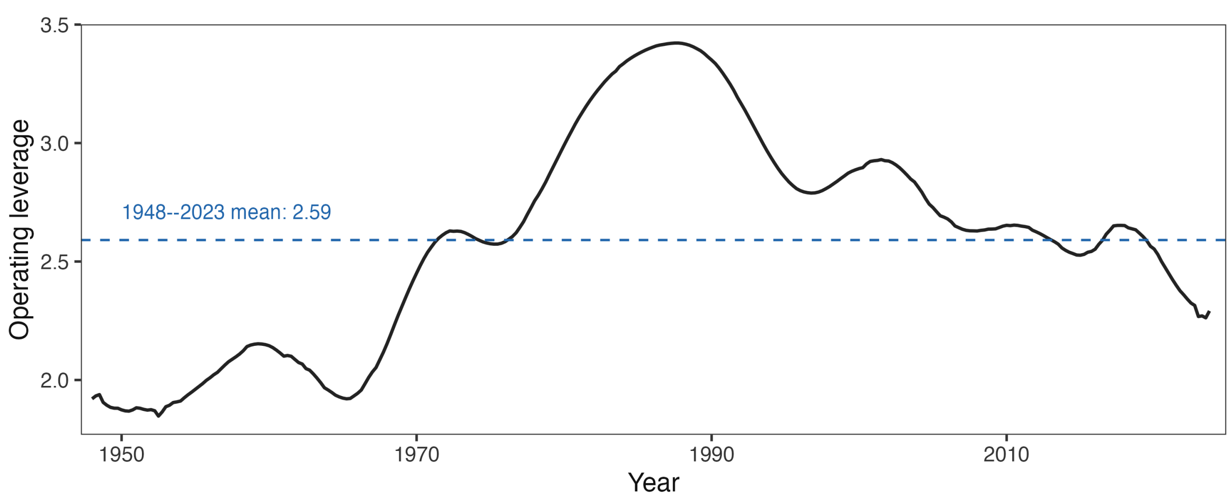
$$\varepsilon_{LL} \approx \varepsilon_{YL} \qquad \text{and} \qquad \varepsilon_{LC} \approx \varepsilon_{YC}.$$

18. All that we require for the argument to work is that some portion of the cost structure is predetermined. This is almost trivially satisfied. Capital is subject to fixed adjustment costs, with the majority of firms holding their capital stock fixed in any given quarter (Winberry 2021; Koby and Wolf 2020). Employment adjusts in infrequent, discrete changes rather than continuously (Caballero, Engel, and Haltiwanger 1997). Leases, debt service, and administrative overhead are fixed by construction.

Both γ and s_π are directly observable from NIPA Table 1.14 (Gross Value Added of Domestic Corporate Business): γ is compensation of employees divided by GVA, and s_π is corporate profits with inventory valuation and capital consumption adjustments divided by GVA. Figure D.3 plots the leverage series over time. It ranges from approximately 1.9 to 3.4, with a postwar mean of 2.59; the empirical leverage from the extended specification ($\varepsilon_{CL}/\varepsilon_{YL} = 9.3$) lies above it at every date, consistent with wage stickiness raising effective leverage above the frictionless benchmark. Leverage was lowest in the early postwar period and peaked in the 1980s. Table 2 compares the frictionless predictions to the data, using the postwar mean leverage of 2.59.

The frictionless model matches the qualitative ordering $\varepsilon_{CL} > \varepsilon_{CC} > \varepsilon_{LL} > \varepsilon_{LC}$ and delivers a cross-elasticity of the right sign and order of magnitude, $2.59 \times 0.96 \approx 2.5$ against a baseline estimate of 8.89, using only national accounts arithmetic and the output elasticities from the extended specification. The frictionless prediction is a floor rather than a point prediction, and the gap between it and the data is the part attributable to sluggish wage adjustment. The systematic pattern in the residuals confirms the wage stickiness channel introduced above. The frictionless model overpredicts personal base elasticities and underpredicts corporate base elasticities in every entry—exactly the signature of slow wage adjustment. Empirical leverage exceeds the frictionless benchmark for both shocks: 9.3 versus 2.59 for the personal tax shock and 4.8 versus 2.59 for the corporate tax shock. The gap is larger for the personal shock because its output response is more front-loaded, so high short-run leverage has more to amplify. A model with partial wage adjustment would close the remaining gap between predictions and data. The frictionless benchmark is not intended as a complete account; it is intended to show that the cross-elasticity is a predictable consequence of operating leverage and the output multiplier asymmetry, not of exotic behavioral responses.

Figure D.3: Leverage in the National Income and Product Accounts



Note: Leverage is computed using Table 1.14 from the National Income and Product Accounts. I compute the compensation share γ as compensation of employees (line 7) divided by gross value added of corporate business (line 1), and the profit share s_π as corporate profits with inventory valuation and capital consumption adjustments (line 14) divided by gross value added of corporate business. Raw leverage is $(1 - \gamma)/s_\pi$. I remove fluctuations at business cycle frequencies (2–40 quarters) using the Christiano and Fitzgerald (2003) filter and plot the resulting trend.

E Robustness of the Progressivity Semi-Elasticities

Table E.1: Semi-Elasticities under Alternative System Methods

	Bottom 99	Top 1	Corporate base	Tariff base
BLP	-0.22	-2.22	4.03	7.58
	(-0.69, 0.36)	(-3.69, -1.02)	(1.98, 10.47)	(4.63, 17.46)
LP	-0.11	-1.58	4.02	7.91
	(-0.79, 1.34)	(-3.03, 3.42)	(1.85, 12.93)	(3.44, 29.06)
Smooth LP	-0.11	-1.58	4.02	7.91
	(-0.79, 1.34)	(-3.03, 3.42)	(1.85, 12.93)	(3.44, 29.06)
Bias-corrected LP	0.18	-1.31	2.26	6.54
	(-3.30, 4.07)	(-11.76, 11.00)	(-9.63, 14.27)	(-26.58, 46.41)
BVAR	-0.11	-2.51	-0.25	1.90
	(-0.28, 0.17)	(-3.15, -2.16)	(-1.19, 0.36)	(1.19, 3.00)
SVAR-IV	-0.04	-1.70	5.63	5.58
	(-5.78, 8.27)	(-13.68, 15.46)	(-31.59, 61.79)	(-44.05, 72.43)

Note: Posterior medians with 90-percent credible intervals in parentheses. Entries are the percentage change in the discounted five-year response of each tax base to a one-percentage-point increase in the top-1-minus-bottom-99 average marginal tax-rate gap, holding initial personal tax payments fixed, evaluated draw by draw at each year's observed rates and exposure weights and averaged over 1948–2023. The personal block has two groups, identified by two narrative instruments over nine reform years, conditioning on the annual corporate and tariff narrative measures. The LP rows hold the first-stage impact basis fixed and vary only propagation. SVAR-IV is a separate VARX identified by the Mertens–Montiel Olea triangular restriction, with a wild bootstrap.

Table E.2: Semi-Elasticities under Direct LP–IV

	Bottom 99	Top 1	Corporate base	Tariff base
BLP	-0.22	-2.22	4.03	7.58
	(-0.69, 0.36)	(-3.69, -1.02)	(1.98, 10.47)	(4.63, 17.46)
LP-IV	0.62	-1.76	-1.94	-6.66
	(-5.20, 6.05)	(-16.53, 13.22)	(-42.20, 54.33)	(-54.32, 73.26)
Smooth LP-IV	0.62	-1.76	-1.94	-6.66
	(-5.20, 6.05)	(-16.53, 13.22)	(-42.20, 54.33)	(-54.32, 73.26)
Bias-corrected LP-IV	0.74	-1.35	-1.32	-6.60
	(-4.43, 6.26)	(-13.31, 11.37)	(-39.20, 46.39)	(-73.01, 57.09)

Note: Posterior medians with 90-percent credible intervals in parentheses. Entries are the percentage change in the discounted five-year response of each tax base to a one-percentage-point increase in the top-1-minus-bottom-99 average marginal tax-rate gap, holding initial personal tax payments fixed, evaluated draw by draw at each year's observed rates and exposure weights and averaged over 1948–2023. The personal block has two groups, identified by two narrative instruments over nine reform years, conditioning on the annual corporate and tariff narrative measures. Each row instruments the two personal marginal-rate equations directly with the narrative measures; no first-stage impact basis is reused.

Table E.3: Semi-Elasticities under Alternative System and Deterministic Specifications

	Bottom 99	Top 1	Corporate base	Tariff base
BLP	-0.22	-2.22	4.03	7.58
	(-0.69, 0.36)	(-3.69, -1.02)	(1.98, 10.47)	(4.63, 17.46)
Expanded macro system	-0.17	-2.18	2.27	4.33
	(-0.46, 0.10)	(-2.92, -1.62)	(1.41, 3.71)	(3.02, 6.88)
MMO trends and dummies	-0.41	-2.27	2.44	8.30
	(-1.71, 0.56)	(-5.27, 0.54)	(-1.08, 11.45)	(2.24, 36.89)
Macro + MMO trends	-0.19	-0.51	2.00	6.83
	(-1.13, 0.44)	(-1.91, 1.59)	(0.20, 6.87)	(3.87, 20.43)
Two annual lags	-0.10	-1.98	3.83	11.25
	(-5.53, 4.45)	(-9.84, 6.32)	(-36.08, 31.77)	(-92.26, 79.84)

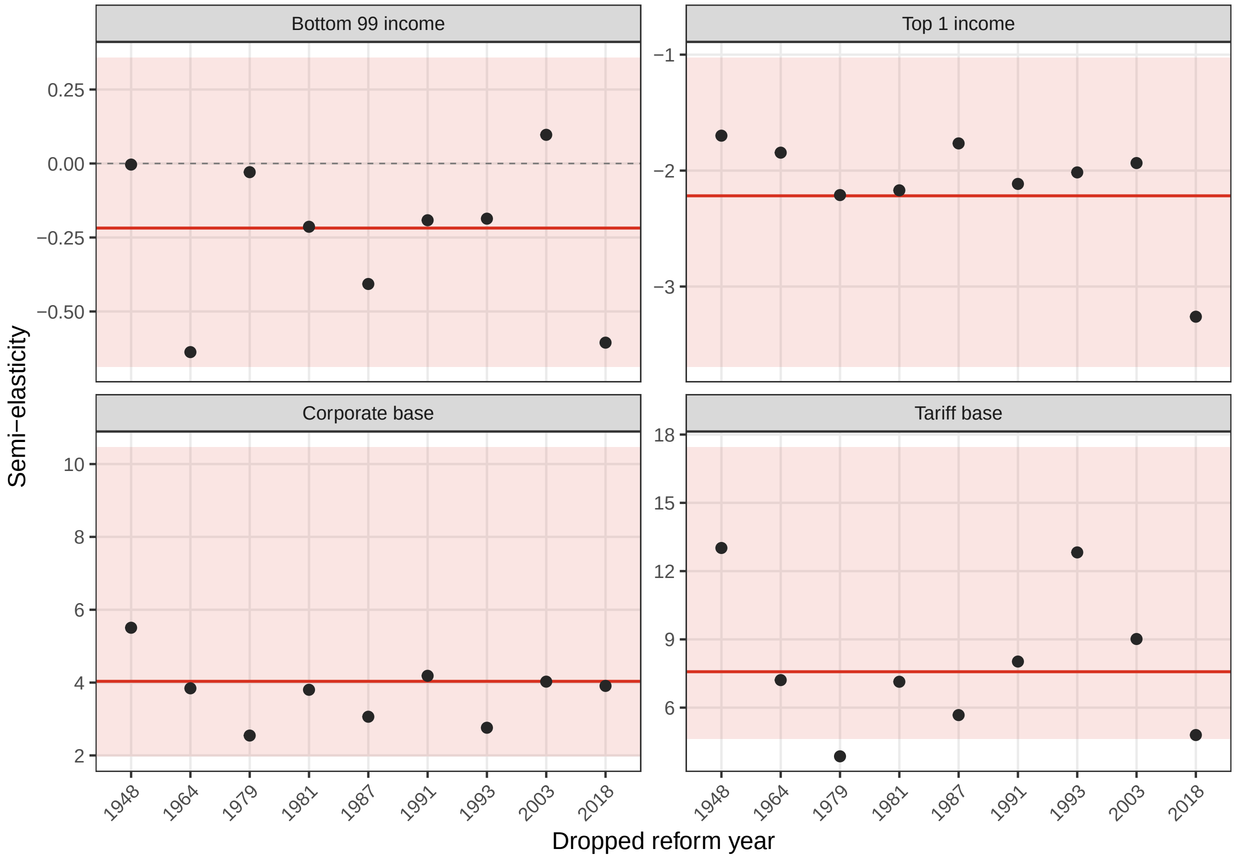
Note: Posterior medians with 90-percent credible intervals in parentheses. Entries are the percentage change in the discounted five-year response of each tax base to a one-percentage-point increase in the top-1-minus-bottom-99 average marginal tax-rate gap, holding initial personal tax payments fixed, evaluated draw by draw at each year's observed rates and exposure weights and averaged over 1948–2023. The personal block has two groups, identified by two narrative instruments over nine reform years, conditioning on the annual corporate and tariff narrative measures. The macro system adds government spending, GDP, and federal debt to the endogenous state. The MMO rows add linear and quadratic trends and 1949 and 2008 dummies in both stages.

Table E.4: Semi-Elasticities under S-Corporation Reallocation

	Bottom 99	Top 1	Corporate base	Tariff base
BLP	-0.22	-2.22	4.03	7.58
	(-0.69, 0.36)	(-3.69, -1.02)	(1.98, 10.47)	(4.63, 17.46)
S-corporation	0.49	-1.47	7.49	7.24
	(0.01, 1.80)	(-2.73, 0.57)	(4.26, 19.83)	(4.05, 19.92)

Note: Posterior medians with 90-percent credible intervals in parentheses. Entries are the percentage change in the discounted five-year response of each tax base to a one-percentage-point increase in the top-1-minus-bottom-99 average marginal tax-rate gap, holding initial personal tax payments fixed, evaluated draw by draw at each year's observed rates and exposure weights and averaged over 1948–2023. The personal block has two groups, identified by two narrative instruments over nine reform years, conditioning on the annual corporate and tariff narrative measures. The second row reallocates S-corporation income between the personal and corporate bases before estimation.

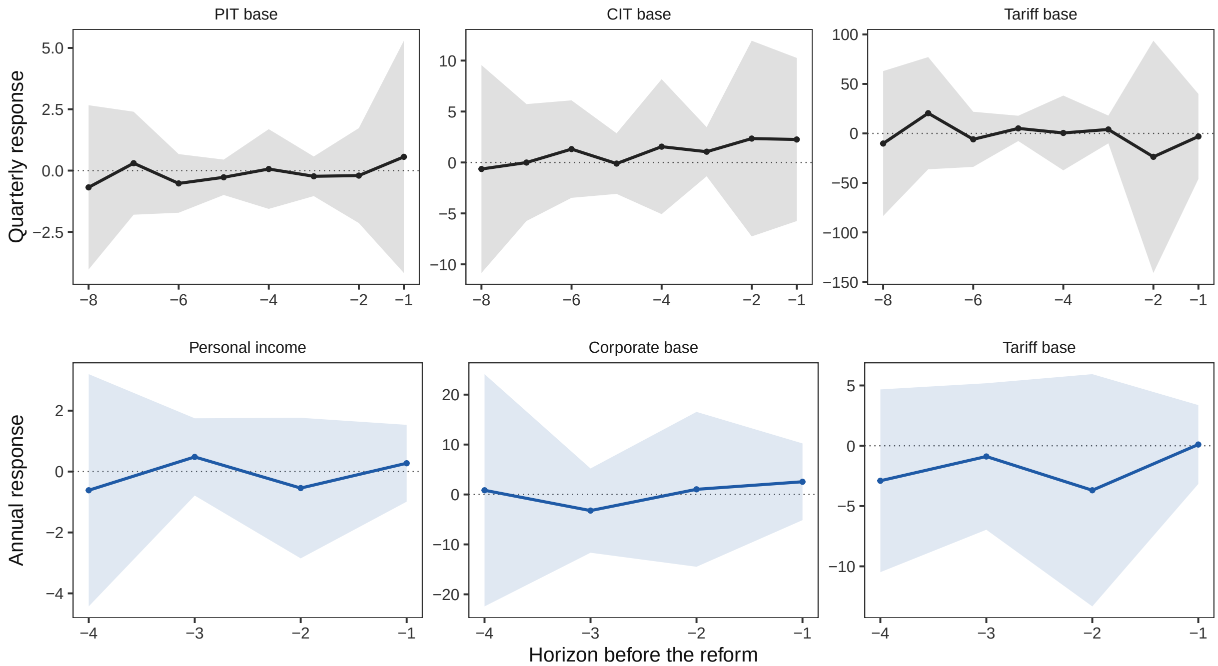
Figure E.1: Progressivity Semi-Elasticity Leave-One-Out



Note: This figure shows the semi-elasticity of tax bases to the progressivity gradient leaving out one reform at a time.

F Identification Diagnostics

Figure F.1: Future-Shock Placebos



Note: The top row reports quarterly own-base placebos for the personal, corporate, and tariff directions. The bottom row reports annual placebos for the retained personal progressivity direction. Lines are point estimates. Shaded regions are 90-percent blockwise simultaneous max-t bands from 1,000 Gaussian wild-bootstrap draws using common multipliers across the displayed coefficients, conditional on the observed joint first stage.

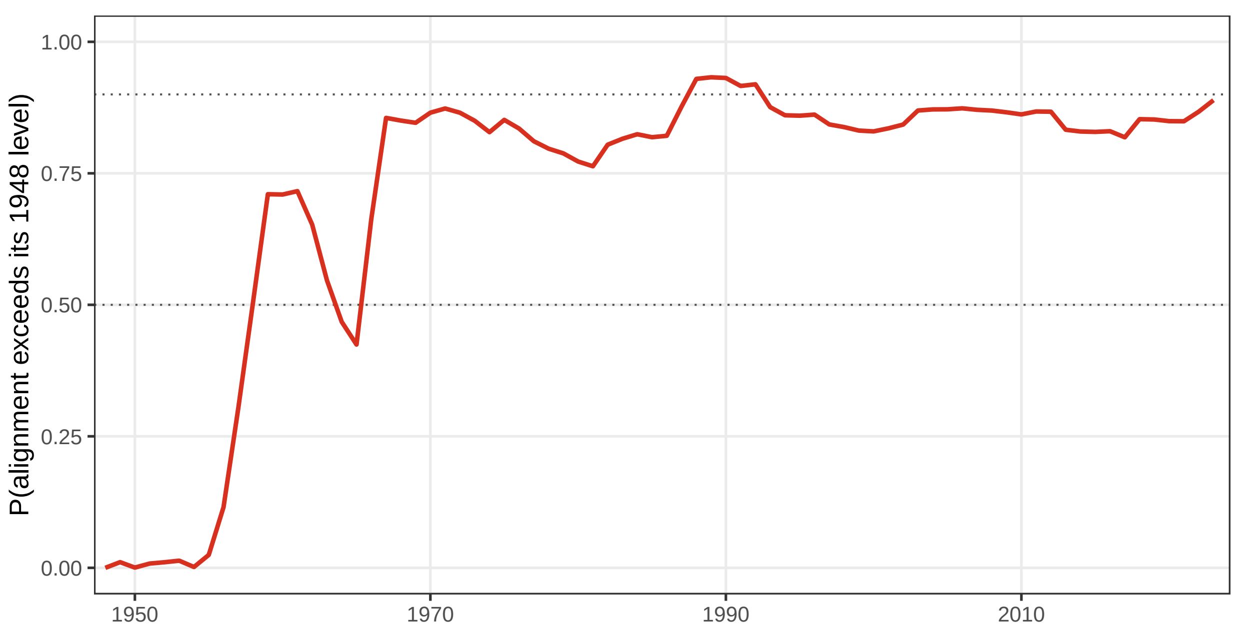
Table F.1: First-Stage Strength of Tax Instruments

	Effective F
Panel A. Quarterly broad-tax structural shocks	
Personal income tax	120.5 [37.5, 406.9]
Corporate income tax	9.3 [5.8, 15.8]
Tariff	11.0 [8.0, 16.7]
Panel B. Annual personal-tax rate innovations	
Bottom 99 percent	12.1 [8.7, 16.1]
Top 1 percent	137.4 [93.0, 157.6]

Note: Posterior medians, with 5th–95th posterior percentiles in brackets. Panel A reports the Montiel-Olea–Pflueger effective F for each recovered Mertens–Ravn structural shock, using one lag of the six quarterly states and Bartlett HAC(20). Panel B reports two-proxy effective F statistics for the annual BLP first-stage personal-rate innovations, whose rate coordinates are the bottom-99 and top-1 average marginal tax rates. It conditions the innovations and the two raw MMO measures on the annual corporate and tariff narrative measures and uses Bartlett HAC(5).

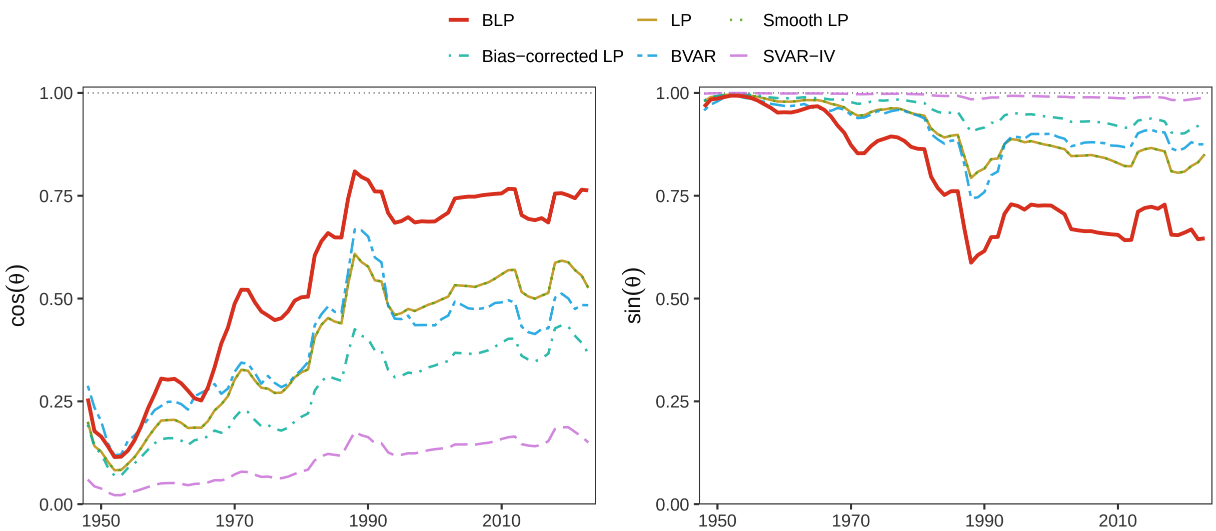
G Alignment Metric Robustness

Figure G.1: Posterior Probability that Alignment Exceeds Its 1948 Level



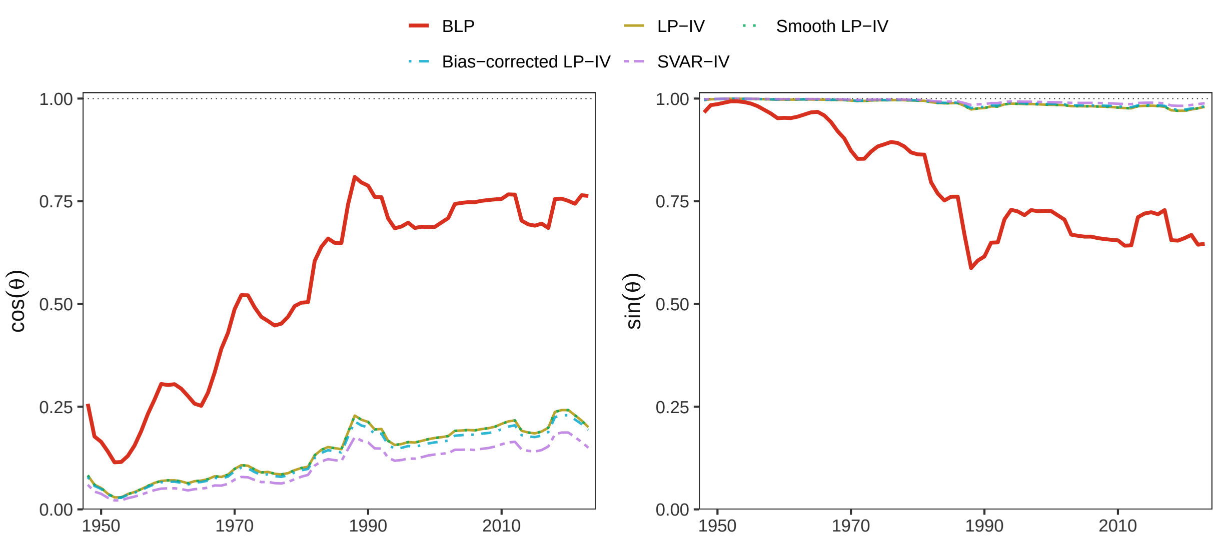
Note: At each date, the figure reports the share of posterior draws for which full alignment exceeds its value in 1948. The comparison is made draw by draw, retaining the joint uncertainty in the broad and progressivity components of the revenue gradient.

Figure G.2: Alignment under Alternative Dynamic Estimators



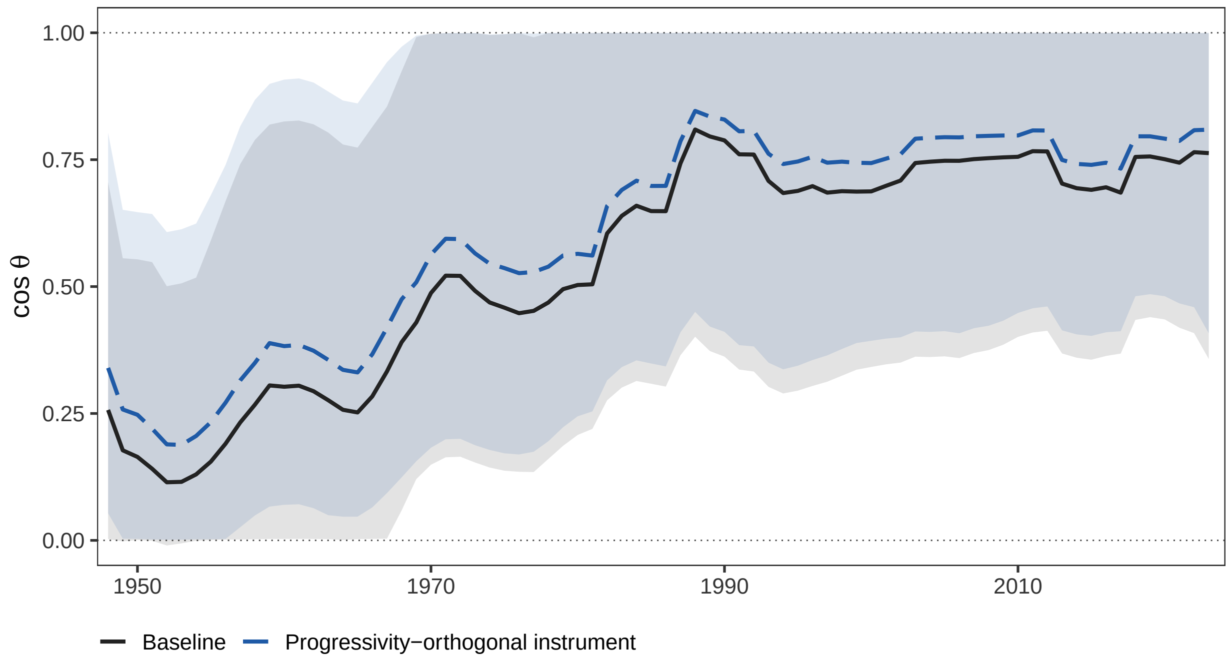
Note: The figure compares the baseline Bayesian local projection estimator with conventional, smooth, and bias-corrected local projections, a Bayesian VAR, and a proxy SVAR. The shaded region around the baseline is its 90-percent posterior credible interval; other series are posterior medians. Panel A reports $\cos\theta$ and Panel B reports $\sin\theta$.

Figure G.3: Alignment under Direct IV Estimation



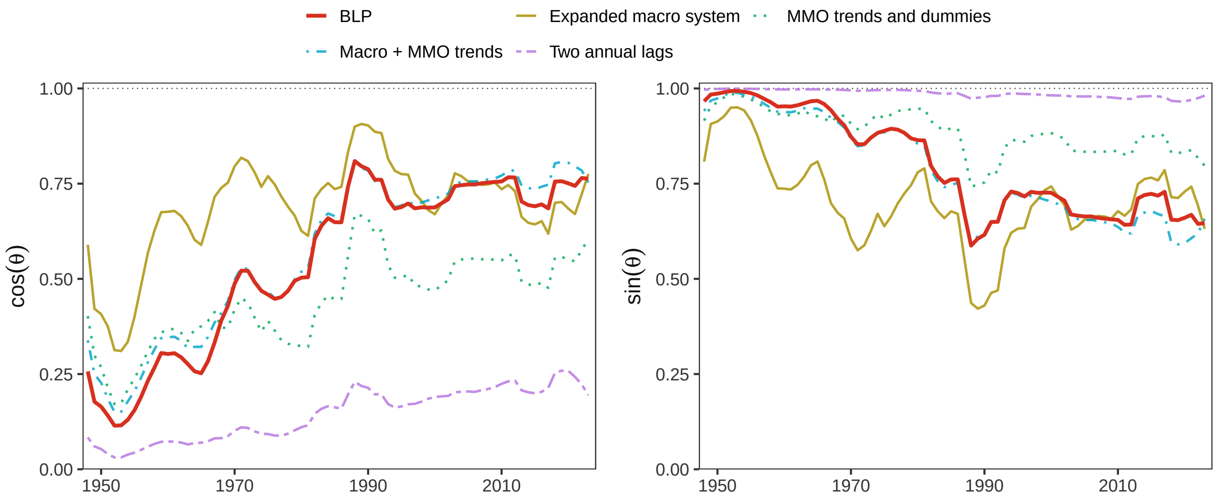
Note: The figure compares the baseline Bayesian local projection estimates with direct LP–IV, bias-corrected LP–IV, smooth LP–IV, and proxy-SVAR estimates. These alternatives address the possibility that the tax shocks are not invertible from the reduced-form innovations. The shaded region is the baseline 90-percent posterior credible interval; alternative series are posterior medians.

Figure G.4: Alignment under a Progressivity-Orthogonal Personal Instrument



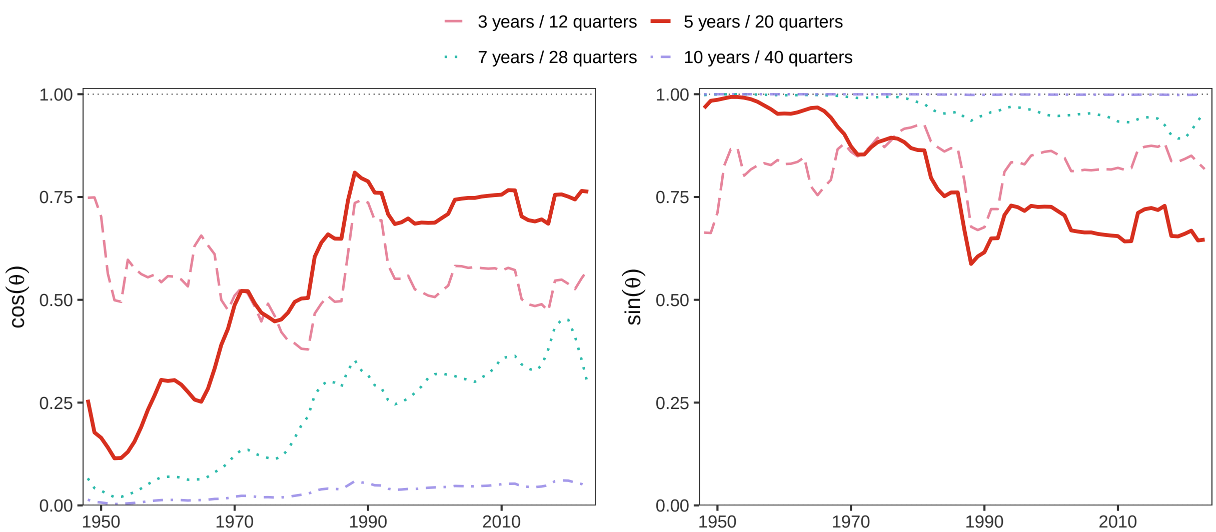
Note: The figure plots posterior medians and 90-percent credible intervals of $\cos\theta$ under the baseline quarterly instrument and under a personal narrative purged of its progressivity loading. The alternative instrument is the residual from projecting the quarterly personal narrative on the top-minus-lower bracket gap of the annual narrative record, without an intercept, which removes 36 percent of its variation and lowers the first-stage effective F for the personal coordinate from 120.5 to 50.4. Nine of the seventeen quarterly narrative events have an annual bracket record and receive their measured gap; the remaining eight have none and are assigned a loading of zero. The orthogonalized path lies above the baseline in all 76 years, with the paired posterior difference positive with probability at least 0.95 in every year.

Figure G.5: Alignment under Alternative Annual Specifications



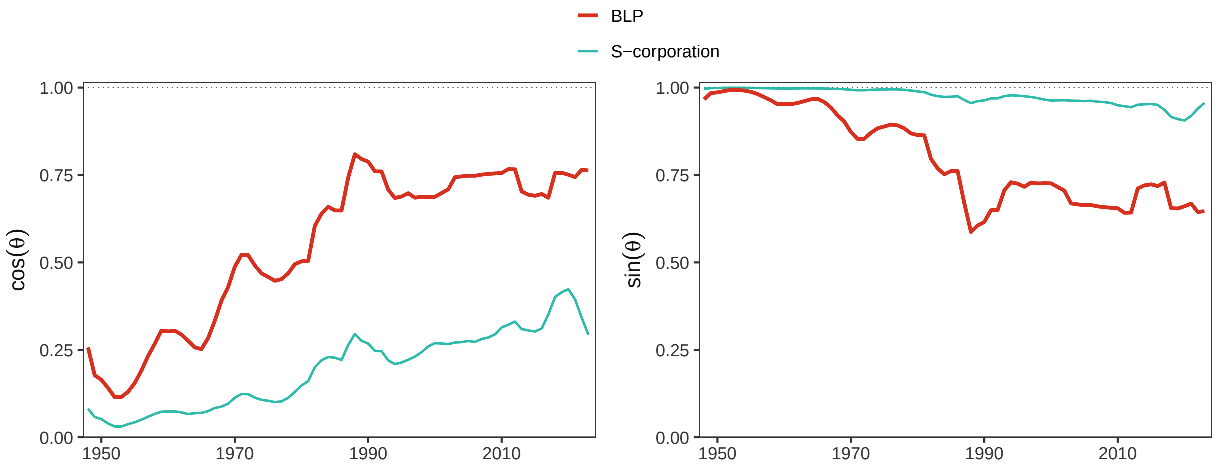
Note: The figure varies the specification of the annual progressivity block by adding aggregate controls, including the deterministic terms used by Mertens and Montiel Olea (2018), and adding a second annual lag. The baseline shaded region is its 90-percent posterior credible interval; alternative series are posterior medians.

Figure G.6: Alignment under Matched Present-Value Horizons



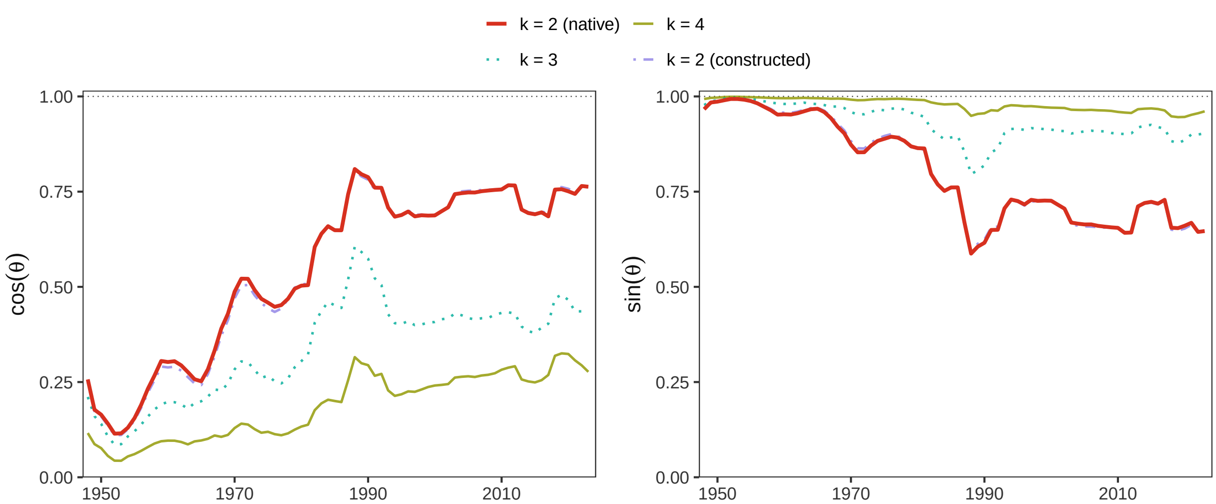
Note: The annual and quarterly response windows are varied together, so that each series uses the same present-value horizon in both empirical blocks. The baseline uses five annual observations and twenty quarterly observations. The shaded region is the baseline 90-percent posterior credible interval; alternative series are posterior medians.

Figure G.7: Alignment under Alternative Treatment of S-Corporation Income



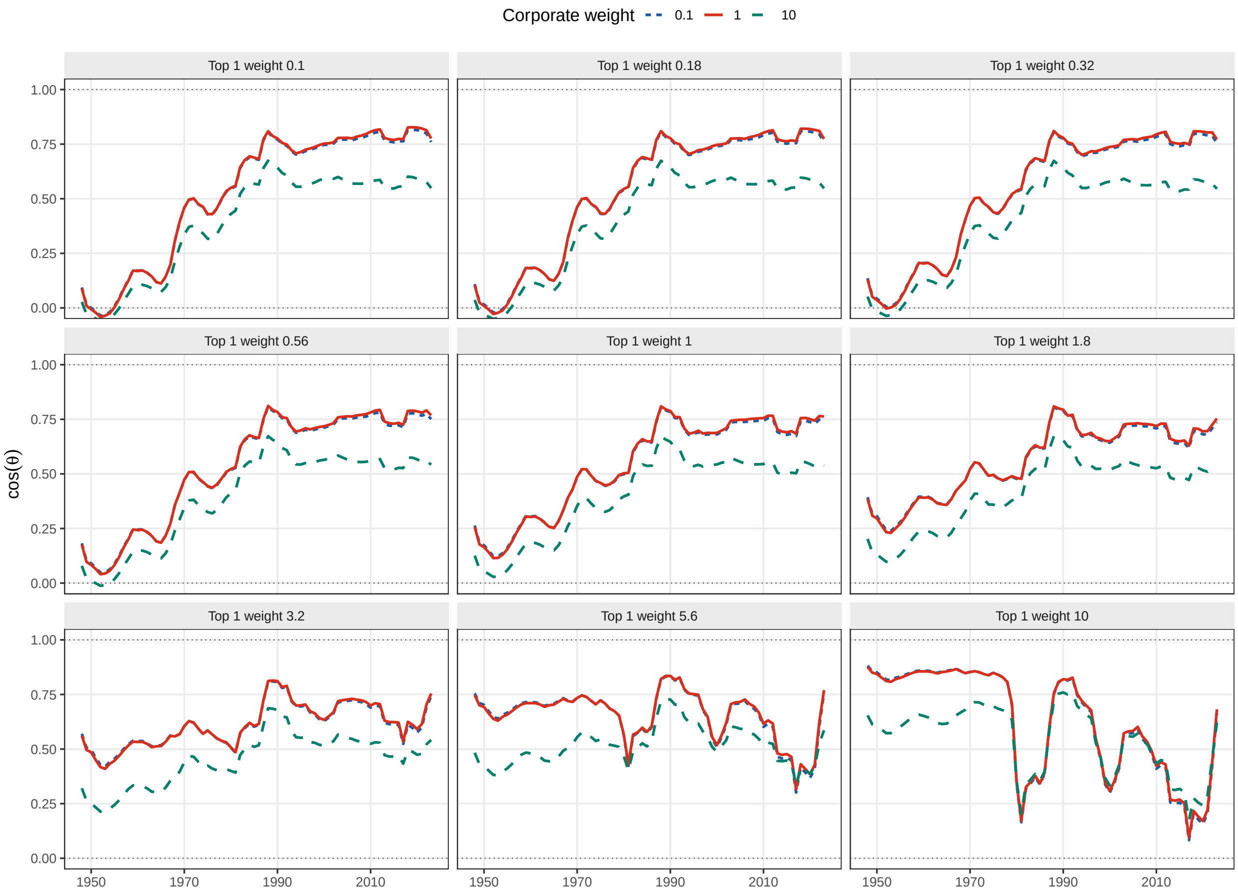
Note: The alternative series reallocates S-corporation income from the measured corporate base to the personal base before constructing the revenue gradient. The comparison assesses whether the baseline treatment of pass-through income affects the alignment calculation. The shaded region is the baseline 90-percent posterior credible interval.

Figure G.8: Alignment across Alternative Income-Group Partitions



Note: The figure compares the preferred native two-group representation of the personal tax schedule with a constructed two-group representation and with three- and four-group partitions. Each specification defines progressivity as a revenue-neutral redistribution of initial personal tax payments within its respective partition. The shaded region is the preferred specification's 90-percent posterior credible interval.

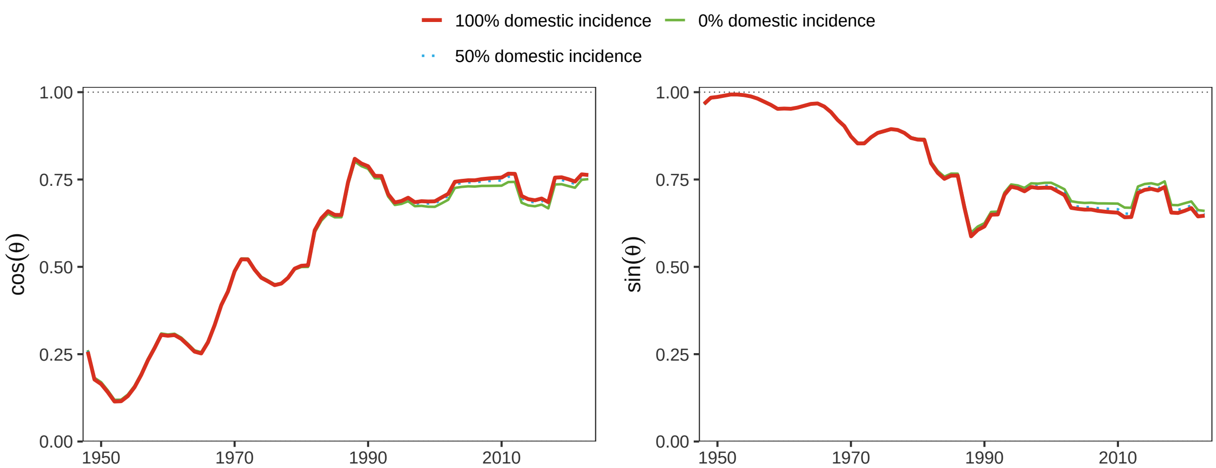
Figure G.9: Alignment under Alternative Social Welfare Weights



Note: Each panel fixes the social welfare weight on the top 1 percent relative to the lower 99 percent. Within each panel, the lines vary the corporate-income weight relative to the personal-income weight. The figure reports posterior medians; intervals are omitted to keep the full welfare-weight grid legible.

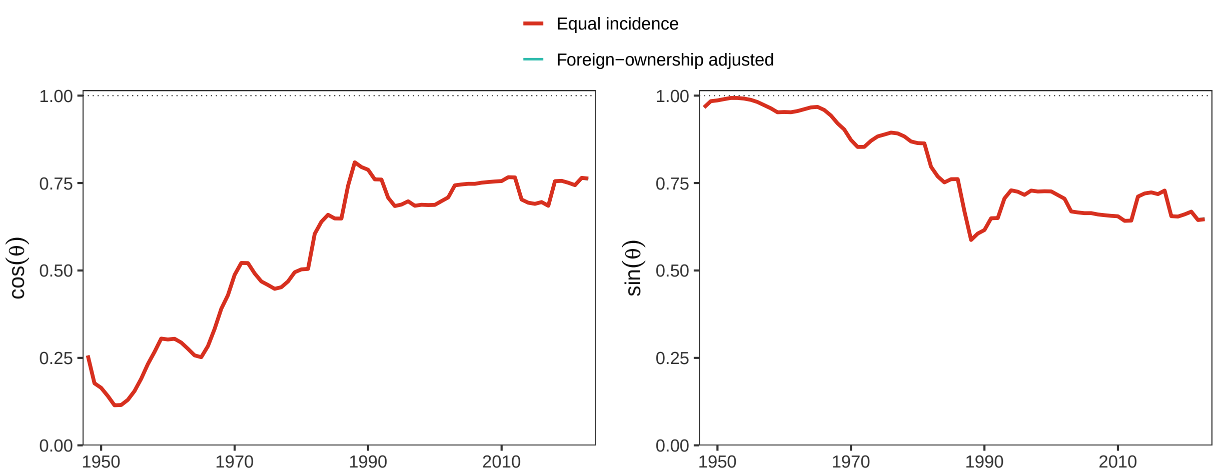
H Additional Robustness Figures

Figure H.1: Alignment under Alternative Assumptions about Tariff Incidence



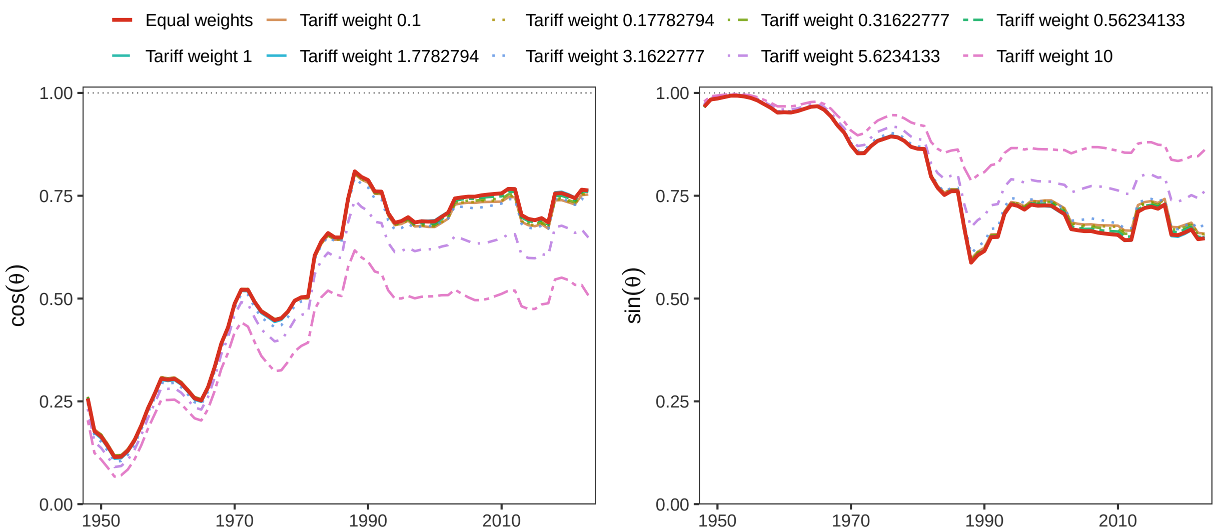
Note: The figure varies the fraction of the marginal tariff burden borne by domestic groups included in welfare. The three series correspond to zero, one-half, and full domestic incidence. The shaded region is the full-domestic-incidence baseline's 90-percent posterior credible interval.

Figure H.2: Alignment under an Adjustment for Foreign Corporate Ownership



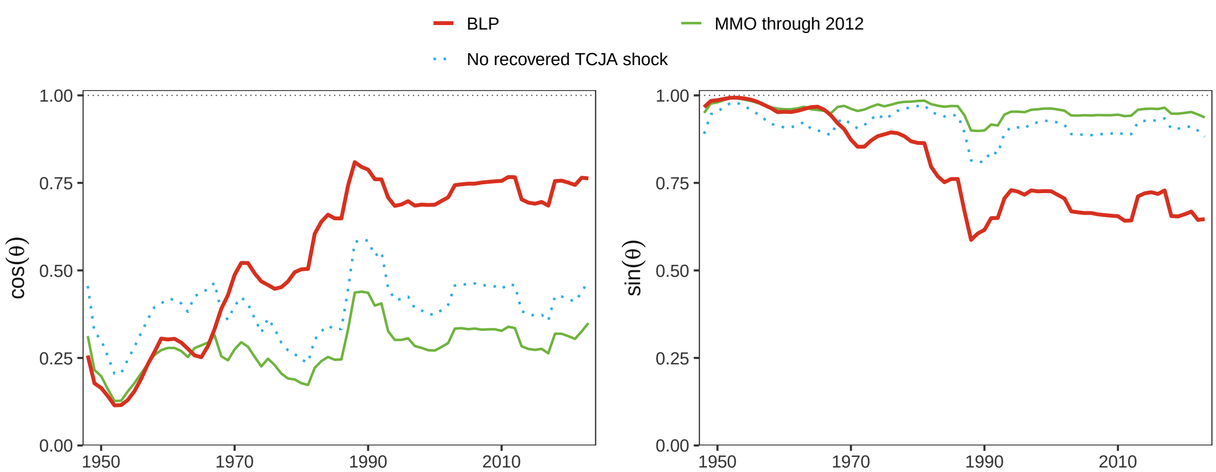
Note: The alternative series adjusts the welfare gradient for the share of corporate income accruing to foreign owners. The comparison isolates this incidence adjustment while holding the estimated revenue responses fixed. The shaded region is the equal-incidence baseline's 90-percent posterior credible interval.

Figure H.3: Alignment under Alternative Welfare Weights on Tariff Burdens



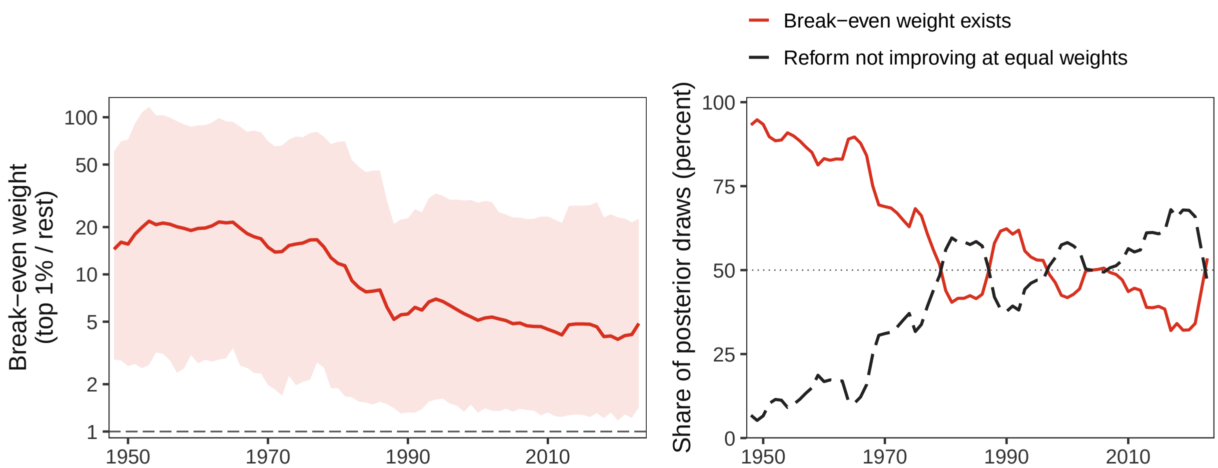
Note: The figure varies the social welfare weight on the tariff burden relative to the equal-weight benchmark, holding the personal and corporate weights fixed. The shaded region is the equal-weight baseline's 90-percent posterior credible interval; alternative series are posterior medians.

Figure H.4: Sensitivity to the Annual Extension and Recovered TCJA Shock



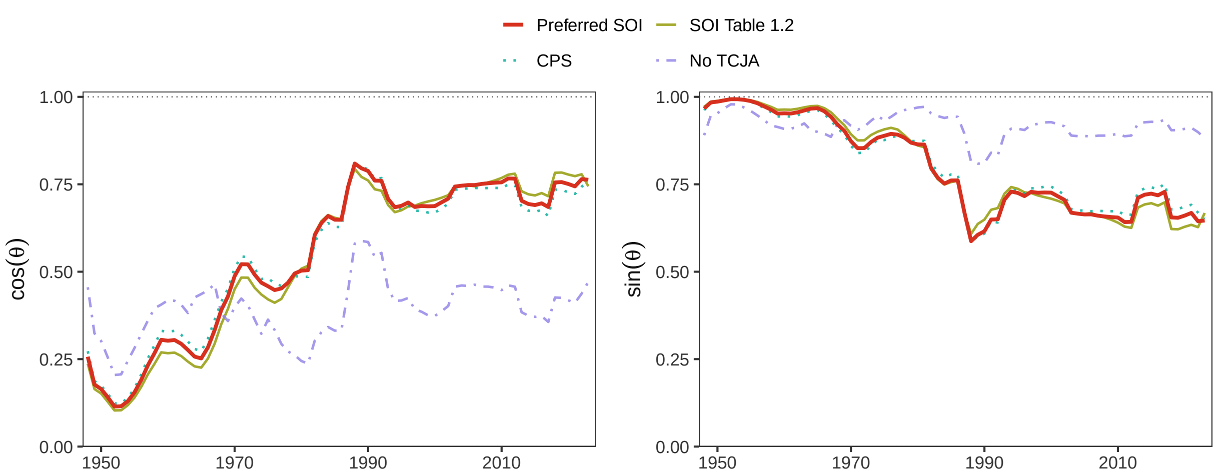
Note: The figure compares the preferred annual progressivity design with one restricted to the original Mertens–Montiel Olea sample through 2012 and with one that omits the recovered TCJA narrative measure. The shaded region is the preferred specification's 90-percent posterior credible interval.

Figure H.5: Welfare-Weight Threshold for the Progressivity Component



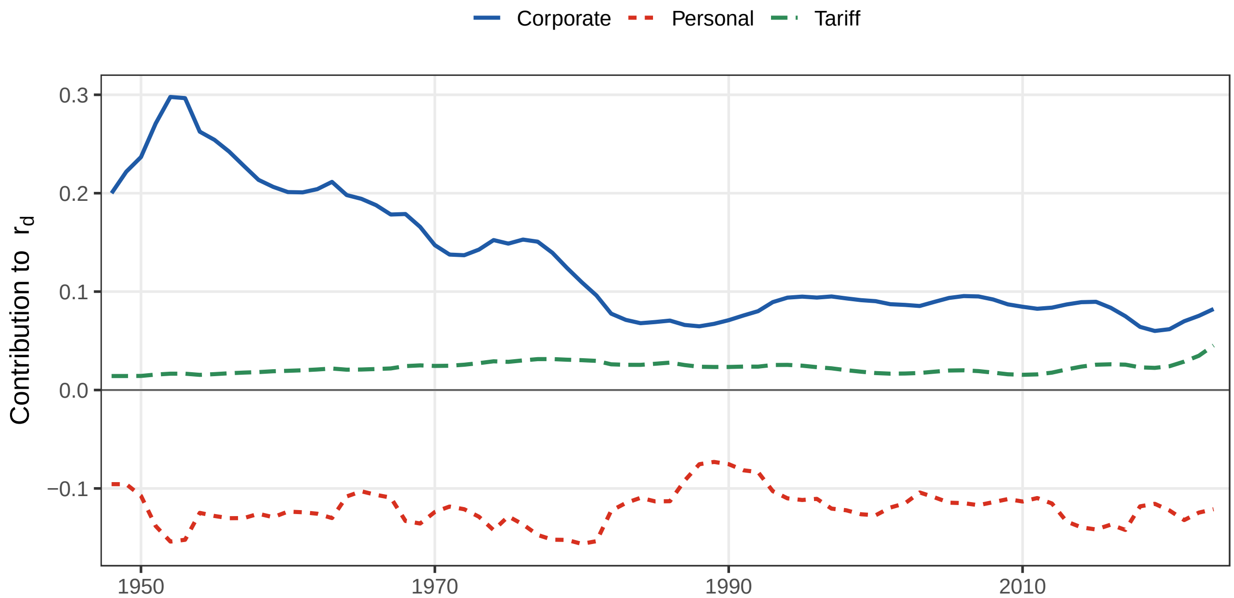
Note: Panel A reports the top-1-to-lower-99 welfare weight at which the locally optimal reform has a zero progressivity component, where such a threshold exists. The horizontal line marks equal welfare weights. Panel B reports the share of posterior draws with a finite threshold and the share for which the reform is not progressive under equal weights. The shaded regions in Panel A are 68-percent and 90-percent posterior credible intervals.

Figure H.6: Sensitivity to the Measurement of the TCJA Progressivity Shock



Note: The figure compares the preferred Statistics of Income construction of the 2018 TCJA measure with alternative SOI and Current Population Survey constructions, as well as a specification that omits the recovered TCJA measure. The shaded region is the preferred specification's 90-percent posterior credible interval.

Figure H.7: Revenue Channels of Personal Progressivity



Note: The figure decomposes r_d into personal, corporate, and tariff base contributions. Lines are posterior medians.